Modern Practical Control Systems

Section 1

System Modeling with Ideal Components

J.C. Vaught

Department of Mechanical Engineering
University of South Carolina

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Contents

Chapter 1

The Art and Science of Modeling

Learning Objectives

After completing this chapter, you will be able to:

- Understand the purpose and process of mathematical modeling
- Apply ideal sensor and actuator assumptions appropriately
- Derive differential equations from physical principles
- Convert between different model representations
- Recognize analogies between different physical domains

The foundation of control systems engineering is the mathematical model—a set of equations that describe how a system behaves. Before we can analyze stability, design controllers, or implement feedback, we must first capture the essential dynamics of the system we wish to control.

This chapter introduces systematic approaches to modeling physical systems across diverse engineering domains. We begin with ideal assumptions: perfect sensors that measure exactly what we want, and ideal actuators that respond instantly with unlimited capability. These assumptions let us focus on the fundamental physics before adding real-world complications in later chapters.

1.1 Why Model?

Mathematical models serve several critical purposes in control engineering:

1. **Prediction**: Models let us simulate system behavior before building hardware, saving time and money.
2. **Analysis**: We can mathematically determine stability, performance limits, and sensitivities.
3. **Design**: Controllers are designed based on models—better models lead to better controllers.
4. **Understanding**: The modeling process itself reveals what physical parameters matter most.

> *"All models are wrong, but some are useful."* — George E.P. Box

A model is always an approximation. The goal is not perfect fidelity but rather capturing the dynamics relevant to control at the frequencies and amplitudes of interest.

Historical Context: Watt's Flyball Governor and Early Feedback Control

The first automatic feedback controller was James Watt's flyball governor (1788), a mechanical device that regulated the speed of steam engines by adjusting fuel flow based on centrifugal force from rotating weights. When engine speed increased, the weights flew outward, reducing fuel flow; when speed decreased, they fell inward, increasing flow. This elegant mechanical feedback system enabled the Industrial Revolution by allowing steam engines to operate autonomously without constant human adjustment. Remarkably, no mathematical model existed for analyzing this system's stability—early engineers relied on trial-and-error design.

The mathematical analysis of feedback control began over 80 years later when James Clerk Maxwell, famous for electromagnetism, published a seminal paper in 1868 analyzing the stability of the flyball governor using differential equations. Maxwell showed that the governor's stability depended on system parameters and introduced the concept that a feedback system could become unstable if parameters were chosen incorrectly. His 1868 paper laid the mathematical foundation for all subsequent control theory, proving that governing equations could predict whether a system would oscillate or diverge—a revolutionary insight that transformed control from an art into a science.

1.2 The Modeling Process

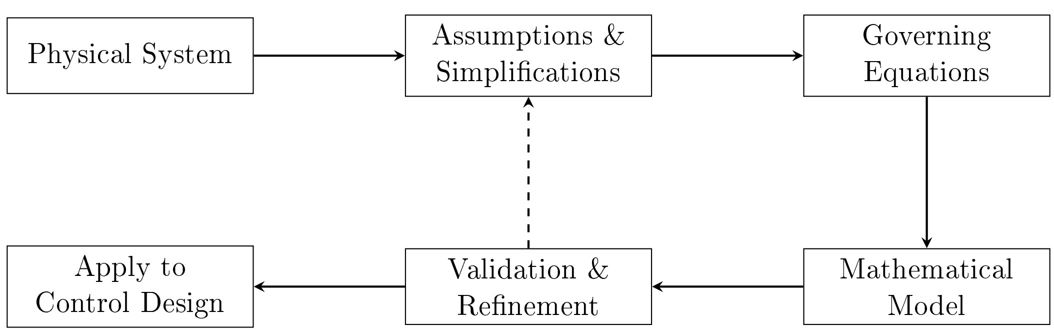

Figure 1.1: The iterative modeling process.

The modeling process typically involves:

1. **Define the system boundary**: What is included? What is external?
2. **Identify inputs and outputs**: What can we manipulate? What do we measure?
3. **List assumptions**: What simplifications are appropriate?
4. **Apply physical laws**: Conservation of energy, mass, momentum, charge, etc.
5. **Derive equations**: Combine physical laws into mathematical form.
6. **Validate**: Compare model predictions with experimental data.
7. **Refine**: Add complexity only where needed.

Chapter 2

Ideal Assumptions

Throughout this textbook, we begin with two fundamental idealizations that simplify analysis while capturing essential dynamics. Understanding these assumptions—and their limitations—is crucial for bridging the gap between elegant theory and robust practical implementation. This chapter establishes the ideal sensor and actuator models, contrasts them with real-world behavior, and examines the practical consequences of these idealizations.

Historical Context

The practice of idealizing components dates back to the earliest days of control theory. James Clerk Maxwell, in his 1868 paper "On Governors," analyzed Watt's centrifugal governor by assuming perfect mechanical linkages and instantaneous response. This approach—simplify to understand, then add complexity—remains the foundation of engineering analysis. Harold Black's invention of the negative feedback amplifier in 1927 similarly relied on ideal assumptions about amplifier gain to derive the fundamental feedback equation $H = G/(1 + GH)$.

2.1 The Role of Idealizations in Control Systems

Control system design follows a hierarchical approach: first, understand the fundamental dynamics using ideal assumptions; second, analyze the impact of non-idealities; third, design controllers robust to these imperfections. This methodology allows engineers to develop intuition about system behavior before confronting the full complexity of real implementations.

The Idealization Philosophy

Ideal assumptions serve three purposes in control engineering:

1. **Analytical tractability**: Simplified models yield closed-form solutions that reveal fundamental relationships.
2. **Design insight**: Ideal models expose the essential dynamics without obscuring them with secondary effects.
3. **Performance benchmarks**: Ideal behavior represents the best achievable performance, providing targets for real implementations.

The feedback control loop shown in Figure 2.1 contains three main components: the plant (physical system to be controlled), the sensor (measurement device), and the actuator (device that applies control action). In this chapter, we focus on idealizing the sensor and actuator, leaving plant dynamics as the primary subject of analysis.

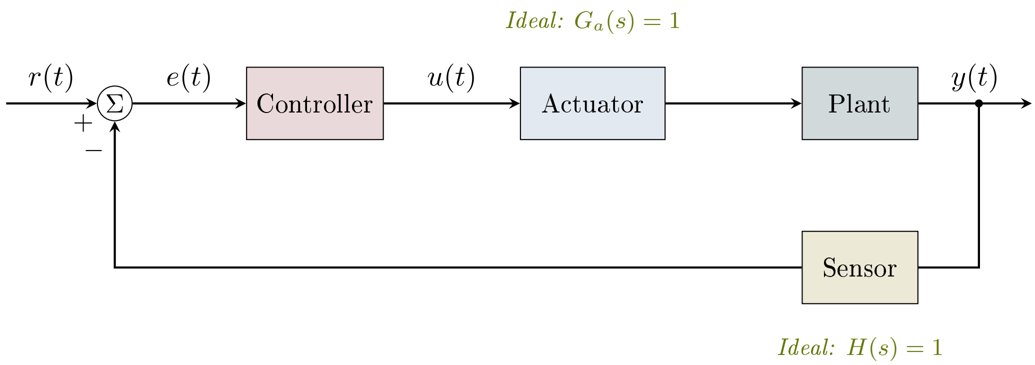

Figure 2.1: Standard feedback control loop showing the plant, controller, actuator, and sensor. Under ideal assumptions, the actuator and sensor transfer functions are unity.

2.2 Ideal Sensors

An **ideal sensor** provides a perfect measurement of the quantity of interest. This idealization assumes that the measurement process itself does not corrupt or delay the information being gathered.

2.2.1 Mathematical Definition

For an ideal sensor measuring quantity $x(t)$, the sensor output is:

$$\boxed{y_{\text{measured}}(t) = x(t)} \tag{2.1}$$

This deceptively simple equation embodies several profound assumptions.

2.2.2 Properties of Ideal Sensors

- **No delay**: The measurement is instantaneous ($\tau = 0$)
- **No noise**: The measurement contains no random errors ($n(t) = 0$)
- **No bias**: The measurement has no systematic offset ($b = 0$)
- **Infinite bandwidth**: All frequencies are captured equally
- **Infinite resolution**: Any value can be measured with infinite precision
- **No loading**: The sensor does not affect the measured quantity
- **Linear response**: Output is directly proportional to input

2.2.3 Transfer Function Representation

In the Laplace domain, an ideal sensor has the transfer function:

$$H_{\text{sensor}}(s) = \frac{Y_{\text{measured}}(s)}{X(s)} = 1 \tag{2.2}$$

This unity transfer function means the sensor has infinite bandwidth (all frequencies pass equally) and zero phase shift (no delay at any frequency).

2.2.4 Real Sensor Model

In contrast to the ideal model, a real sensor introduces several non-idealities. The general real sensor model is:

$$\boxed{y_{\text{measured}}(t) = G_s \cdot x(t-\tau) + b + n(t)} \tag{2.3}$$

where:

- G_s is the sensor gain (ideally 1, but may vary with temperature, age, or operating point)
- τ is the measurement delay (time lag between true value and measurement)
- b is the bias or offset (systematic error)
- $n(t)$ is measurement noise (random error, often modeled as Gaussian)

Example 2.1: Temperature Sensor Comparison

Consider measuring the temperature of a chemical reactor at $T = 150°\text{C}$.
Ideal sensor:

$$T_{\text{measured}}(t) = 150°\text{C} \quad \text{(exactly, instantaneously)} \tag{2.4}$$

Real thermocouple (Type K):

- Gain error: $\pm 0.75\%$ of reading $\Rightarrow G_s \in [0.9925, 1.0075]$
- Time constant: $\tau \approx 0.5$ to 5 seconds (depending on sheath)
- Bias: $\pm 2.2°\text{C}$ or $\pm 0.75\%$ (whichever is greater)
- Noise: Typically 0.1–0.5°C RMS from electromagnetic interference

The real measurement might read:

$$T_{\text{measured}}(t) = 1.005 \times 150 \times (1 - \mathrm{e}^{-t/2}) + 1.5 + 0.3\sin(120\pi t) \tag{2.5}$$

where the exponential term represents first-order dynamics with a 2-second time constant, 1.5°C bias, and 60 Hz electrical noise.

2.2.5 Frequency Response of Real Sensors

Real sensors exhibit bandwidth limitations. A first-order sensor model has the transfer function:

$$H_{\text{sensor}}(s) = \frac{G_s}{1+\tau_s s}\mathrm{e}^{-\tau_d s} \tag{2.6}$$

where τ_s is the sensor time constant and τ_d is pure time delay. The magnitude response rolls off at high frequencies:

$$|H_{\text{sensor}}(j\omega)| = \frac{G_s}{\sqrt{1 + (\omega\tau_s)^2}} \tag{2.7}$$

The -3 dB bandwidth (where magnitude drops to $1/\sqrt{2}$ of DC value) is:

$$\omega_{BW} = \frac{1}{\tau_s} \quad \text{or} \quad f_{BW} = \frac{1}{2\pi\tau_s} \tag{2.8}$$

Example 2.2: Accelerometer Bandwidth Requirements

A vibration monitoring system must accurately measure oscillations up to 500 Hz. What sensor time constant is required?

Solution:

For accurate measurement, the sensor bandwidth should be at least 5–10 times the highest frequency of interest (to minimize phase distortion):

$$f_{BW} \geq 5 \times 500 = 2500 \text{ Hz} \tag{2.9}$$

Required time constant:

$$\tau_s \leq \frac{1}{2\pi \times 2500} = 63.7\ \mu\text{s} \tag{2.10}$$

A piezoelectric accelerometer with $\tau_s < 50\ \mu$s would be appropriate. A strain-gauge accelerometer with $\tau_s \approx 1$ ms would be inadequate, as it would attenuate and phase-shift the high-frequency components.

2.3 Ideal Actuators

An **ideal actuator** produces exactly the commanded output without limitation or delay. This idealization assumes infinite power, infinite bandwidth, and perfect mechanical transmission.

2.3.1 Mathematical Definition

For an ideal actuator receiving command $u(t)$, the actual output is:

$$\boxed{F_{\text{actual}}(t) = u(t)} \tag{2.11}$$

where the output may be force, torque, voltage, current, flow rate, or any other physical quantity depending on the actuator type.

2.3.2 Properties of Ideal Actuators

- **No delay**: Response is instantaneous
- **No dynamics**: No transient behavior in the actuator itself
- **Infinite bandwidth**: Can track any commanded signal perfectly
- **Unlimited magnitude**: No saturation limits on force/torque/power

- **Unlimited rate**: Can change output arbitrarily fast
- **No backlash or friction**: Perfect mechanical transmission
- **Bidirectional**: Can push and pull equally well

2.3.3 Transfer Function Representation

In the Laplace domain, an ideal actuator has the transfer function:

$$G_{\text{actuator}}(s) = \frac{F_{\text{actual}}(s)}{U(s)} = 1 \tag{2.12}$$

2.3.4 Real Actuator Limitations

Real actuators face fundamental physical constraints that limit their performance.

Saturation

Every actuator has maximum output limits. The saturation function is:

$$\boxed{F_{\text{actual}} = \text{sat}(u, F_{\min}, F_{\max}) = \begin{cases} F_{\max} & \text{if } u > F_{\max} \\ u & \text{if } F_{\min} \leq u \leq F_{\max} \\ F_{\min} & \text{if } u < F_{\min} \end{cases}} \tag{2.13}$$

For symmetric saturation limits ($F_{\max} = -F_{\min} = F_{\text{sat}}$):

$$F_{\text{actual}} = F_{\text{sat}} \cdot \text{sat}\left(\frac{u}{F_{\text{sat}}}\right) = F_{\text{sat}} \cdot \text{sgn}(u) \cdot \min\left(1, \left|\frac{u}{F_{\text{sat}}}\right|\right) \tag{2.14}$$

Rate Limiting

Actuators cannot change their output infinitely fast. Rate limiting is modeled as:

$$\boxed{\dot{F}_{\text{actual}} = \text{sat}\left(\frac{u - F_{\text{actual}}}{\tau_a}, -R_{\max}, R_{\max}\right)} \tag{2.15}$$

where $R_{\max}$ is the maximum slew rate and τ_a is the actuator time constant.

Combined Dynamics Model

A comprehensive real actuator model includes both dynamics and limits:

$$G_{\text{actuator}}(s) = \frac{K_a}{(1 + \tau_1 s)(1 + \tau_2 s)} \mathrm{e}^{-\tau_d s} \tag{2.16}$$

subject to saturation and rate limits applied to the output.

Example 2.3: DC Motor Actuator Limitations

A DC motor positioning system has the following specifications:

- Maximum torque: $\tau_{\max} = \pm 5$ N·m
- Maximum speed: $\omega_{\max} = 3000$ rpm $= 314$ rad/s
- Electrical time constant: $\tau_e = 2$ ms
- Mechanical time constant: $\tau_m = 50$ ms
- Maximum torque rate: $\dot{\tau}_{\max} = 500$ N·m/s

Analysis:
The transfer function from commanded to actual torque is approximately:

$$G_{\text{motor}}(s) = \frac{1}{(1 + 0.002s)(1 + 0.05s)} \approx \frac{1}{1 + 0.052s} \tag{2.17}$$

The bandwidth is:

$$f_{BW} = \frac{1}{2\pi \times 0.052} = 3.06 \text{ Hz} \tag{2.18}$$

For a step command from 0 to 5 N·m:

- Ideal actuator: Instantaneous step to 5 N·m
- Real actuator with dynamics only: $\tau(t) = 5(1 - \mathrm{e}^{-t/0.052})$ N·m
- Real actuator with rate limit: Rise time limited to $5/500 = 10$ ms minimum

The rate limit dominates for large step commands, while dynamics dominate for small signal tracking.

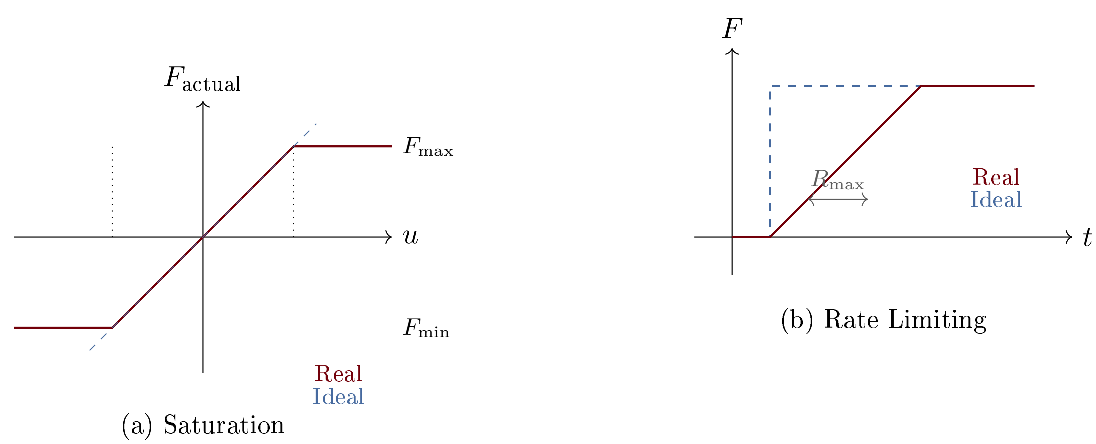

Figure 2.2: Actuator nonlinearities: (a) Saturation limits the maximum output magnitude regardless of command size. (b) Rate limiting restricts how fast the output can change, causing lag during rapid command changes.

2.4 Comparison: Ideal vs. Real Components

Understanding the gap between ideal and real component behavior is essential for practical control design. Table 2.1 summarizes key differences.

Property	**Ideal**	**Real**
Sensors		
Transfer function	$H(s) = 1$	$H(s) = \frac{G_s}{1+\tau s} e^{-\tau_d s}$
Bandwidth	∞	Finite (Hz to MHz)
Noise	None	Present (thermal, quantization)
Resolution	Infinite	Limited by ADC bits
Actuators		
Transfer function	$G(s) = 1$	$G(s) = \frac{K_a}{(1+\tau s)} e^{-\tau_d s}$
Output range	$\pm\infty$	$[F_{\min}, F_{\max}]$
Slew rate	∞	$\pm R_{\max}$
Power	Unlimited	Limited by source

Table 2.1: Comparison of ideal and real sensor/actuator characteristics.

2.4.1 When Ideal Assumptions Are Valid

The ideal assumptions become reasonable approximations when:

1. **Plant dynamics are slow**: If the plant time constants are much larger than sensor/actuator time constants, the latter can be neglected.
2. **Operating within linear region**: When commands stay well within saturation limits, nonlinearities are avoided.
3. **Signal-to-noise ratio is high**: When measured signals are large compared to noise.
4. **Sampling is fast**: When digital sampling rate is much higher than signal bandwidth.

Rule of Thumb for Neglecting Dynamics

A dynamic element can often be approximated as ideal if its bandwidth is at least 5–10 times higher than the closed-loop bandwidth of the control system. For example, if a position control loop has a 10 Hz bandwidth, sensors and actuators with bandwidth exceeding 50–100 Hz can typically be modeled as ideal.

Example 2.4: Validity of Ideal Assumptions

Consider a temperature control system for a large industrial furnace:

- Furnace thermal time constant: $\tau_{\text{furnace}} = 30$ minutes $= 1800$ s
- Thermocouple time constant: $\tau_{\text{sensor}} = 5$ s
- Gas valve time constant: $\tau_{\text{actuator}} = 0.5$ s

Analysis:

Ratio of plant to sensor dynamics: $1800/5 = 360$
Ratio of plant to actuator dynamics: $1800/0.5 = 3600$
Both ratios greatly exceed 10, so ideal sensor and actuator assumptions are excellent approximations for this system. The furnace dynamics completely dominate the closed-loop behavior.
Contrast with a fast system:
For a hard disk drive head positioning system:

- Mechanical resonance: $f_n = 5$ kHz $\Rightarrow \tau_{\text{mech}} \approx 32\ \mu$s
- Position sensor (optical encoder): $\tau_{\text{sensor}} = 10\ \mu$s
- Voice coil actuator: $\tau_{\text{actuator}} = 50\ \mu$s

Ratios: $32/10 = 3.2$ and $32/50 = 0.64$
Here, sensor and actuator dynamics are comparable to plant dynamics and **cannot** be neglected. Ideal assumptions would lead to significant design errors.

2.5 Practical Consequences of Ideal Assumptions

Using ideal assumptions during design, then encountering real components during implementation, can lead to several practical problems.

2.5.1 Stability Margin Erosion

Phase lag from sensor and actuator dynamics reduces stability margins. A system designed with 45° phase margin assuming ideal components may have only 20° margin (or become unstable) when real component dynamics are included.

Important Note

Controllers designed assuming ideal sensors and actuators almost always require detuning (reducing gains) when implemented with real components. A common rule: design for 60° phase margin with ideal components to achieve 30–45° margin in practice.

2.5.2 Integrator Windup

When actuators saturate, integral control action continues accumulating error, causing large overshoots when the system returns to the linear region. This phenomenon, called **integrator windup**, does not occur with ideal (unsaturated) actuators.

2.5.3 Noise Amplification

High-gain controllers designed for ideal (noiseless) sensors will amplify measurement noise when implemented with real sensors, potentially causing actuator chatter and component wear.

2.5.4 Limit Cycling

The combination of saturation, rate limiting, and time delays can cause sustained oscillations (limit cycles) that would not occur with ideal components.

Practical Considerations

When transitioning from ideal models to real implementations:

1. Add sensor and actuator dynamics to your simulation model
2. Include saturation and rate limits
3. Inject realistic noise levels
4. Verify stability margins with the augmented model
5. Implement anti-windup schemes for integral control
6. Consider adding filtering to reduce noise sensitivity

2.6 Summary

Ideal sensor and actuator assumptions are powerful tools for initial control system analysis and design. The ideal sensor ($y = x$) and ideal actuator ($F = u$) models allow focus on plant dynamics without the complexity of auxiliary components. However, engineers must recognize when these assumptions break down:

- When plant dynamics are comparable to sensor/actuator dynamics
- When operating near saturation limits
- When noise levels are significant relative to signal levels
- When precise timing or phase relationships matter

The remainder of this textbook primarily uses ideal sensor and actuator assumptions, explicitly noting when non-idealities become important. Later chapters on robustness, implementation, and advanced control techniques address methods for handling real-world component limitations.

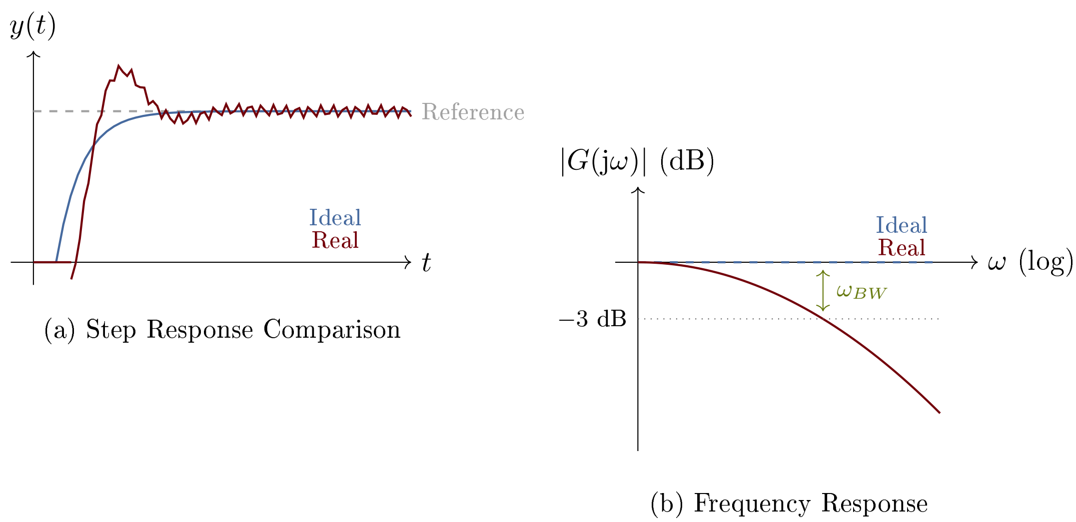

Figure 2.3: Comparison of ideal and real sensor/actuator behavior: (a) Time-domain step response showing delay, overshoot, and noise in the real system. (b) Frequency response showing the bandwidth limitation of real components versus the infinite bandwidth of ideal components.

Chapter 3

Physical Laws for Dynamic Systems

The mathematical models that describe dynamic systems arise from a remarkably small set of physical principles. These conservation laws—governing momentum, charge, energy, and mass—provide the foundation for deriving the differential equations that characterize system behavior. Mastering these principles enables engineers to model systems across mechanical, electrical, thermal, and fluid domains using a unified framework.

3.1 Conservation Laws Framework

All physical systems obey fundamental conservation principles. In control systems engineering, we exploit these principles to derive mathematical models that predict system behavior. The key insight is that conservation laws take a universal form: the rate of change of a conserved quantity equals the net flow into the system plus any internal generation.

The Four Conservation Laws

The dynamics of virtually all engineering systems can be derived from four conservation principles:

1. **Conservation of Momentum**: $\dfrac{\mathrm{d}p}{\mathrm{d}t} = \sum F$ (mechanical systems)
2. **Conservation of Charge**: $\dfrac{\mathrm{d}q}{\mathrm{d}t} = \sum i$ (electrical systems)
3. **Conservation of Energy**: $\dfrac{\mathrm{d}E}{\mathrm{d}t} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} + \dot{W}$ (thermal systems)
4. **Conservation of Mass**: $\dfrac{\mathrm{d}m}{\mathrm{d}t} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}}$ (fluid systems)

Each law states that the rate of accumulation of a conserved quantity equals the net inflow plus any sources.

The general form of any conservation law can be expressed as:

$$\frac{\mathrm{d}}{\mathrm{d}t}(\text{stored quantity}) = (\text{rate in}) - (\text{rate out}) + (\text{generation rate}) \tag{3.1}$$

This framework provides a systematic approach to modeling: identify the conserved quantities, draw system boundaries, and apply the appropriate conservation law to each element or node.

3.2 Newton's Laws for Mechanical Systems

Mechanical systems are governed by Newton's laws of motion, which relate forces to the acceleration of masses. These laws form the basis for modeling translational and rotational mechanical systems.

3.2.1 Translational Motion

Newton's Second Law for translational motion states that the sum of forces equals mass times acceleration:

$$\sum F = ma = m\frac{\mathrm{d}v}{\mathrm{d}t} = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} \tag{3.2}$$

In terms of momentum $p = mv$:

$$\sum F = \frac{\mathrm{d}p}{\mathrm{d}t} \tag{3.3}$$

This formulation is more general and applies even when mass varies with time.

3.2.2 Rotational Motion

The rotational analog of Newton's second law relates torques to angular acceleration:

$$\sum \tau = J\alpha = J\frac{\mathrm{d}\omega}{\mathrm{d}t} = J\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} \tag{3.4}$$

where J is the moment of inertia (kg·m^2), α is angular acceleration (rad/s^2), ω is angular velocity (rad/s), and θ is angular displacement (rad).

In terms of angular momentum $L = J\omega$:

$$\sum \tau = \frac{\mathrm{d}L}{\mathrm{d}t} \tag{3.5}$$

3.2.3 d'Alembert's Principle

An alternative formulation treats the inertial force as a pseudo-force, converting dynamics problems into equivalent statics problems:

$$\sum F - ma = 0 \quad \Rightarrow \quad \sum F + F_{\text{inertia}} = 0 \tag{3.6}$$

where $F_{\text{inertia}} = -ma$ is the inertial force. This principle is particularly useful for complex mechanical systems and forms the basis for Lagrangian mechanics.

Example 3.1: Free Body Diagram Analysis

A mass $m = 2$ kg rests on a surface with kinetic friction coefficient $\mu_k = 0.3$. A horizontal force $F(t)$ is applied. Derive the equation of motion.

Solution:

Drawing the free body diagram, the forces acting on the mass are:

- Applied force: $F(t)$ (horizontal, positive direction)
- Friction force: $f = \mu_k N = \mu_k mg$ (opposing motion)
- Normal force: $N = mg$ (vertical, balances weight)

- Weight: $W = mg$ (vertical, downward)

Applying Newton's second law in the horizontal direction:

$$m\ddot{x} = F(t) - \mu_k mg$$

Substituting values:

$$2\ddot{x} = F(t) - (0.3)(2)(9.81) = F(t) - 5.89$$

The equation of motion is: $\ddot{x} = 0.5F(t) - 2.94\ \mathrm{m/s^2}$

3.3 Constitutive Relations

Conservation laws alone are insufficient to model systems—we also need *constitutive relations* that describe how individual components behave. These relations connect effort variables (forces, voltages, pressures) to flow variables (velocities, currents, volumetric flow rates).

3.3.1 Mechanical Constitutive Relations

Linear Spring (stores potential energy):

$$F = k(x_1 - x_2) \quad \text{or} \quad F = kx \tag{3.7}$$

where k is the spring constant (N/m).

Viscous Damper (dissipates energy):

$$F = b(\dot{x}_1 - \dot{x}_2) \quad \text{or} \quad F = b\dot{x} \tag{3.8}$$

where b is the damping coefficient (N·s/m).

Mass (stores kinetic energy):

$$F = m\ddot{x} \tag{3.9}$$

3.3.2 Electrical Constitutive Relations

Resistor (dissipates energy):

$$v = iR \quad \text{or} \quad i = \frac{v}{R} = Gv \tag{3.10}$$

where R is resistance (Ω) and $G = 1/R$ is conductance (S).

Capacitor (stores energy in electric field):

$$i = C\frac{\mathrm{d}v}{\mathrm{d}t} \quad \text{or} \quad v = \frac{1}{C}\int i\,\mathrm{d}t \tag{3.11}$$

where C is capacitance (F).

Inductor (stores energy in magnetic field):

$$v = L\frac{\mathrm{d}i}{\mathrm{d}t} \quad \text{or} \quad i = \frac{1}{L}\int v\,\mathrm{d}t \tag{3.12}$$

where L is inductance (H).

3.3.3 Thermal Constitutive Relations

Thermal Resistance (Fourier's law for conduction):

$$\dot{Q} = \frac{T_1 - T_2}{R_{\text{th}}} = \frac{\Delta T}{R_{\text{th}}} \tag{3.13}$$

where R_{th} is thermal resistance (K/W).

Thermal Capacitance:

$$\dot{Q} = C_{\text{th}} \frac{\mathrm{d}T}{\mathrm{d}t} = m C_p \frac{\mathrm{d}T}{\mathrm{d}t} \tag{3.14}$$

where $C_{\text{th}} = mC_p$ is thermal capacitance (J/K).

3.3.4 Fluid Constitutive Relations

Fluid Resistance (laminar flow):

$$Q = \frac{P_1 - P_2}{R_f} = \frac{\Delta P}{R_f} \tag{3.15}$$

where R_f is fluid resistance (Pa·s/m^3) and Q is volumetric flow rate (m^3/s).

Fluid Capacitance (tank):

$$Q = C_f \frac{\mathrm{d}P}{\mathrm{d}t} \quad \text{where} \quad C_f = \frac{A}{\rho g} \tag{3.16}$$

where A is the tank cross-sectional area.

Example 3.2: Domain Analogies

A mechanical mass-spring-damper system has: $m = 1$ kg, $k = 100$ N/m, $b = 10$ N·s/m. Find the analogous electrical RLC circuit parameters.
Solution:
Using the force-voltage analogy:

Mechanical	$\leftrightarrow$	Electrical
Mass m	$\leftrightarrow$	Inductance L
Damping b	$\leftrightarrow$	Resistance R
Stiffness k	$\leftrightarrow$	Inverse capacitance $1/C$

The mechanical equation: $m\ddot{x} + b\dot{x} + kx = F$
The analogous electrical equation: $L\ddot{q} + R\dot{q} + \frac{1}{C}q = v$
Using impedance scaling with $\omega_n = \sqrt{k/m} = 10$ rad/s:
If we choose $L = 1$ H, then $R = 10\ \Omega$ and $C = 0.01$ F $= 10$ mF.
Both systems have identical natural frequency $\omega_n = 10$ rad/s and damping ratio $\zeta = 0.5$.

3.4 Kirchhoff's Laws for Electrical Systems

Electrical circuits are governed by Kirchhoff's laws, which express conservation of charge and energy in electrical networks.

3.4.1 Kirchhoff's Current Law (KCL)

The algebraic sum of currents entering any node equals zero:

$$\sum_{\text{node}} i_k = 0 \tag{3.17}$$

This is a direct consequence of conservation of charge—charge cannot accumulate at a node (except in capacitors, which are modeled separately).

3.4.2 Kirchhoff's Voltage Law (KVL)

The algebraic sum of voltages around any closed loop equals zero:

$$\sum_{\text{loop}} v_k = 0 \tag{3.18}$$

This expresses conservation of energy—the work done moving a charge around a closed path must be zero.

3.4.3 Systematic Circuit Analysis

For circuits with n nodes and b branches:

- Apply KCL at $(n-1)$ independent nodes
- Apply KVL around $(b-n+1)$ independent loops
- Use constitutive relations for each element

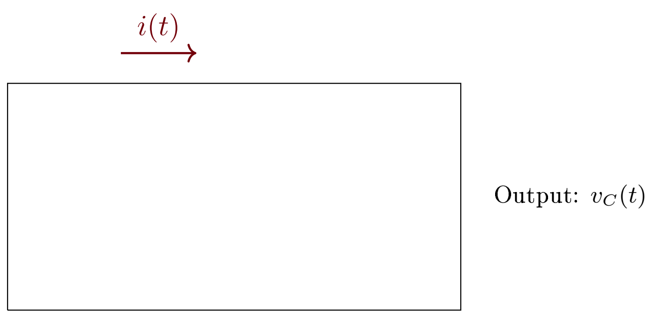

Figure 3.1: Series RLC circuit with input voltage $v_{\text{in}}(t)$ and output voltage $v_C(t)$ across the capacitor. Applying KVL: $v_{\text{in}} = v_R + v_L + v_C$.

Example 3.3: RLC Circuit Analysis

Derive the differential equation for the series RLC circuit shown in Figure 9.9.
Solution:
Applying KVL around the loop:

$$v_{\text{in}}(t) - v_R - v_L - v_C = 0$$

Substituting constitutive relations with $i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$:

$$v_{\text{in}} = iR + L\frac{\mathrm{d}i}{\mathrm{d}t} + v_C$$

Since $i = C\frac{\mathrm{d}v_C}{\mathrm{d}t}$:

$$v_{\text{in}} = RC\frac{\mathrm{d}v_C}{\mathrm{d}t} + LC\frac{\mathrm{d}^2 v_C}{\mathrm{d}t^2} + v_C$$

Rearranging into standard form:

$$\boxed{LC\frac{\mathrm{d}^2 v_C}{\mathrm{d}t^2} + RC\frac{\mathrm{d}v_C}{\mathrm{d}t} + v_C = v_{\text{in}}(t)}$$

In terms of natural frequency and damping ratio:

$$\omega_n = \frac{1}{\sqrt{LC}}, \quad \zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$$

3.5 Conservation of Energy

Energy conservation provides an alternative approach to deriving system equations and is essential for thermal systems and energy-based modeling methods.

3.5.1 General Energy Balance

For any system, the first law of thermodynamics states:

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} + \dot{W}_{\text{in}} - \dot{W}_{\text{out}} \tag{3.19}$$

where E is total stored energy, $\dot{Q}$ is heat transfer rate, and $\dot{W}$ is work rate (power).

3.5.2 Thermal Systems

For a thermal mass with no work interactions:

$$mC_p\frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} \tag{3.20}$$

Combining with Fourier's law for heat conduction:

$$mC_p\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{T_\infty - T}{R_{\text{th}}} \tag{3.21}$$

This yields a first-order system with time constant $\tau = mC_pR_{\text{th}} = C_{\text{th}}R_{\text{th}}$.

3.5.3 Energy Storage in System Elements

Each system element stores or dissipates energy:

Element	Energy Type	Expression	Storage/Dissipation
Mass	Kinetic	$E = \frac{1}{2}mv^2$	Storage
Spring	Potential	$E = \frac{1}{2}kx^2$	Storage
Damper	—	$P = bv^2$	Dissipation
Inductor	Magnetic	$E = \frac{1}{2}Li^2$	Storage
Capacitor	Electric	$E = \frac{1}{2}Cv^2$	Storage
Resistor	—	$P = i^2R$	Dissipation

Table 3.1: Energy storage and dissipation in common elements.

Example 3.4: Thermal System Modeling

A metal block ($m = 5$ kg, $C_p = 450$ J/(kg·K)) is placed in a furnace at $T_\infty = 500$°C. The thermal resistance between block and furnace is $R_{\text{th}} = 0.2$ K/W. Find the time constant and the time to reach 450°C from 25°C.

Solution:

The governing equation:

$$mC_p\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{T_\infty - T}{R_{\text{th}}}$$

Rearranging:

$$\tau\frac{\mathrm{d}T}{\mathrm{d}t} + T = T_\infty$$

Time constant:

$$\tau = mC_pR_{\text{th}} = (5)(450)(0.2) = 450 \text{ s} = 7.5 \text{ min}$$

The solution for step response:

$$T(t) = T_\infty - (T_\infty - T_0)\mathrm{e}^{-t/\tau} = 500 - 475\mathrm{e}^{-t/450}$$

To reach 450°C:

$$450 = 500 - 475\mathrm{e}^{-t/450} \quad \Rightarrow \quad \mathrm{e}^{-t/450} = \frac{50}{475} = 0.105$$

$$t = -450\ln(0.105) = 1013 \text{ s} \approx 16.9 \text{ min}$$

3.6 Conservation of Mass

Mass conservation governs fluid systems and chemical processes. The principle states that mass can neither be created nor destroyed (in non-nuclear processes).

3.6.1 General Mass Balance

For a control volume:

$$\frac{\mathrm{d}m}{\mathrm{d}t} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} \tag{3.22}$$

For incompressible fluids with constant density ρ:

$$\rho\frac{\mathrm{d}V}{\mathrm{d}t} = \rho Q_{\text{in}} - \rho Q_{\text{out}} \tag{3.23}$$

where Q is the volumetric flow rate (m^3/s).

3.6.2 Tank Level Systems

For a tank with cross-sectional area A and liquid height h:

$$A\frac{\mathrm{d}h}{\mathrm{d}t} = Q_{\text{in}} - Q_{\text{out}} \tag{3.24}$$

If outflow depends on head (gravity-driven through a valve):

$$Q_{\text{out}} = \frac{h}{R_f} \quad \text{or} \quad Q_{\text{out}} = C_d A_o \sqrt{2gh} \tag{3.25}$$

The linearized model for small deviations around operating point h_0:

$$A\frac{\mathrm{d}\Delta h}{\mathrm{d}t} = \Delta Q_{\text{in}} - \frac{\Delta h}{R_f} \tag{3.26}$$

Example 3.5: Two-Tank System

Two tanks are connected in series. Tank 1 has area $A_1 = 1\ \mathrm{m}^2$ and outflow resistance $R_1 = 100\ \mathrm{s/m}^2$. Tank 2 has area $A_2 = 2\ \mathrm{m}^2$ and outflow resistance $R_2 = 200\ \mathrm{s/m}^2$. Derive the transfer function from inlet flow Q_{in} to tank 2 level h_2.

Solution:

For Tank 1:

$$A_1\frac{\mathrm{d}h_1}{\mathrm{d}t} = Q_{\text{in}} - \frac{h_1}{R_1}$$

For Tank 2:

$$A_2\frac{\mathrm{d}h_2}{\mathrm{d}t} = \frac{h_1}{R_1} - \frac{h_2}{R_2}$$

Taking Laplace transforms:

$$H_1(s) = \frac{R_1}{A_1R_1s+1}Q_{\text{in}}(s) = \frac{100}{100s+1}Q_{\text{in}}(s)$$

$$H_2(s) = \frac{R_2}{A_2R_2s+1}\cdot\frac{H_1(s)}{R_1} = \frac{1}{A_2R_2s+1}H_1(s)$$

The overall transfer function:

$$\frac{H_2(s)}{Q_{\text{in}}(s)} = \frac{R_2}{(A_1R_1s+1)(A_2R_2s+1)} = \frac{200}{(100s+1)(400s+1)}$$

Time constants: $\tau_1 = A_1R_1 = 100$ s, $\tau_2 = A_2R_2 = 400$ s.

3.7 Energy-Based Derivation Methods

For complex systems, energy-based methods provide systematic approaches that automatically satisfy constraints and handle coupled dynamics.

3.7.1 Lagrangian Mechanics

The Lagrangian $\mathcal{L}$ is defined as the difference between kinetic and potential energy:

$$\mathcal{L} = T - V \tag{3.27}$$

The equations of motion are obtained from the Euler-Lagrange equations:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i \tag{3.28}$$

where q_i are generalized coordinates and Q_i are generalized non-conservative forces.

Lagrange and Hamilton: Energy Methods in Mechanics

Joseph-Louis Lagrange (1736–1813) revolutionized mechanics by reformulating Newton's laws in terms of energy. His *Mécanique Analytique* (1788) introduced the Lagrangian formulation, which elegantly handles constraints and generalizes to any coordinate system. Rather than tracking forces and accelerations directly, Lagrange's approach requires only the kinetic and potential energies as functions of generalized coordinates.

William Rowan Hamilton (1805–1865) further extended these ideas, developing Hamiltonian mechanics based on the total energy (the Hamiltonian). Hamilton's formulation, using position and momentum as independent variables, proved essential for quantum mechanics and modern control theory. The Hamilton-Jacobi equation connects classical mechanics to optimal control through dynamic programming, while the Hamiltonian structure underlies passivity-based control methods used today.

3.7.2 Rayleigh Dissipation Function

To include viscous damping in Lagrangian mechanics, we use the Rayleigh dissipation function:

$$\mathcal{D} = \frac{1}{2}\sum_i b_i \dot{q}_i^2 \tag{3.29}$$

The modified Euler-Lagrange equation becomes:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} + \frac{\partial \mathcal{D}}{\partial \dot{q}_i} = Q_i \tag{3.30}$$

3.7.3 Hamiltonian Formulation

The Hamiltonian $\mathcal{H}$ represents total system energy:

$$\mathcal{H} = T + V = \sum_i p_i \dot{q}_i - \mathcal{L} \tag{3.31}$$

where $p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$ is the generalized momentum. Hamilton's equations are:

$$\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial \mathcal{H}}{\partial q_i} \tag{3.32}$$

Example 3.6: Lagrangian Derivation of Pendulum Equation

Use Lagrangian mechanics to derive the equation of motion for a simple pendulum of length ℓ and mass m.

Solution:

Choose θ (angle from vertical) as the generalized coordinate.

Kinetic energy (mass moving in arc):

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m(\ell\dot{\theta})^2 = \frac{1}{2}m\ell^2\dot{\theta}^2$$

Potential energy (height above lowest point):

$$V = mg\ell(1 - \cos\theta)$$

Lagrangian:

$$\mathcal{L} = T - V = \frac{1}{2}m\ell^2\dot{\theta}^2 - mg\ell(1 - \cos\theta)$$

Applying the Euler-Lagrange equation:

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m\ell^2\dot{\theta}, \quad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) = m\ell^2\ddot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -mg\ell\sin\theta$$

The equation of motion:

$$m\ell^2\ddot{\theta} + mg\ell\sin\theta = 0 \quad \Rightarrow \quad \boxed{\ddot{\theta} + \frac{g}{\ell}\sin\theta = 0}$$

This agrees with the result from Newton's laws but was derived systematically from energy considerations.

Practical Considerations

When deriving equations of motion:

- For systems with few degrees of freedom and simple geometry, Newton's laws (free body diagrams) are often quickest.
- For systems with constraints, multiple bodies, or complex geometry, Lagrangian methods avoid constraint forces entirely.
- For port-Hamiltonian and passivity-based control design, the Hamiltonian formulation provides the natural framework.
- Always verify your model by checking units, limiting cases, and energy conservation (for conservative systems).

3.8 Summary: A Unified Modeling Approach

The physical laws presented in this chapter can be unified into a systematic modeling procedure:

1. **Identify the domain**: Determine whether the system is mechanical, electrical, thermal, fluid, or multi-domain.
2. **Define system boundaries**: Establish control volumes or free body diagrams.
3. **Select state variables**: Choose energy storage elements (masses, springs, inductors, capacitors, thermal masses, tank levels).
4. **Apply conservation laws**: Use Newton's laws, Kirchhoff's laws, or energy/mass balances as appropriate.
5. **Incorporate constitutive relations**: Apply element-specific laws (Hooke's law, Ohm's law, Fourier's law, etc.).
6. **Combine and simplify**: Eliminate algebraic variables to obtain differential equations in state variables.
7. **Linearize if necessary**: For nonlinear systems, linearize about the operating point.

This systematic approach enables modeling of systems across all physical domains using the same fundamental principles, forming the foundation for the analysis and control design methods developed in subsequent chapters.

Chapter 4

Model Representations

The same dynamic system can be represented in several equivalent mathematical forms, each offering unique advantages for analysis and design. Understanding these representations and the ability to convert between them is fundamental to control systems engineering. The choice of representation often determines which analysis tools are most effective and can significantly impact the complexity of controller design.

In this chapter, we explore four primary representations: differential equations, transfer functions, state-space models, and block diagrams. Each captures the same underlying dynamics but emphasizes different aspects of system behavior. Differential equations arise naturally from physical laws and are ideal for simulation. Transfer functions provide frequency-domain insight and simplify interconnection analysis. State-space representations handle multiple-input, multiple-output (MIMO) systems elegantly and enable modern control techniques. Block diagrams offer visual clarity for understanding signal flow and system architecture.

Mastering the conversions between these forms allows engineers to leverage the strengths of each representation, selecting the most appropriate tool for each task in the analysis and design process.

4.1 Differential Equation Form

The differential equation is the most fundamental representation, derived directly from physical laws such as Newton's laws of motion, Kirchhoff's circuit laws, or conservation principles. For a linear time-invariant (LTI) system with input $u(t)$ and output $y(t)$, the general form is:

$$\boxed{a_n \frac{\mathrm{d}^n y}{\mathrm{d}t^n} + a_{n-1} \frac{\mathrm{d}^{n-1} y}{\mathrm{d}t^{n-1}} + \cdots + a_1 \frac{\mathrm{d}y}{\mathrm{d}t} + a_0 y = b_m \frac{\mathrm{d}^m u}{\mathrm{d}t^m} + \cdots + b_1 \frac{\mathrm{d}u}{\mathrm{d}t} + b_0 u} \tag{4.1}$$

where n is the system order (the highest derivative of the output) and $m \leq n$ for physical realizability.

Proper and Strictly Proper Systems

A system is **proper** if $m \leq n$ (the degree of the numerator does not exceed that of the denominator). A system is **strictly proper** if $m < n$. Most physical systems are strictly proper because they cannot respond instantaneously to input changes.

Example 4.1: RC Circuit Differential Equation

Consider a series RC circuit with input voltage $v_{in}(t)$ and output voltage $v_C(t)$ across the capacitor.
Applying Kirchhoff's voltage law:

$$v_{in}(t) = Ri(t) + v_C(t) = RC\frac{\mathrm{d}v_C}{\mathrm{d}t} + v_C(t) \tag{4.2}$$

Rearranging into standard form:

$$\tau\frac{\mathrm{d}v_C}{\mathrm{d}t} + v_C = v_{in} \tag{4.3}$$

where $\tau = RC$ is the time constant. This is a first-order system ($n = 1$, $m = 0$).
For $R = 10$ kΩ and $C = 100$ μF: $\tau = 1$ second.

The differential equation form is essential for:

- Deriving models from first principles (physics-based modeling)
- Numerical simulation using ODE solvers
- Understanding initial condition effects
- Nonlinear system analysis before linearization

4.2 Transfer Function Form

The transfer function is obtained by taking the Laplace transform of the differential equation with zero initial conditions. It represents the ratio of output to input in the frequency domain.

Definition 4.1 (Transfer Function)**.** For an LTI system, the transfer function $G(s)$ is defined as:

$$\boxed{G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1}s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}} \tag{4.4}$$

The transfer function can be factored into pole-zero form:

$$G(s) = K\frac{(s-z_1)(s-z_2)\cdots(s-z_m)}{(s-p_1)(s-p_2)\cdots(s-p_n)} = K\frac{\prod_{i=1}^{m}(s-z_i)}{\prod_{j=1}^{n}(s-p_j)} \tag{4.5}$$

where z_i are the **zeros** (roots of the numerator), p_j are the **poles** (roots of the denominator), and K is the gain.

Poles and Zeros Determine System Behavior

- **Poles** determine stability and natural response modes (exponential, oscillatory)
- **Zeros** affect the response amplitude and shape
- Poles in the right half-plane ($\mathrm{Re}\{p_j\} > 0$) cause instability
- Zeros in the right half-plane create non-minimum phase behavior

Example 4.2: Mass-Spring-Damper Transfer Function

For the mass-spring-damper system with equation $m\ddot{x} + b\dot{x} + kx = F(t)$:
Taking the Laplace transform with zero initial conditions:

$$ms^2X(s) + bsX(s) + kX(s) = F(s) \tag{4.6}$$

Solving for the transfer function:

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} = \frac{1/m}{s^2 + (b/m)s + k/m} \tag{4.7}$$

In standard second-order form with $\omega_n = \sqrt{k/m}$ and $\zeta = b/(2\sqrt{km})$:

$$G(s) = \frac{\omega_n^2/k}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{4.8}$$

The poles are at $s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$.

4.2.1 Closed-Loop Transfer Functions

For feedback systems, the closed-loop transfer function is derived using block diagram algebra.

$$\boxed{T(s) = \frac{G(s)}{1 + G(s)H(s)}} \tag{4.9}$$

where $G(s)$ is the forward-path transfer function and $H(s)$ is the feedback-path transfer function. For unity feedback ($H(s) = 1$):

$$T(s) = \frac{G(s)}{1 + G(s)} \tag{4.10}$$

4.3 State-Space Form

State-space representation describes the system using first-order differential equations with state variables that capture the system's internal dynamics.

Definition 4.2 (State-Space Representation)**.** A continuous-time LTI system in state-space form is:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad \text{(State equation)} \tag{4.11}$$
$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad \text{(Output equation)} \tag{4.12}$$

where:

- $\mathbf{x} \in \mathbb{R}^n$ is the state vector
- $\mathbf{u} \in \mathbb{R}^p$ is the input vector
- $\mathbf{y} \in \mathbb{R}^q$ is the output vector
- $\mathbf{A} \in \mathbb{R}^{n \times n}$ is the system matrix

- $\mathbf{B} \in \mathbb{R}^{n\times p}$ is the input matrix
- $\mathbf{C} \in \mathbb{R}^{q\times n}$ is the output matrix
- $\mathbf{D} \in \mathbb{R}^{q\times p}$ is the feedthrough matrix

4.3.1 State Variables

State variables represent the minimum set of information needed to completely describe a system's future behavior, given the input. Common choices include:

- Physical variables: positions, velocities, currents, voltages
- Phase variables: output and its derivatives
- Canonical forms: controllable or observable canonical form

Example 4.3: Mass-Spring-Damper State-Space Model

For $m\ddot{x} + b\dot{x} + kx = F$, define states $x_1 = x$ (position) and $x_2 = \dot{x}$ (velocity).
Then $\dot{x}_1 = x_2$ and $\dot{x}_2 = \ddot{x} = \frac{1}{m}(F - bx_2 - kx_1)$.
State-space matrices:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -k/m & -b/m \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1/m \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \end{bmatrix} \tag{4.13}$$

For $m = 1$ kg, $b = 3$ N·s/m, $k = 2$ N/m:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \end{bmatrix} \tag{4.14}$$

4.3.2 Controllable Canonical Form

For a transfer function $G(s) = \frac{b_1 s + b_0}{s^2 + a_1 s + a_0}$, the controllable canonical form is:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -a_0 & -a_1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} b_0 & b_1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \end{bmatrix} \tag{4.15}$$

4.3.3 Observable Canonical Form

The observable canonical form is the transpose of the controllable form:

$$\mathbf{A} = \begin{bmatrix} 0 & -a_0 \\ 1 & -a_1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \end{bmatrix} \tag{4.16}$$

State-Space Advantages

State-space representation offers several advantages:

- Natural handling of MIMO systems
- Direct representation of internal system states

- Foundation for modern control techniques (LQR, Kalman filtering)
- Easy incorporation of initial conditions
- Computational efficiency for simulation

4.4 Block Diagram Form

Block diagrams provide a graphical representation of system interconnections, showing signal flow between components. They are particularly useful for understanding complex system architectures and deriving overall transfer functions.

4.4.1 Block Diagram Elements

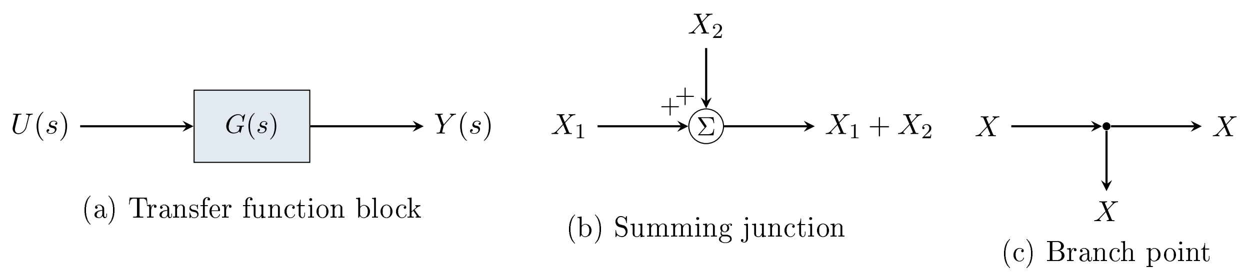

Figure 4.1: Basic block diagram elements: (a) transfer function block, (b) summing junction, (c) branch point.

4.4.2 Block Diagram Reduction Rules

Configuration	**Diagram**	**Equivalent**
Series (cascade)	$G_1 \to G_2$	G_1G_2
Parallel	$G_1 \parallel G_2$	$G_1 + G_2$
Negative feedback	$G/(1+GH)$ loop	$\dfrac{G}{1+GH}$
Positive feedback	$G/(1-GH)$ loop	$\dfrac{G}{1-GH}$

Table 4.1: Block diagram reduction rules.

Example 4.4: Feedback System Block Diagram

Consider a unity feedback system with controller $C(s) = K$ and plant $G(s) = \frac{1}{s+1}$. The open-loop transfer function is:

$$L(s) = C(s)G(s) = \frac{K}{s+1} \tag{4.17}$$

The closed-loop transfer function is:

$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{K/(s+1)}{1 + K/(s+1)} = \frac{K}{s + 1 + K} \tag{4.18}$$

The closed-loop pole is at $s = -(1 + K)$, showing that increasing gain moves the pole further into the left half-plane, improving stability margins but potentially affecting transient response.

4.5 Conversion Between Representations

Converting between representations is a critical skill. This section provides systematic procedures for all common conversions.

4.5.1 State-Space to Transfer Function

The transfer function can be obtained from state-space matrices using:

$$\boxed{G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}} \tag{4.19}$$

For SISO systems, this yields a scalar transfer function. For MIMO systems, it produces a transfer function matrix.

Example 4.5: State-Space to Transfer Function Conversion

Given the state-space model:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \end{bmatrix} \tag{4.20}$$

Step 1: Compute $(s\mathbf{I} - \mathbf{A})$:

$$s\mathbf{I} - \mathbf{A} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix} \tag{4.21}$$

Step 2: Find the inverse using $\mathbf{M}^{-1} = \frac{1}{\det(\mathbf{M})}\text{adj}(\mathbf{M})$:

$$\det(s\mathbf{I} - \mathbf{A}) = s(s+3) + 2 = s^2 + 3s + 2 \tag{4.22}$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \tag{4.23}$$

Step 3: Compute $G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$:

$$G(s) = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{s^2 + 3s + 2} \tag{4.24}$$

Factoring: $G(s) = \frac{1}{(s+1)(s+2)}$, with poles at $s = -1$ and $s = -2$.

4.5.2 Transfer Function to State-Space

Multiple state-space realizations exist for any transfer function. The controllable canonical form provides a systematic approach.

For $G(s) = \frac{b_{n-1}s^{n-1}+\cdots+b_1 s+b_0}{s^n+a_{n-1}s^{n-1}+\cdots+a_1 s+a_0}$:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \tag{4.25}$$

$$\mathbf{C} = \begin{bmatrix} b_0 & b_1 & \cdots & b_{n-1} \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \end{bmatrix} \tag{4.26}$$

Example 4.6: Transfer Function to State-Space

Convert $G(s) = \frac{2s+5}{s^2+4s+3}$ to controllable canonical form.
Comparing with the general form: $a_0 = 3$, $a_1 = 4$, $b_0 = 5$, $b_1 = 2$.

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 5 & 2 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \end{bmatrix} \tag{4.27}$$

Verification: The eigenvalues of $\mathbf{A}$ are the roots of $\det(s\mathbf{I}-\mathbf{A}) = s^2+4s+3 = (s+1)(s+3)$, matching the poles at $s=-1$ and $s=-3$.

4.5.3 Differential Equation to Transfer Function

Take the Laplace transform of both sides assuming zero initial conditions, then solve for $Y(s)/U(s)$.

Example 4.7: ODE to Transfer Function

Convert $2\ddot{y} + 5\dot{y} + 3y = 4\dot{u} + u$ to transfer function form.
Taking the Laplace transform:

$$2s^2Y(s) + 5sY(s) + 3Y(s) = 4sU(s) + U(s) \tag{4.28}$$

Solving:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{4s+1}{2s^2+5s+3} \tag{4.29}$$

4.5.4 Differential Equation to State-Space

Define state variables as the output and its derivatives (phase variables), then express the highest derivative in terms of lower-order terms.

Example 4.8: ODE to State-Space (No Input Derivatives)

Convert $\ddot{y} + 3\dot{y} + 2y = u$ to state-space form.
Define states: $x_1 = y$, $x_2 = \dot{y}$.
Then: $\dot{x}_1 = x_2$ and $\dot{x}_2 = \ddot{y} = u - 3\dot{y} - 2y = -2x_1 - 3x_2 + u$.

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \end{bmatrix} \tag{4.30}$$

Handling Input Derivatives

When the differential equation contains derivatives of the input (e.g., $\ddot{y} + \dot{y} = \dot{u} + u$), the phase variable approach requires modification. Use either:

1. The controllable canonical form directly from the transfer function
2. A change of variables to eliminate input derivatives

Never differentiate the input signal in implementation—this amplifies noise.

4.5.5 Transfer Function to Differential Equation

Cross-multiply and take the inverse Laplace transform.

For $G(s) = \frac{b_1 s + b_0}{a_2 s^2 + a_1 s + a_0}$:

$$(a_2 s^2 + a_1 s + a_0)Y(s) = (b_1 s + b_0)U(s) \tag{4.31}$$

Inverse Laplace transform yields:

$$a_2 \ddot{y} + a_1 \dot{y} + a_0 y = b_1 \dot{u} + b_0 u \tag{4.32}$$

Software Tools for Model Conversion

Both MATLAB and Python provide functions for converting between representations:

MATLAB (Control System Toolbox):

```
% Transfer function to state-space
sys_tf = tf([2 5], [1 4 3]);
sys_ss = ss(sys_tf);

% State-space to transfer function
sys_tf2 = tf(sys_ss);

% Access matrices
[A, B, C, D] = ssdata(sys_ss);
[num, den] = tfdata(sys_tf, 'v');
```

Python (control library):

```
import control as ct
import numpy as np

# Transfer function to state-space
sys_tf = ct.tf([2, 5], [1, 4, 3])
sys_ss = ct.tf2ss(sys_tf)

# State-space to transfer function
sys_tf2 = ct.ss2tf(sys_ss)

# Access matrices
A, B, C, D = sys_ss.A, sys_ss.B, sys_ss.C, sys_ss.D
```

4.6 Practical Example: DC Motor in All Forms

The DC motor provides an excellent example for demonstrating all four representations. Consider an armature-controlled DC motor with the following parameters:

- Armature resistance $R_a = 2\ \Omega$
- Armature inductance $L_a = 0.5$ H
- Motor constant $K_m = 0.1$ N·m/A (also equals back-EMF constant K_e)
- Rotor inertia $J = 0.01$ kg·m^2
- Viscous friction $b = 0.1$ N·m·s/rad

4.6.1 Differential Equation Form

The electrical and mechanical equations are:

$$L_a \frac{\mathrm{d}i_a}{\mathrm{d}t} + R_a i_a = V_a - K_e \omega \tag{4.33}$$

$$J \frac{\mathrm{d}\omega}{\mathrm{d}t} + b\omega = K_m i_a \tag{4.34}$$

Substituting values:

$$0.5 \frac{\mathrm{d}i_a}{\mathrm{d}t} + 2i_a = V_a - 0.1\omega \tag{4.35}$$

$$0.01 \frac{\mathrm{d}\omega}{\mathrm{d}t} + 0.1\omega = 0.1 i_a \tag{4.36}$$

4.6.2 Transfer Function Form

Eliminating i_a between the Laplace-transformed equations:

$$G(s) = \frac{\Omega(s)}{V_a(s)} = \frac{K_m}{(L_a s + R_a)(Js + b) + K_m K_e} = \frac{0.1}{0.005s^2 + 0.07s + 0.21} \tag{4.37}$$

Simplifying:

$$G(s) = \frac{20}{s^2 + 14s + 42} \tag{4.38}$$

Poles: $s = -7 \pm j\sqrt{42 - 49} = -7 \pm j\sqrt{-7}$... Computing correctly: $s = \frac{-14 \pm \sqrt{196-168}}{2} = -7 \pm 2.65$
So poles at $s = -4.35$ and $s = -9.65$.

4.6.3 State-Space Form

Choosing states $x_1 = i_a$ (armature current) and $x_2 = \omega$ (angular velocity):

$$\mathbf{A} = \begin{bmatrix} -R_a/L_a & -K_e/L_a \\ K_m/J & -b/J \end{bmatrix} = \begin{bmatrix} -4 & -0.2 \\ 10 & -10 \end{bmatrix} \tag{4.39}$$

$$\mathbf{B} = \begin{bmatrix} 1/L_a \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} 0 \end{bmatrix} \tag{4.40}$$

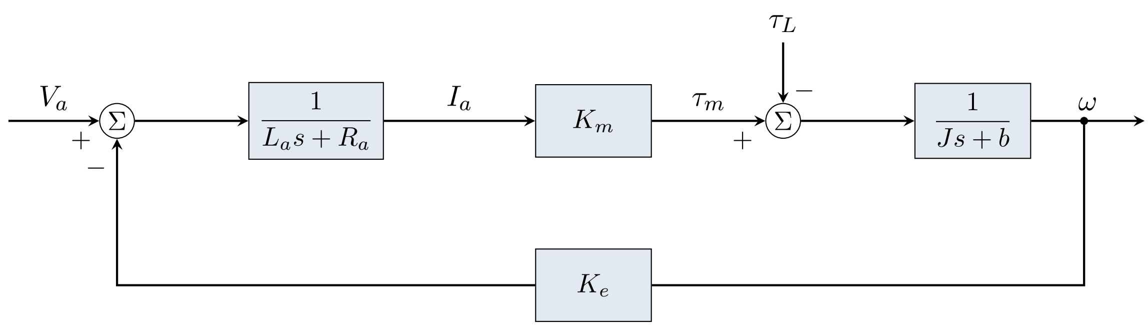

Figure 4.2: Block diagram of armature-controlled DC motor showing electrical-mechanical coupling and back-EMF feedback.

4.6.4 Block Diagram Form

The DC motor block diagram shows the coupling between electrical and mechanical subsystems.

4.7 Selecting the Right Representation

Each representation has strengths that make it preferred for specific tasks.

Representation	**Best Used For**
Differential Equation	Physics-based derivation, nonlinear systems, simulation with general ODE solvers
Transfer Function	Frequency response analysis (Bode, Nyquist), classical controller design (root locus, lead-lag), stability analysis
State-Space	MIMO systems, modern control (LQR, LQG), observer design, numerical simulation, controllability/observability analysis
Block Diagram	System architecture visualization, subsystem interconnection, deriving overall transfer functions

Table 4.2: Guidelines for selecting system representation.

Representation Selection Summary

- **Start with differential equations** when deriving models from physics
- **Use transfer functions** for SISO classical control design and frequency analysis
- **Use state-space** for MIMO systems, modern control techniques, and simulation
- **Use block diagrams** to understand and communicate system structure
- **Convert freely** between representations as needed for the task at hand

Historical Development of System Representations

The differential equation form dates back to Newton and Leibniz in the 17th century. Transfer functions emerged in the late 19th century with Heaviside's operational calculus and were formalized with the Laplace transform. State-space methods were developed in the 1950s-1960s by Kalman and others, enabling the "modern control" revolution. Block diagrams evolved from signal flow graphs developed by Mason in the 1950s. Today, all four representations remain essential tools, each serving different aspects of control system analysis and design.

Chapter 5

System Analogies

5.1 Introduction to Domain Analogies

Remarkably, systems from different physical domains—mechanical, electrical, thermal, hydraulic, and acoustic—often share the same mathematical structure. This deep connection arises because energy storage, energy dissipation, and power flow obey analogous relationships across domains. By recognizing these analogies, engineers can transfer intuition and analysis techniques developed in one domain to problems in another.

The practical benefits of system analogies are substantial. Electrical engineers have developed powerful circuit analysis tools—mesh analysis, nodal analysis, impedance methods, and simulation software like SPICE—that can be applied to mechanical, thermal, and fluid systems once the appropriate analogous quantities are identified. Furthermore, many modern systems span multiple domains (electromechanical actuators, thermoelectric devices, piezoelectric sensors), and analogies provide a unified framework for their analysis.

Power-Conjugate Variables

In any physical domain, power is the product of two conjugate variables:

$$P = (\text{effort}) \times (\text{flow}) \tag{5.1}$$

The **effort** variable (also called "across" variable) represents potential or driving force, while the **flow** variable (also called "through" variable) represents rate of motion or current. Identifying these conjugate pairs is the key to establishing analogies.

5.2 Force-Voltage Analogy

The **force-voltage analogy** (also called the Maxwell analogy or impedance analogy) maps mechanical force to electrical voltage and mechanical velocity to electrical current. This is the most commonly used analogy in control systems because mechanical impedance directly corresponds to electrical impedance.

Concept	Mechanical (Trans.)	Electrical	Units
Effort (across)	Force F	Voltage V	N, V
Flow (through)	Velocity v	Current i	m/s, A
Displacement	Position x	Charge q	m, C
Momentum	Momentum $p = mv$	Flux linkage $\lambda = Li$	N·s, Wb
Resistance	Damper b	Resistor R	N·s/m, Ω
Capacitance	Compliance $1/k$	Capacitor C	m/N, F
Inductance	Mass m	Inductor L	kg, H

Table 5.1: Force-voltage analogy: mechanical and electrical quantities.

5.2.1 Element Relationships

For the **mass** (stores kinetic energy):

$$F = m\frac{\mathrm{d}v}{\mathrm{d}t} \quad \longleftrightarrow \quad V = L\frac{\mathrm{d}i}{\mathrm{d}t} \tag{5.2}$$

For the **spring** (stores potential energy):

$$F = kx = k\int v\,\mathrm{d}t \quad \longleftrightarrow \quad V = \frac{1}{C}\int i\,\mathrm{d}t = \frac{q}{C} \tag{5.3}$$

For the **damper** (dissipates energy):

$$F = bv \quad \longleftrightarrow \quad V = Ri \tag{5.4}$$

5.2.2 Series and Parallel Connections

In the force-voltage analogy, elements connected in **series mechanically** (same velocity) correspond to elements in **parallel electrically** (same current). Conversely, elements in **parallel mechanically** (same force) correspond to elements in **series electrically** (same voltage).

Example 5.1: Mass-Spring-Damper to Series RLC Circuit

Consider a mass-spring-damper system with the equation of motion:

$$m\ddot{x} + b\dot{x} + kx = F(t) \tag{5.5}$$

Substituting $v = \dot{x}$ and integrating the spring term:

$$m\frac{\mathrm{d}v}{\mathrm{d}t} + bv + k\int v\,\mathrm{d}t = F(t) \tag{5.6}$$

This is mathematically identical to the series RLC circuit equation:

$$L\frac{\mathrm{d}i}{\mathrm{d}t} + Ri + \frac{1}{C}\int i\,\mathrm{d}t = V(t) \tag{5.7}$$

The analogous parameters are:

$$\begin{aligned} m = 2\ \text{kg} &\quad\longleftrightarrow\quad L = 2\ \text{H} \\ b = 8\ \text{N·s/m} &\quad\longleftrightarrow\quad R = 8\ \Omega \\ k = 50\ \text{N/m} &\quad\longleftrightarrow\quad C = 1/50 = 0.02\ \text{F} \end{aligned}$$

Both systems have the same transfer function form:

$$G(s) = \frac{V(s)/X(s)}{F(s)/V(s)} = \frac{1}{2s^2 + 8s + 50} \tag{5.8}$$

with natural frequency $\omega_n = 5$ rad/s and damping ratio $\zeta = 0.4$.

5.3 Force-Current Analogy

The **force-current analogy** (also called the Firestone analogy or mobility analogy) maps mechanical force to electrical current and velocity to voltage. While less intuitive, this analogy preserves the topology of interconnections: series mechanical connections map to series electrical connections.

Concept	**Mechanical (Trans.)**	**Electrical**	**Units**
Effort (across)	Velocity v	Voltage V	m/s, V
Flow (through)	Force F	Current i	N, A
Resistance	Compliance $1/b$	Resistor R	m/(N·s), Ω
Capacitance	Mass m	Capacitor C	kg, F
Inductance	Compliance $1/k$	Inductor L	m/N, H

Table 5.2: Force-current analogy: mechanical and electrical quantities.

5.3.1 Element Relationships

For the **mass**:

$$v = \frac{1}{m}\int F\,\mathrm{d}t \quad \longleftrightarrow \quad V = \frac{1}{C}\int i\,\mathrm{d}t \tag{5.9}$$

For the **spring**:

$$v = \frac{1}{k}\frac{\mathrm{d}F}{\mathrm{d}t} \quad \longleftrightarrow \quad V = L\frac{\mathrm{d}i}{\mathrm{d}t} \tag{5.10}$$

For the **damper**:

$$v = \frac{F}{b} \quad \longleftrightarrow \quad V = Ri \tag{5.11}$$

Choosing the Right Analogy

Force-voltage analogy: Use when you want mechanical impedance to equal electrical impedance. Preferred for impedance matching problems and when using network analysis tools.

Force-current analogy: Use when you want to preserve circuit topology (series ↔ series, parallel ↔ parallel). Preferred when physical layout matters or when using bond graph methods.

5.4 Impedance Methods for Multi-Domain Systems

The concept of **impedance**—the ratio of effort to flow in the Laplace domain—provides a powerful unified framework for analyzing systems across domains.

5.4.1 Mechanical Impedance

For translational mechanical elements, the mechanical impedance $Z_m(s)$ is defined as:

$$Z_m(s) = \frac{F(s)}{V(s)} \tag{5.12}$$

where $V(s) = sX(s)$ is the Laplace transform of velocity.

Mechanical Impedances

$$\text{Mass:} \quad Z_m(s) = ms \tag{5.13}$$
$$\text{Damper:} \quad Z_m(s) = b \tag{5.14}$$
$$\text{Spring:} \quad Z_m(s) = \frac{k}{s} \tag{5.15}$$

These combine exactly like electrical impedances:

- **Series connection** (elements sharing the same velocity): $Z_{\text{total}} = Z_1 + Z_2 + \cdots$
- **Parallel connection** (elements sharing the same force): $\frac{1}{Z_{\text{total}}} = \frac{1}{Z_1} + \frac{1}{Z_2} + \cdots$

Example 5.2: Impedance Analysis of a Speaker System

A loudspeaker consists of a voice coil (electrical) driving a cone (mechanical) coupled to an acoustic load. The mechanical subsystem has:

- Cone mass: $m = 0.015$ kg
- Suspension compliance (inverse stiffness): $C_m = 1/k = 0.001$ m/N
- Mechanical damping: $R_m = b = 2$ N·s/m

The total mechanical impedance is:

$$Z_m(s) = ms + R_m + \frac{1}{C_m s} = 0.015s + 2 + \frac{1000}{s} \tag{5.16}$$

At a driving frequency ω, replacing $s = \mathrm{j}\omega$:

$$Z_m(\mathrm{j}\omega) = \mathrm{j}(0.015\omega) + 2 - \mathrm{j}\frac{1000}{\omega} \tag{5.17}$$

The mechanical resonance occurs when the imaginary part vanishes:

$$0.015\omega_0 = \frac{1000}{\omega_0} \quad \Rightarrow \quad \omega_0 = \sqrt{\frac{1000}{0.015}} = 258 \text{ rad/s} = 41 \text{ Hz} \tag{5.18}$$

At resonance, the impedance is purely resistive: $Z_m(\mathrm{j}\omega_0) = R_m = 2$ N·s/m, and the cone moves with maximum velocity for a given force.

5.4.2 Electrical Equivalent Circuits

Using the force-voltage analogy, any mechanical system can be represented as an equivalent electrical circuit. This allows powerful circuit simulation tools to analyze mechanical dynamics.

Example 5.3: Quarter-Car Electrical Equivalent

The quarter-car suspension model (sprung mass m_s, unsprung mass m_u, suspension stiffness k_s, damping b_s, tire stiffness k_t) can be represented as an equivalent electrical circuit.
Using the force-voltage analogy with the following correspondences:

$$\begin{aligned}
\text{Road velocity } \dot{x}_r &\longleftrightarrow \text{Voltage source } V_r \\
\text{Body mass } m_s &\longleftrightarrow \text{Inductor } L_s = m_s \\
\text{Wheel mass } m_u &\longleftrightarrow \text{Inductor } L_u = m_u \\
\text{Suspension spring } k_s &\longleftrightarrow \text{Capacitor } C_s = 1/k_s \\
\text{Tire spring } k_t &\longleftrightarrow \text{Capacitor } C_t = 1/k_t \\
\text{Damper } b_s &\longleftrightarrow \text{Resistor } R_s = b_s
\end{aligned}$$

With typical values ($m_s = 250$ kg, $m_u = 35$ kg, $k_s = 16{,}000$ N/m, $b_s = 1{,}000$ N·s/m, $k_t = 160{,}000$ N/m):

$$\begin{aligned}
L_s &= 250 \text{ H} \quad C_s = 62.5\ \mu\text{F} \quad R_s = 1000\ \Omega \\
L_u &= 35 \text{ H} \quad C_t = 6.25\ \mu\text{F}
\end{aligned}$$

This equivalent circuit can be simulated in SPICE to analyze ride quality and suspension response.

Gabriel Kron and Unified Network Theory

Gabriel Kron (1901–1968), a Hungarian-American engineer at General Electric, revolutionized the analysis of complex electromechanical systems through his development of **tensor network analysis**. In the 1930s and 1940s, Kron showed that electrical machines, power systems, and mechanical structures could all be analyzed using a unified mathematical framework based on network topology and tensor calculus.
Kron's key insight was that the equations governing different physical systems—electrical circuits, mechanical linkages, and rotating machines—share a common mathematical structure that can be expressed in terms of generalized coordinates and network connections. His methods, published in works like *Tensors for Circuits* (1942) and *Diakoptics* (1963), laid the foundation for modern computer-aided analysis of multi-domain systems and influenced the development of bond graph theory, finite element methods, and object-oriented modeling languages like Modelica.

5.5 Thermal System Analogies

Thermal systems exhibit first-order dynamics (no thermal equivalent of inductance) and map naturally to RC electrical circuits.

Concept	Thermal	Electrical	Units
Effort (across)	Temperature T	Voltage V	K (or °C), V
Flow (through)	Heat flow $\dot{Q}$	Current i	W, A
Stored quantity	Thermal energy Q	Charge q	J, C
Resistance	Thermal resistance R_t	Resistor R	K/W, Ω
Capacitance	Thermal capacitance C_t	Capacitor C	J/K, F

Table 5.3: Thermal-electrical analogy.

5.5.1 Thermal Element Relationships

Thermal resistance relates temperature difference to heat flow:

$$\dot{Q} = \frac{T_1 - T_2}{R_t} \quad \longleftrightarrow \quad i = \frac{V_1 - V_2}{R} \tag{5.19}$$

For conduction through a solid: $R_t = \frac{L}{kA}$ where L is length, k is thermal conductivity, and A is cross-sectional area.

For convection: $R_t = \frac{1}{hA}$ where h is the convection coefficient.

Thermal capacitance relates stored energy to temperature:

$$Q = C_t T \quad \text{and} \quad \dot{Q} = C_t \frac{\mathrm{d}T}{\mathrm{d}t} \tag{5.20}$$

where $C_t = mc_p$ (mass times specific heat capacity).

Example 5.4: Building Thermal Model

A simplified single-zone building model has:

- Interior thermal mass: $C_i = 5 \times 10^6$ J/K
- Wall thermal resistance: $R_w = 0.01$ K/W
- Window thermal resistance: $R_{win} = 0.05$ K/W

The equivalent circuit has C_i connected to ambient temperature T_a through parallel resistances R_w and R_{win}.
Total thermal resistance:

$$R_{eq} = \frac{R_w R_{win}}{R_w + R_{win}} = \frac{0.01 \times 0.05}{0.01 + 0.05} = 0.00833 \text{ K/W} \tag{5.21}$$

Time constant:

$$\tau = R_{eq} C_i = 0.00833 \times 5 \times 10^6 = 41{,}667 \text{ s} \approx 11.6 \text{ hours} \tag{5.22}$$

Transfer function from ambient temperature to interior temperature:

$$\frac{T_i(s)}{T_a(s)} = \frac{1}{R_{eq} C_i s + 1} = \frac{1}{41667s + 1} \tag{5.23}$$

5.6 Analogies in Practice: Multi-Domain Systems

Modern engineering systems frequently span multiple physical domains. System analogies enable unified modeling and analysis of these complex systems.

5.6.1 Electromechanical Systems

Motors and generators couple electrical and mechanical domains through electromagnetic interaction.

Example 5.5: DC Motor Complete Model

A DC motor couples electrical and mechanical subsystems through the motor constant K_m.
Electrical subsystem (armature circuit):

$$V_a = L_a \frac{\mathrm{d}i_a}{\mathrm{d}t} + R_a i_a + e_b \tag{5.24}$$

where $e_b = K_m \omega$ is the back-EMF.
Mechanical subsystem (rotor dynamics):

$$J\frac{\mathrm{d}\omega}{\mathrm{d}t} + B\omega = \tau_m - \tau_L \tag{5.25}$$

where $\tau_m = K_m i_a$ is the motor torque.
Coupling equations:

$$e_b = K_m \omega \quad \text{(back-EMF)} \tag{5.26}$$

$$\tau_m = K_m i_a \quad \text{(torque)} \tag{5.27}$$

Combining these with $\tau_L = 0$:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_m}{(L_a s + R_a)(Js + B) + K_m^2} \tag{5.28}$$

For a small DC motor with $L_a = 0.5$ mH, $R_a = 1\ \Omega$, $J = 0.01$ kg·m^2, $B = 0.1$ N·m·s/rad, $K_m = 0.5$ V/(rad/s):

$$\frac{\Omega(s)}{V_a(s)} = \frac{0.5}{0.000005s^2 + 0.0105s + 0.35} = \frac{100{,}000}{s^2 + 2100s + 70{,}000} \tag{5.29}$$

The electrical time constant $\tau_e = L_a/R_a = 0.5$ ms is much faster than the mechanical time constant $\tau_m = J/B = 0.1$ s, justifying the common approximation of neglecting armature inductance.

5.6.2 Comprehensive Analogy Table

Table 5.4 summarizes analogies across mechanical (translational and rotational), electrical, thermal, and fluid domains.

Quantity	Mech. (Trans.)	Mech. (Rot.)	Electrical	Thermal
Effort	Force F	Torque τ	Voltage V	Temp. T
Flow	Velocity v	Angular vel. ω	Current i	Heat flow $\dot{Q}$
Displacement	Position x	Angle θ	Charge q	Heat Q
Inertia	Mass m	Inertia J	Inductor L	—
Compliance	$1/k$ (spring)	$1/\kappa$	Capacitor C	Thermal cap. C_t
Dissipation	Damper b	Damper B	Resistor R	Thermal res. R_t
Kinetic energy	$\frac{1}{2}mv^2$	$\frac{1}{2}J\omega^2$	$\frac{1}{2}Li^2$	—
Potential energy	$\frac{1}{2}kx^2$	$\frac{1}{2}\kappa\theta^2$	$\frac{1}{2}CV^2$	—
Power dissipation	bv^2	$B\omega^2$	Ri^2	$\dot{Q}^2R_t$

Table 5.4: Comprehensive system analogies across physical domains.

5.6.3 Unified State-Space Modeling

For multi-domain systems, state-space representation provides a unified framework. Energy storage elements contribute state variables:

- Inductors/masses/inertias: state = current/velocity/angular velocity
- Capacitors/springs: state = voltage/displacement/angle
- Thermal capacitances: state = temperature

Multi-Domain Simulation Tools

Modern simulation environments handle multi-domain systems directly:

- **Simulink/Simscape** (MathWorks): Physical modeling with mechanical, electrical, thermal, and hydraulic libraries
- **Modelica** (open standard): Equation-based, object-oriented modeling language
- **SPICE variants**: Electrical circuit simulators that can model other domains via equivalent circuits
- **Bond graphs**: Unified graphical representation preserving power flow

5.7 Key Concepts and Limitations

Limitations of System Analogies

While system analogies are powerful tools, they have important limitations:

1. **Linear systems only:** The analogies presented assume linear, time-invariant elements. Nonlinear effects (saturation, backlash, friction) require extended models.
2. **Lumped parameter assumption:** We assume properties are concentrated at discrete points. Distributed systems (transmission lines, flexible beams) require partial differential equations or discretization.

3. **No thermal inductance:** Thermal systems have no equivalent of mass or inductance—there is no "thermal momentum." Thermal dynamics are inherently first-order.
4. **Sign conventions:** Different domains use different sign conventions. Be careful when combining domains to maintain consistent power flow directions.
5. **Frequency limitations:** At high frequencies, parasitic effects (stray capacitance, lead inductance, radiation) become significant and must be modeled explicitly.

Chapter Summary: System Analogies

1. Physical systems from different domains share common mathematical structures based on energy storage, dissipation, and power flow.
2. The **force-voltage analogy** maps $F \leftrightarrow V$, $v \leftrightarrow i$, and is preferred for impedance-based analysis.
3. The **force-current analogy** maps $F \leftrightarrow i$, $v \leftrightarrow V$, and preserves circuit topology.
4. **Mechanical impedance** ($Z_m = ms$, $Z_m = b$, $Z_m = k/s$) allows circuit analysis methods to be applied to mechanical systems.
5. **Thermal systems** are first-order (RC equivalent) with no inductive element.
6. Multi-domain systems (electromechanical, thermoelectric) can be analyzed using unified state-space or equivalent circuit methods.
7. Analogies assume linear, lumped-parameter systems—extend with care to nonlinear or distributed systems.

Chapter 6

Test Inputs and System Response

Understanding how systems respond to standard test inputs is fundamental to control system analysis and design. Rather than testing systems with arbitrary signals, engineers use a small set of mathematically tractable inputs that reveal essential dynamic characteristics. These standard inputs—impulse, step, ramp, and sinusoidal—each expose different aspects of system behavior and together provide a complete picture of system performance.

The choice of test input depends on what information we seek. The impulse response reveals the system's fundamental dynamics and completely characterizes linear time-invariant (LTI) systems. The step response shows transient behavior and steady-state accuracy. The ramp response exposes tracking capability for changing references. The sinusoidal response, when applied across frequencies, maps the complete frequency response. Mastery of these test inputs and their associated performance metrics is essential for any control engineer.

Why Standard Test Inputs?

Standard test inputs serve three critical purposes:

- **Mathematical tractability**: Simple Laplace transforms enable analytical solutions
- **Reproducibility**: Standardized tests allow comparison across different systems
- **Complete characterization**: Together, they reveal all essential dynamic behavior

6.1 Impulse Input and Response

The impulse function, also called the Dirac delta function, represents an idealized instantaneous pulse of infinite amplitude and infinitesimal duration, but with unit area:

$$r(t) = \delta(t) = \begin{cases} \infty & t = 0 \\ 0 & t \neq 0 \end{cases}, \quad \int_{-\infty}^{\infty} \delta(t)\,\mathrm{d}t = 1 \tag{6.1}$$

The Laplace transform of the impulse function is remarkably simple:

$$\mathcal{L}\{\delta(t)\} = 1 \tag{6.2}$$

This simplicity makes the impulse response particularly valuable. For a system with transfer function $G(s)$, the output to an impulse input is:

$$Y(s) = G(s) \cdot 1 = G(s) \quad \Rightarrow \quad y(t) = h(t) = \mathcal{L}^{-1}\{G(s)\} \tag{6.3}$$

The impulse response $h(t)$ is the inverse Laplace transform of the transfer function itself, making it the system's fundamental characterization.

Impulse Response

The **impulse response** $h(t)$ of an LTI system is the output when the input is a unit impulse $\delta(t)$. It completely characterizes the system because any output can be computed via convolution:

$$y(t) = \int_0^t h(\tau)r(t-\tau)\,\mathrm{d}\tau = h(t) * r(t) \tag{6.4}$$

For a second-order underdamped system with transfer function:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{6.5}$$

The impulse response is:

$$h(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} \mathrm{e}^{-\zeta\omega_n t} \sin(\omega_d t), \quad \omega_d = \omega_n\sqrt{1-\zeta^2} \tag{6.6}$$

Example 6.1: Impulse Response of a First-Order System

Find the impulse response of the first-order system $G(s) = \dfrac{K}{\tau s + 1}$.

Solution:
Rewrite in standard form:

$$G(s) = \frac{K/\tau}{s + 1/\tau} \tag{6.7}$$

Taking the inverse Laplace transform:

$$h(t) = \frac{K}{\tau}\mathrm{e}^{-t/\tau}, \quad t \geq 0 \tag{6.8}$$

The impulse response is an exponential decay with time constant τ. The initial value $h(0^+) = K/\tau$ depends on the DC gain and time constant, while the decay rate is determined solely by τ.

6.2 Step Input and Response

The unit step function represents an instantaneous change from zero to a constant value:

$$r(t) = u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases} \tag{6.9}$$

The Laplace transform is:

$$\mathcal{L}\{u(t)\} = \frac{1}{s} \tag{6.10}$$

The step response is perhaps the most commonly used test because it reveals both transient and steady-state behavior. For a system $G(s)$:

$$Y(s) = G(s) \cdot \frac{1}{s} \tag{6.11}$$

6.2.1 Time-Domain Performance Specifications

The step response defines several critical performance metrics that engineers use to specify system requirements.

Step Response Specifications

For a second-order underdamped system responding to a unit step:

- **Rise time** t_r: Time to go from 10% to 90% of final value
- **Peak time** t_p: Time to reach the first (maximum) peak
- **Percent overshoot** %OS: Amount the response exceeds the final value
- **Settling time** t_s: Time to remain within a specified band (typically 2%) of the final value

For a standard second-order system, these specifications have closed-form expressions:

$$t_r \approx \frac{1.8}{\omega_n} \quad \text{(approximation for } 0.3 < \zeta < 0.8) \tag{6.12}$$

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} \tag{6.13}$$

$$\%\text{OS} = 100 \cdot \mathrm{e}^{-\pi\zeta/\sqrt{1-\zeta^2}} \tag{6.14}$$

$$t_s \approx \frac{4}{\zeta\omega_n} \quad \text{(2\% criterion)} \tag{6.15}$$

Design Trade-offs

The step response specifications reveal fundamental trade-offs:

- **Speed vs. overshoot**: Faster rise time typically increases overshoot
- **Damping ratio** ζ: Higher ζ reduces overshoot but slows response
- **Natural frequency** ω_n: Higher ω_n speeds up all time metrics

The "sweet spot" for many applications is $\zeta \approx 0.7$, giving about 5% overshoot with reasonably fast response.

Example 6.2: Step Response Specifications

A position control system has the closed-loop transfer function:

$$G(s) = \frac{100}{s^2 + 10s + 100} \tag{6.16}$$

Find the step response specifications.
Solution:

Comparing with the standard form $\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$:

$$\omega_n^2 = 100 \quad \Rightarrow \quad \omega_n = 10 \text{ rad/s} \tag{6.17}$$

$$2\zeta\omega_n = 10 \quad \Rightarrow \quad \zeta = 0.5 \tag{6.18}$$

The damped natural frequency:

$$\omega_d = \omega_n\sqrt{1-\zeta^2} = 10\sqrt{1-0.25} = 8.66 \text{ rad/s} \tag{6.19}$$

Computing the specifications:

$$t_r \approx \frac{1.8}{10} = 0.18 \text{ s} \tag{6.20}$$

$$t_p = \frac{\pi}{8.66} = 0.363 \text{ s} \tag{6.21}$$

$$\%\text{OS} = 100 \cdot \mathrm{e}^{-\pi(0.5)/\sqrt{0.75}} = 100 \cdot \mathrm{e}^{-1.814} = 16.3\% \tag{6.22}$$

$$t_s \approx \frac{4}{0.5 \times 10} = 0.8 \text{ s} \tag{6.23}$$

6.3 Ramp Input and Response

The ramp input represents a constantly increasing signal:

$$r(t) = t \cdot u(t) \tag{6.24}$$

The Laplace transform is:

$$\mathcal{L}\{t \cdot u(t)\} = \frac{1}{s^2} \tag{6.25}$$

Ramp inputs test a system's ability to track continuously changing references, such as a robot arm following a trajectory or an antenna tracking a moving satellite.

6.3.1 Steady-State Error and Velocity Error Constant

For ramp inputs, the steady-state error depends on the system type. The **velocity error constant** K_v characterizes this behavior:

$$K_v = \lim_{s \to 0} s \cdot G(s)H(s) \tag{6.26}$$

For a unity feedback system with open-loop transfer function $G(s)$:

$$e_{ss} = \frac{1}{K_v} \tag{6.27}$$

System Type and Ramp Response

The steady-state error to a ramp input depends on the system type (number of free integrators):

- **Type 0**: $K_v = 0$, infinite steady-state error (cannot track ramp)

- **Type 1**: $K_v =$ finite, constant steady-state error $e_{ss} = 1/K_v$
- **Type 2 or higher**: $K_v = \infty$, zero steady-state error

Example 6.3: Ramp Response and Velocity Error

A unity feedback system has the open-loop transfer function:

$$G(s) = \frac{50}{s(s+5)} \tag{6.28}$$

Find the velocity error constant and steady-state error to a unit ramp input.
Solution:
This is a Type 1 system (one free integrator in $G(s)$).
The velocity error constant:

$$K_v = \lim_{s\to 0} s \cdot \frac{50}{s(s+5)} = \lim_{s\to 0} \frac{50}{s+5} = \frac{50}{5} = 10 \tag{6.29}$$

The steady-state error to a unit ramp:

$$e_{ss} = \frac{1}{K_v} = \frac{1}{10} = 0.1 \tag{6.30}$$

The system tracks the ramp with a constant position error of 0.1 units—it always lags behind the reference by this amount.

6.4 Sinusoidal Input and Frequency Response

The sinusoidal input is fundamental to frequency-domain analysis:

$$r(t) = A\sin(\omega t) \tag{6.31}$$

The Laplace transform is:

$$\mathcal{L}\{A\sin(\omega t)\} = \frac{A\omega}{s^2+\omega^2} \tag{6.32}$$

For stable LTI systems, a sinusoidal input produces a sinusoidal output at the same frequency, but with different amplitude and phase. This steady-state response is characterized by the **frequency response** $G(\mathrm{j}\omega)$.

6.4.1 Frequency Response Function

The frequency response is obtained by substituting $s = \mathrm{j}\omega$ into the transfer function:

$$G(\mathrm{j}\omega) = |G(\mathrm{j}\omega)| \cdot \mathrm{e}^{\mathrm{j}\angle G(\mathrm{j}\omega)} \tag{6.33}$$

where:

$$|G(\mathrm{j}\omega)| = \sqrt{[\mathrm{Re}\{G(\mathrm{j}\omega)\}]^2 + [\mathrm{Im}\{G(\mathrm{j}\omega)\}]^2} \quad \text{(magnitude)} \tag{6.34}$$

$$\angle G(\mathrm{j}\omega) = \arctan\left(\frac{\mathrm{Im}\{G(\mathrm{j}\omega)\}}{\mathrm{Re}\{G(\mathrm{j}\omega)\}}\right) \quad \text{(phase)} \tag{6.35}$$

For a sinusoidal input $r(t) = A\sin(\omega t)$, the steady-state output is:

$$y_{ss}(t) = A|G(\mathrm{j}\omega)|\sin(\omega t + \angle G(\mathrm{j}\omega)) \tag{6.36}$$

Frequency Response Interpretation

The frequency response $G(\mathrm{j}\omega)$ provides:

- **Magnitude** $|G(\mathrm{j}\omega)|$: Ratio of output amplitude to input amplitude
- **Phase** $\angle G(\mathrm{j}\omega)$: Phase shift between output and input

Plotting these versus frequency yields Bode plots, which are essential tools for control design.

Example 6.4: Frequency Response Calculation

Find the frequency response of the first-order system $G(s) = \dfrac{10}{s+2}$ at $\omega = 2$ rad/s.

Solution:

Substitute $s = \mathrm{j}\omega = 2\mathrm{j}$:

$$G(\mathrm{j}2) = \frac{10}{2\mathrm{j}+2} = \frac{10}{2(1+\mathrm{j})} = \frac{5}{1+\mathrm{j}} \tag{6.37}$$

Rationalize by multiplying by the complex conjugate:

$$G(\mathrm{j}2) = \frac{5(1-\mathrm{j})}{(1+\mathrm{j})(1-\mathrm{j})} = \frac{5(1-\mathrm{j})}{2} = 2.5 - 2.5\mathrm{j} \tag{6.38}$$

Magnitude:

$$|G(\mathrm{j}2)| = \sqrt{2.5^2 + 2.5^2} = \sqrt{12.5} = 3.54 \tag{6.39}$$

Phase:

$$\angle G(\mathrm{j}2) = \arctan\left(\frac{-2.5}{2.5}\right) = \arctan(-1) = -45 \tag{6.40}$$

For a 1 V amplitude sinusoid at 2 rad/s, the output would have amplitude 3.54 V and lag the input by 45°.

6.5 Comparison of Standard Test Inputs

The four standard test inputs each serve distinct purposes in system analysis. Table 6.1 summarizes their properties.

6.5.1 Relationships Between Test Inputs

The test inputs are mathematically related through differentiation and integration:

$$\delta(t) \xleftarrow{\text{differentiate}} u(t) \xleftarrow{\text{differentiate}} t \cdot u(t) \tag{6.41}$$

In the Laplace domain, this corresponds to multiplication/division by s:

$$1 \xleftarrow{\times s} \frac{1}{s} \xleftarrow{\times s} \frac{1}{s^2} \tag{6.42}$$

Input	Time Domain	Laplace Transform	Primary Use
Impulse	$\delta(t)$	1	System identification
Step	$u(t)$	$\frac{1}{s}$	Transient response
Ramp	$t \cdot u(t)$	$\frac{1}{s^2}$	Tracking accuracy
Sinusoidal	$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$	Frequency response

Table 6.1: Summary of standard test inputs and their applications.

This relationship means that the step response can be obtained by integrating the impulse response, and the ramp response by integrating the step response.

Practical Implementation

In practice, true impulse functions cannot be generated. Engineers approximate impulses using:

- Short-duration rectangular pulses
- Hammer strikes (mechanical systems)
- Brief voltage spikes (electrical systems)

The approximation is valid when the pulse duration is much shorter than the system's fastest time constant.

6.6 Historical Context

Historical Context

The development of test input theory parallels the evolution of control systems analysis itself. Oliver Heaviside (1850–1925) introduced operational calculus in the 1880s, providing the mathematical foundation for analyzing step responses in electrical circuits. His "operational methods" predated and inspired the formal development of Laplace transforms for engineering applications.

Harry Nyquist (1889–1976) at Bell Labs revolutionized frequency-domain analysis in 1932 with his stability criterion, which relies on sinusoidal test inputs to determine closed-loop stability from open-loop frequency response. His work showed that sweeping through frequencies could reveal system behavior that step responses might miss.

Hendrik Bode (1905–1982), also at Bell Labs, further developed frequency response methods in the 1930s and 1940s. Bode plots—logarithmic graphs of magnitude and phase versus frequency—became the standard tool for control system design. His work demonstrated that sinusoidal testing at multiple frequencies provides complete system characterization.

Together, these pioneers established that a small set of mathematically elegant test inputs could fully characterize any linear system, enabling the systematic design methods that control engineers use today.

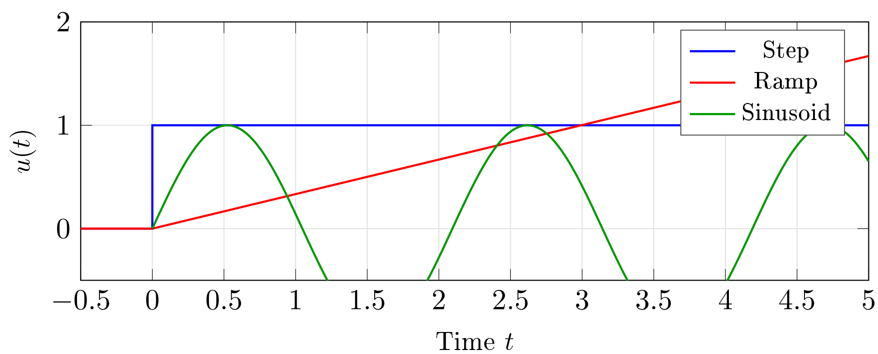

Figure 6.1: Standard test inputs for system characterization.

Chapter 7

First and Second Order System Responses

The vast majority of physical systems encountered in control engineering can be approximated by first or second-order models. Understanding these canonical forms provides the foundation for analyzing and designing control systems across all engineering domains.

7.1 First-Order Systems

7.1.1 Standard Form and Parameters

The standard first-order transfer function is:

$$G(s) = \frac{K}{\tau s + 1} \tag{7.1}$$

where:

- K = DC gain (steady-state gain), dimensionless or with appropriate units
- τ = time constant (seconds)

The single pole is located at $s = -1/\tau$, which lies on the negative real axis for stable systems ($\tau > 0$).

7.1.2 Step Response Derivation

For a unit step input $R(s) = 1/s$, the output in the Laplace domain is:

$$Y(s) = G(s) \cdot R(s) = \frac{K}{\tau s + 1} \cdot \frac{1}{s} = \frac{K}{s(\tau s + 1)} \tag{7.2}$$

Using partial fraction expansion:

$$Y(s) = \frac{K}{s(\tau s + 1)} = \frac{A}{s} + \frac{B}{\tau s + 1} \tag{7.3}$$

Solving for coefficients: $A = K$ (setting $s = 0$) and $B = -K\tau$ (setting $s = -1/\tau$).
Taking the inverse Laplace transform:

$$\boxed{y(t) = K\left(1 - \mathrm{e}^{-t/\tau}\right), \quad t \geq 0} \tag{7.4}$$

First-Order Step Response Characteristics

- At $t = \tau$: output reaches 63.2% of final value ($1 - e^{-1} \approx 0.632$)
- At $t = 2\tau$: reaches 86.5% of final value
- At $t = 3\tau$: reaches 95.0% of final value
- At $t = 4\tau$: reaches 98.2% of final value (practical settling time)
- At $t = 5\tau$: reaches 99.3% of final value
- Initial slope: $\mathrm{d}y/\mathrm{d}t|_{t=0} = K/\tau$
- Final value: $y(\infty) = K$

7.2 Second-Order Systems

7.2.1 Standard Form and Parameters

The standard second-order transfer function is:

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{7.5}$$

where:

- K = DC gain (steady-state value for unit step input)
- ω_n = undamped natural frequency (rad/s)
- ζ = damping ratio (dimensionless)

Definition: Damping Ratio

The damping ratio ζ quantifies the level of damping relative to critical damping. It determines the qualitative nature of the system response:

- $\zeta = 0$: Undamped (perpetual oscillation)
- $0 < \zeta < 1$: Underdamped (decaying oscillation)
- $\zeta = 1$: Critically damped (fastest non-oscillatory response)
- $\zeta > 1$: Overdamped (sluggish, non-oscillatory response)

7.2.2 Pole Locations

The characteristic equation $s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$ yields poles at:

$$\boxed{s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}} \tag{7.6}$$

For the underdamped case ($0 < \zeta < 1$), the poles are complex conjugates:

$$s_{1,2} = -\zeta\omega_n \pm \mathrm{j}\omega_n\sqrt{1 - \zeta^2} = -\sigma \pm \mathrm{j}\omega_d \tag{7.7}$$

where $\sigma = \zeta\omega_n$ is the decay rate and $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ is the damped natural frequency.

7.3 Underdamped Response ($0 < \zeta < 1$)

7.3.1 Complete Step Response Derivation

For a unit step input, the Laplace-domain output is:

$$Y(s) = \frac{K\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \tag{7.8}$$

Using partial fractions with complex poles $s = -\sigma \pm \mathrm{j}\omega_d$:

$$Y(s) = \frac{K}{s} - K\frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{7.9}$$

Completing the square in the denominator and manipulating:

$$Y(s) = \frac{K}{s} - K\frac{(s+\sigma) + \sigma}{(s+\sigma)^2 + \omega_d^2} \tag{7.10}$$

$$Y(s) = \frac{K}{s} - K\frac{s+\sigma}{(s+\sigma)^2 + \omega_d^2} - K\frac{\sigma}{\omega_d}\frac{\omega_d}{(s+\sigma)^2 + \omega_d^2} \tag{7.11}$$

Taking the inverse Laplace transform:

$$y(t) = K\left[1 - \mathrm{e}^{-\sigma t}\cos(\omega_d t) - \frac{\sigma}{\omega_d}\mathrm{e}^{-\sigma t}\sin(\omega_d t)\right] \tag{7.12}$$

Using the identity $\cos\theta + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\theta = \frac{1}{\sqrt{1-\zeta^2}}\sin(\theta + \phi)$ where $\phi = \arctan\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right) = \arccos(\zeta)$:

$$\boxed{y(t) = K\left[1 - \frac{\mathrm{e}^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)\right], \quad \phi = \arccos(\zeta)} \tag{7.13}$$

An equivalent form using cosine is:

$$y(t) = K\left[1 - \frac{\mathrm{e}^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\cos(\omega_d t - \psi)\right], \quad \psi = \arctan\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right) \tag{7.14}$$

7.4 Critically Damped Response ($\zeta = 1$)

When $\zeta = 1$, the poles are real and repeated at $s = -\omega_n$:

$$G(s) = \frac{K\omega_n^2}{(s+\omega_n)^2} \tag{7.15}$$

For a unit step input:

$$Y(s) = \frac{K\omega_n^2}{s(s+\omega_n)^2} = \frac{K}{s} - \frac{K}{s+\omega_n} - \frac{K\omega_n}{(s+\omega_n)^2} \tag{7.16}$$

Taking the inverse Laplace transform:

$$\boxed{y(t) = K\left[1 - (1 + \omega_n t)\mathrm{e}^{-\omega_n t}\right]} \tag{7.17}$$

Critical damping represents the boundary between oscillatory and non-oscillatory behavior, providing the fastest approach to steady state without overshoot.

7.5 Overdamped Response ($\zeta > 1$)

When $\zeta > 1$, the poles are real and distinct:

$$s_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}, \quad s_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1} \tag{7.18}$$

Let $\alpha = \omega_n(\zeta - \sqrt{\zeta^2 - 1})$ and $\beta = \omega_n(\zeta + \sqrt{\zeta^2 - 1})$, so $s_1 = -\alpha$ and $s_2 = -\beta$. The step response becomes:

$$\boxed{y(t) = K\left[1 - \frac{\beta}{\beta - \alpha}\mathrm{e}^{-\alpha t} + \frac{\alpha}{\beta - \alpha}\mathrm{e}^{-\beta t}\right]} \tag{7.19}$$

The response is dominated by the slower pole (smaller α), resulting in sluggish behavior.

7.6 Pole-Zero Map and Damping Relationship

The location of poles in the complex s-plane directly determines system behavior. For second-order systems, key geometric relationships exist.

7.6.1 Geometric Interpretation

For underdamped systems, the poles lie on a circle of radius ω_n centered at the origin:

$$|s| = \sqrt{\sigma^2 + \omega_d^2} = \sqrt{(\zeta\omega_n)^2 + (\omega_n\sqrt{1 - \zeta^2})^2} = \omega_n \tag{7.20}$$

The angle θ from the negative real axis satisfies:

$$\cos\theta = \zeta, \quad \sin\theta = \sqrt{1 - \zeta^2} \tag{7.21}$$

Pole Location and System Behavior

- **Distance from origin** (ω_n): Determines speed of response
- **Real part** ($-\sigma = -\zeta\omega_n$): Determines decay rate of oscillations
- **Imaginary part** ($\pm\omega_d$): Determines frequency of oscillation
- **Angle from negative real axis**: $\theta = \arccos(\zeta)$ determines overshoot
- Lines of constant ζ are radial lines from the origin
- Lines of constant ω_n are circular arcs centered at the origin
- Lines of constant σ are vertical lines parallel to the imaginary axis

7.7 Time-Domain Specifications

Control system design often begins with time-domain specifications that must be translated into requirements on ζ and ω_n.

7.7.1 Rise Time

Rise time t_r is typically defined as the time to go from 10% to 90% of the final value. For underdamped second-order systems, an accurate approximation is:

$$\boxed{t_r \approx \frac{1.8}{\omega_n}} \quad \text{(for } 0.3 < \zeta < 0.8\text{)} \tag{7.22}$$

A more precise formula valid for broader ranges:

$$t_r \approx \frac{1 + 1.1\zeta + 1.4\zeta^2}{\omega_n} \tag{7.23}$$

7.7.2 Peak Time

Peak time t_p is when the first overshoot occurs. Setting $\mathrm{d}y/\mathrm{d}t = 0$ for the underdamped response:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{K\omega_n}{\sqrt{1-\zeta^2}} \mathrm{e}^{-\zeta\omega_n t} \sin(\omega_d t) = 0 \tag{7.24}$$

This occurs when $\sin(\omega_d t) = 0$, giving $\omega_d t = n\pi$. The first peak is at:

$$\boxed{t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}} \tag{7.25}$$

7.7.3 Percent Overshoot

The maximum overshoot M_p is:

$$M_p = y(t_p) - K = K\frac{\mathrm{e}^{-\zeta\omega_n t_p}}{\sqrt{1-\zeta^2}} \sin(\omega_d t_p + \phi) - K \tag{7.26}$$

Substituting $t_p = \pi/\omega_d$ and simplifying:

$$\boxed{\%OS = 100 \cdot \mathrm{e}^{-\pi\zeta/\sqrt{1-\zeta^2}}} \tag{7.27}$$

Inverting this relationship to find ζ from a desired overshoot:

$$\boxed{\zeta = \frac{-\ln(\%OS/100)}{\sqrt{\pi^2 + \ln^2(\%OS/100)}}} \tag{7.28}$$

7.7.4 Settling Time

Settling time t_s is when the response remains within a specified percentage of the final value. Using the envelope $\mathrm{e}^{-\zeta\omega_n t}$:

For 2% criterion ($\mathrm{e}^{-\zeta\omega_n t_s} = 0.02$):

$$\boxed{t_s \approx \frac{4}{\zeta\omega_n} = \frac{4}{\sigma}} \tag{7.29}$$

For 5% criterion:

$$t_s \approx \frac{3}{\zeta\omega_n} \tag{7.30}$$

Common Mistakes in Time-Domain Calculations

- Using ω_n instead of ω_d in peak time calculations
- Forgetting that settling time depends on $\zeta\omega_n$, not ω_n alone
- Confusing rise time definitions (0-100% vs 10-90%)
- Applying underdamped formulas to overdamped systems
- Neglecting that overshoot formula only applies for $0 < \zeta < 1$

7.8 Worked Examples

Example 7.1: First-Order Thermal System

A thermal system has a transfer function $G(s) = \frac{50}{10s+1}$ relating heater input (watts) to temperature change (degrees C). Find:

(a) The time constant and DC gain

(b) The response to a 2 W step input

(c) The time to reach 95% of steady state

Solution:
(a) Comparing with standard form $G(s) = K/(\tau s + 1)$:

$$\tau = 10 \text{ s (time constant)} \tag{7.31}$$
$$K = 50\ ^\circ\text{C/W (DC gain)} \tag{7.32}$$

(b) For a 2 W step input, the steady-state temperature rise is:

$$\Delta T_{ss} = K \cdot 2 = 100^\circ\text{C} \tag{7.33}$$

The response is:

$$\Delta T(t) = 100(1 - \mathrm{e}^{-t/10})^\circ\text{C} \tag{7.34}$$

(c) At 95% of steady state: $0.95 = 1 - \mathrm{e}^{-t/10}$, giving:

$$t = -10\ln(0.05) = 30 \text{ s} = 3\tau \tag{7.35}$$

Example 7.2: Second-Order System Analysis

A position control system has transfer function:

$$G(s) = \frac{100}{s^2 + 4s + 100} \tag{7.36}$$

Find: (a) ω_n, ζ, and pole locations, (b) all time-domain specifications, (c) the step response expression.
Solution:

(a) Comparing with $G(s) = K\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$:

$$\omega_n^2 = 100 \quad \Rightarrow \quad \omega_n = 10 \text{ rad/s} \tag{7.37}$$
$$2\zeta\omega_n = 4 \quad \Rightarrow \quad \zeta = 0.2 \tag{7.38}$$
$$K = 1 \tag{7.39}$$

Poles: $s_{1,2} = -2 \pm \mathrm{j}\sqrt{100-4} = -2 \pm \mathrm{j}9.80$
Damped frequency: $\omega_d = 10\sqrt{1-0.04} = 9.80$ rad/s
(b) Time-domain specifications:

$$t_r \approx \frac{1.8}{10} = 0.18 \text{ s} \tag{7.40}$$
$$t_p = \frac{\pi}{9.80} = 0.32 \text{ s} \tag{7.41}$$
$$\%OS = 100\mathrm{e}^{-0.2\pi/\sqrt{1-0.04}} = 100\mathrm{e}^{-0.641} = 52.7\% \tag{7.42}$$
$$t_s \approx \frac{4}{0.2 \times 10} = 2 \text{ s} \tag{7.43}$$

(c) Step response ($\phi = \arccos(0.2) = 78.5° = 1.37$ rad):

$$y(t) = 1 - \frac{\mathrm{e}^{-2t}}{0.98}\sin(9.80t + 1.37) \tag{7.44}$$

Example 7.3: Design for Specified Overshoot

Design a second-order system with no more than 10% overshoot and settling time (2%) of 0.5 seconds.
Solution:
From the overshoot requirement using Equation (7.28):

$$\zeta = \frac{-\ln(0.10)}{\sqrt{\pi^2 + \ln^2(0.10)}} = \frac{2.303}{\sqrt{9.87 + 5.30}} = \frac{2.303}{3.89} = 0.591 \tag{7.45}$$

From the settling time requirement:

$$t_s = \frac{4}{\zeta\omega_n} = 0.5 \quad \Rightarrow \quad \zeta\omega_n = 8 \tag{7.46}$$

Therefore:

$$\omega_n = \frac{8}{0.591} = 13.5 \text{ rad/s} \tag{7.47}$$

The required transfer function is:

$$G(s) = \frac{182.25}{s^2 + 16s + 182.25} \tag{7.48}$$

Verification: $\omega_d = 13.5\sqrt{1-0.35} = 10.9$ rad/s, $t_p = 0.29$ s

Example 7.4: Overdamped System Response

A hydraulic actuator has transfer function:

$$G(s) = \frac{25}{s^2 + 10s + 25} = \frac{25}{(s+5)^2} \tag{7.49}$$

(a) Identify the system type and parameters. (b) Find the step response.
Solution:
(a) Comparing with standard form: $\omega_n^2 = 25 \Rightarrow \omega_n = 5$ rad/s
$2\zeta\omega_n = 10 \Rightarrow \zeta = 1$ (critically damped)
The system has a repeated pole at $s = -5$.
(b) Using Equation (7.17):

$$y(t) = 1 - (1 + 5t)\mathrm{e}^{-5t} \tag{7.50}$$

Settling time (2%): $t_s \approx 4/5 = 0.8$ s (using the approximation for critical damping)

Example 7.5: Comparing Damping Cases

Three systems have the same natural frequency $\omega_n = 4$ rad/s but different damping:

- System A: $\zeta = 0.5$ (underdamped)
- System B: $\zeta = 1.0$ (critically damped)
- System C: $\zeta = 2.0$ (overdamped)

Compare their step responses.
Solution:
System A ($\zeta = 0.5$):

$$\omega_d = 4\sqrt{1 - 0.25} = 3.46 \text{ rad/s} \tag{7.51}$$
$$t_p = \pi/3.46 = 0.91 \text{ s} \tag{7.52}$$
$$\%OS = 100\mathrm{e}^{-0.5\pi/0.866} = 16.3\% \tag{7.53}$$
$$t_s = 4/(0.5 \times 4) = 2 \text{ s} \tag{7.54}$$

System B ($\zeta = 1.0$):

$$\text{No overshoot, poles at } s = -4 \tag{7.55}$$
$$t_s \approx 4/4 = 1 \text{ s} \tag{7.56}$$

System C ($\zeta = 2.0$):

$$\text{Poles: } s_1 = -4(2 - \sqrt{3}) = -1.07, \quad s_2 = -4(2 + \sqrt{3}) = -14.9 \tag{7.57}$$
$$t_s \approx 4/1.07 = 3.7 \text{ s (dominated by slow pole)} \tag{7.58}$$

The critically damped system (B) has the fastest settling time without overshoot. The underdamped system (A) reaches steady state faster initially but overshoots. The overdamped system (C) is slowest overall.

7.9 Practical Considerations

Practical Considerations

Real systems deviate from ideal first and second-order models:

- **Higher-order dynamics**: Most physical systems have additional poles; second-order models capture dominant behavior when other poles are far in the left half-plane.
- **Zeros**: Transfer function zeros affect overshoot and rise time; a zero near a pole can approximately cancel its effect.
- **Nonlinearities**: Saturation, dead zones, and hysteresis cause deviations from linear predictions.
- **Parameter uncertainty**: Actual values of ζ and ω_n may differ from design values; robust designs account for this variation.
- **Time delays**: Transport delays add phase lag not captured by rational transfer functions; they significantly affect stability.

Chapter Summary: Key Takeaways

- **First-order systems** are characterized by a single time constant τ; the step response is exponential with 63.2% rise at $t = \tau$.
- **Second-order systems** are characterized by ω_n and ζ; the damping ratio determines whether the response is underdamped, critically damped, or overdamped.
- **Pole locations** encode all dynamic behavior: distance from origin sets speed, angle from negative real axis sets damping.
- **Time-domain specifications** (t_r, t_p, $\%OS$, t_s) translate directly to requirements on ζ and ω_n.
- **Design process**: Specify time-domain requirements $\rightarrow$ calculate required ζ and ω_n $\rightarrow$ place poles appropriately.

Chapter 8

Mechanical Systems

Mechanical systems form the intuitive foundation for understanding dynamics. The physical concepts of mass, springs, and dampers have direct analogs in other domains and provide excellent intuition for system behavior.

8.1 Translational Mechanical Elements

8.1.1 Mass (Inertia)

Mass resists changes in velocity according to Newton's second law:

$$F = m\frac{\mathrm{d}v}{\mathrm{d}t} = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = m\ddot{x} \tag{8.1}$$

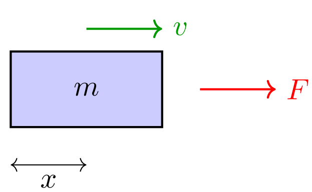

Figure 8.1: Mass element with applied force, coordinate system, and ground reference.

In the Laplace domain:

$$F(s) = ms^2X(s) \quad \Rightarrow \quad \frac{X(s)}{F(s)} = \frac{1}{ms^2} \tag{8.2}$$

8.1.2 Spring (Stiffness)

A linear spring produces a force proportional to displacement:

$$F = kx \quad \text{or} \quad F = k(x_1 - x_2) \tag{8.3}$$

where k is the spring constant (N/m or lb/in).

In the Laplace domain:

$$F(s) = kX(s) \quad \Rightarrow \quad \frac{X(s)}{F(s)} = \frac{1}{k} \tag{8.4}$$

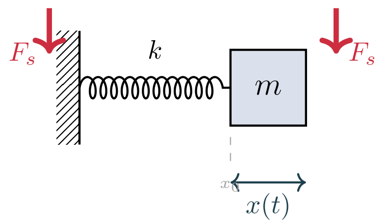

Figure 8.2: Spring element with force arrows on both ends and displacement annotation.

8.1.3 Damper (Viscous Friction)

A viscous damper produces a force proportional to velocity:

$$F = b\dot{x} = b\frac{\mathrm{d}x}{\mathrm{d}t} \quad \text{or} \quad F = b(\dot{x}_1 - \dot{x}_2) \tag{8.5}$$

where b is the damping coefficient (N·s/m or lb·s/in).

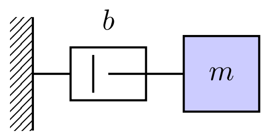

Figure 8.3: Viscous damper element.

In the Laplace domain:

$$F(s) = bsX(s) \quad \Rightarrow \quad \frac{X(s)}{F(s)} = \frac{1}{bs} \tag{8.6}$$

8.2 Mass-Spring-Damper System

The canonical mechanical system combines all three elements.

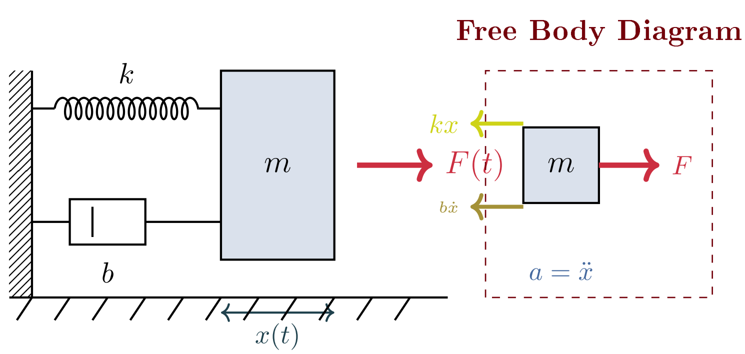

Figure 8.4: Mass-spring-damper system with free body diagram showing all forces.

8.2.1 Equation of Motion

Applying Newton's second law:

$$m\ddot{x} = F(t) - kx - b\dot{x} \tag{8.7}$$

Rearranging:

$$\boxed{m\ddot{x} + b\dot{x} + kx = F(t)} \tag{8.8}$$

8.2.2 Transfer Function

Taking the Laplace transform with zero initial conditions:

$$ms^2X(s) + bsX(s) + kX(s) = F(s) \tag{8.9}$$

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} \tag{8.10}$$

In standard second-order form:

$$G(s) = \frac{1/k}{(m/k)s^2 + (b/k)s + 1} = \frac{1/k}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n}s + 1} \tag{8.11}$$

where:

$$\omega_n = \sqrt{\frac{k}{m}} \quad \text{(natural frequency)} \tag{8.12}$$

$$\zeta = \frac{b}{2\sqrt{km}} \quad \text{(damping ratio)} \tag{8.13}$$

8.2.3 Pole-Zero Map for Different Damping Ratios

The location of poles in the complex s-plane determines the system's response characteristics. For a second-order system, the poles are located at:

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} \tag{8.14}$$

The poles move along a circular locus as the damping ratio changes, creating three distinct response regions.

Key observations from the pole-zero map:

- **Underdamped** ($\zeta < 1$): Complex conjugate poles create oscillatory response with overshoot.
- **Critically damped** ($\zeta = 1$): Repeated real pole gives fastest non-oscillatory response.
- **Overdamped** ($\zeta > 1$): Two distinct real poles result in sluggish response without oscillation.

Example 1.2: Automobile Shock Absorber Test

A shock absorber test rig has $m = 50$ kg, $k = 20{,}000$ N/m, and $b = 1{,}000$ N·s/m. Find:

(a) Natural frequency and damping ratio

(b) The response to a 100 N step force

Solution:

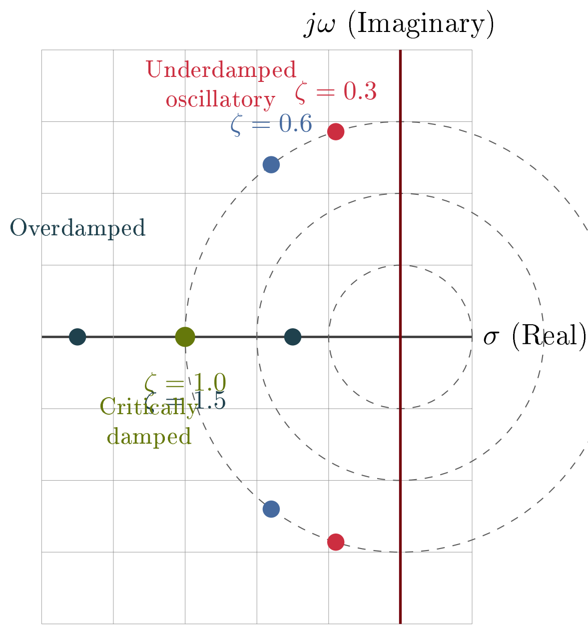

Figure 8.5: Pole-zero map showing pole locations for different damping ratios ($\omega_n = 3$ rad/s fixed). As ζ increases from underdamped to overdamped, poles move from the complex plane onto the real axis.

(a) Natural frequency and damping ratio:

$$\omega_n = \sqrt{\frac{20{,}000}{50}} = \sqrt{400} = 20 \text{ rad/s} = 3.18 \text{ Hz} \tag{8.15}$$

$$\zeta = \frac{1{,}000}{2\sqrt{20{,}000 \times 50}} = \frac{1{,}000}{2{,}000} = 0.5 \tag{8.16}$$

The system is underdamped.
(b) Transfer function:

$$G(s) = \frac{1}{50s^2 + 1000s + 20000} = \frac{0.00005}{s^2 + 20s + 400} \tag{8.17}$$

For a 100 N step: $Y(s) = \frac{100}{s} \cdot G(s)$
Using partial fractions and inverse Laplace:

$$x(t) = \frac{100}{20000}\left[1 - \frac{e^{-10t}}{\sqrt{1-0.25}}\sin(17.32t + 1.047)\right] \tag{8.18}$$

Steady-state displacement: $x_{ss} = \frac{F}{k} = \frac{100}{20000} = 5$ mm

Historical Context: Hooke's Law, Newton's Mechanics, and Vibration Analysis

The spring, one of the most fundamental control components, was mathematically described by Robert Hooke in 1660 when he published his law of elasticity: the force in a spring is proportional to its extension (Hooke stated it in Latin: "Ut tensio, sic vis"—as the extension, so the force). This simple linear relationship, $F = kx$, became the foundation for modeling elastic systems for centuries. Combined with Isaac Newton's laws of motion (published in 1687), Hooke's law enabled the prediction of oscillatory behavior in mechanical systems. The mass-spring system's equation of motion, $m\ddot{x} + kx = F$, is perhaps the most important differential equation in all of engineering.

The systematic analysis of mechanical vibrations emerged in the 18th and 19th centuries as engineers faced resonance problems in machinery, bridges, and vehicles. Understanding that vibrations depended on mass, stiffness, and damping—and that underdamped systems could produce dangerous oscillations—drove the development of stability analysis methods. Vibration analysis, grounded in Hooke's and Newton's laws, ultimately led to the Nyquist and Bode analysis techniques that dominate modern control design, making the humble spring-mass-damper system the pedagogical gateway to control theory.

8.3 Vehicle Suspension System

A quarter-car model represents one corner of a vehicle suspension.

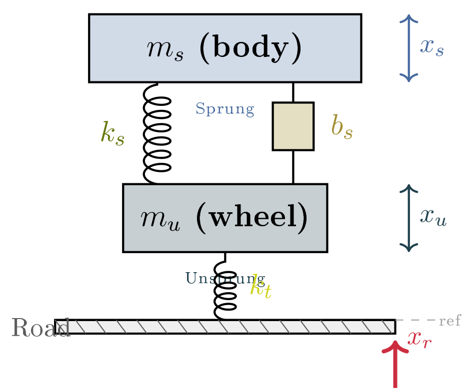

Figure 8.6: Quarter-car suspension model with distinct colors for sprung (body) and unsprung (wheel) masses, and road input visualization.

8.3.1 Equations of Motion

For the sprung mass (body):

$$m_s\ddot{x}_s = -k_s(x_s - x_u) - b_s(\dot{x}_s - \dot{x}_u) \tag{8.19}$$

For the unsprung mass (wheel):

$$m_u\ddot{x}_u = k_s(x_s - x_u) + b_s(\dot{x}_s - \dot{x}_u) - k_t(x_u - x_r) \tag{8.20}$$

8.3.2 State-Space Representation

Defining state variables: $\mathbf{x} = [x_s - x_u, \dot{x}_s, x_u - x_r, \dot{x}_u]^\top$

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -k_s/m_s & -b_s/m_s & 0 & b_s/m_s \\ 0 & 0 & 0 & 1 \\ k_s/m_u & b_s/m_u & -k_t/m_u & -b_s/m_u \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \end{bmatrix} \dot{x}_r \tag{8.21}$$

8.3.3 Transfer Functions

From road input $X_r(s)$ to body displacement $X_s(s)$:

$$\frac{X_s(s)}{X_r(s)} = \frac{(b_s s + k_s)k_t}{m_s m_u s^4 + (m_s + m_u)b_s s^3 + [m_s k_t + (m_s + m_u)k_s]s^2 + b_s k_t s + k_s k_t} \tag{8.22}$$

Example 1.3: Quarter-Car Parameters

Typical values for a passenger car:

- Sprung mass: $m_s = 250$ kg (quarter of body mass)
- Unsprung mass: $m_u = 35$ kg (wheel, brake, suspension links)
- Suspension stiffness: $k_s = 16{,}000$ N/m
- Suspension damping: $b_s = 1{,}000$ N·s/m
- Tire stiffness: $k_t = 160{,}000$ N/m

The body bounce frequency: $f_s = \frac{1}{2\pi}\sqrt{\frac{k_s}{m_s}} = 1.27$ Hz

The wheel hop frequency: $f_u = \frac{1}{2\pi}\sqrt{\frac{k_t}{m_u}} = 10.8$ Hz

Figure 8.7 shows a pickup truck profile that can be analyzed using the quarter-car model at each wheel.

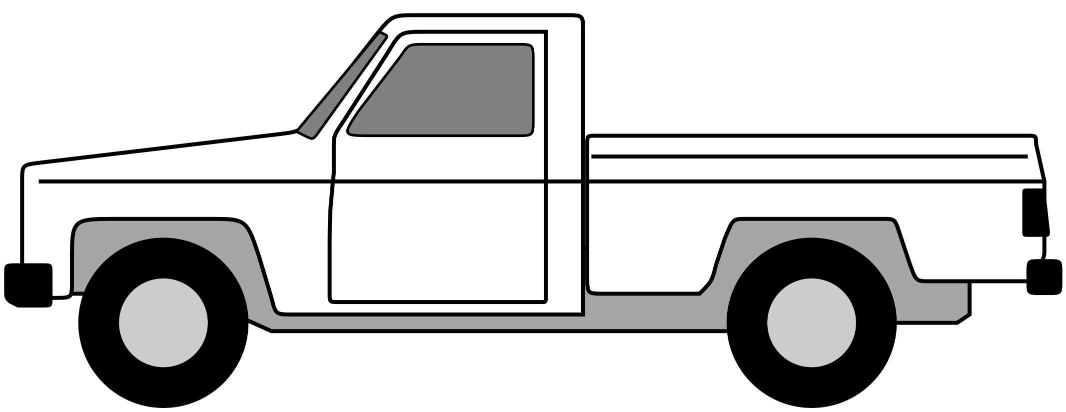

Figure 8.7: Pickup truck profile for quarter-car suspension analysis.

8.4 Rotational Mechanical Systems

Rotational systems follow analogous laws with angular quantities.

Translational	Rotational	Unit
Displacement x	Angle θ	rad
Velocity $v = \dot{x}$	Angular velocity $\omega = \dot{\theta}$	rad/s
Acceleration $a = \ddot{x}$	Angular acceleration $\alpha = \ddot{\theta}$	rad/s^2
Force F	Torque τ	N·m
Mass m	Moment of inertia J	kg·m^2
Spring constant k	Torsional stiffness κ	N·m/rad
Damping b	Rotational damping B	N·m·s/rad

Table 8.1: Translational-rotational analogies.

8.4.1 Rotational Inertia

$$\tau = J\ddot{\theta} = J\dot{\omega} \tag{8.23}$$

8.4.2 Torsional Spring

$$\tau = \kappa\theta \quad \text{or} \quad \tau = \kappa(\theta_1 - \theta_2) \tag{8.24}$$

8.4.3 Rotational Damper

$$\tau = B\dot{\theta} = B\omega \tag{8.25}$$

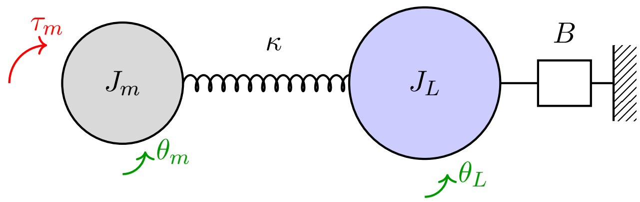

Figure 8.8: Two-inertia system with flexible coupling.

Example 1.4: Rotating Shaft System

A motor drives a load through a flexible shaft (see Figure 8.8). The motor inertia is $J_m = 0.01$ kg·m^2, load inertia is $J_L = 0.1$ kg·m^2, shaft stiffness is $\kappa = 100$ N·m/rad, and bearing friction is $B = 0.5$ N·m·s/rad.
Equations of motion:

$$J_m\ddot{\theta}_m = \tau_m - \kappa(\theta_m - \theta_L) \tag{8.26}$$

$$J_L\ddot{\theta}_L = \kappa(\theta_m - \theta_L) - B\dot{\theta}_L \tag{8.27}$$

Transfer function from motor torque to load angle:

$$\frac{\Theta_L(s)}{\tau_m(s)} = \frac{\kappa}{J_m J_L s^4 + J_m B s^3 + (J_m + J_L)\kappa s^2 + B\kappa s} \tag{8.28}$$

8.5 Gear Trains

Gears transform speed and torque between shafts.

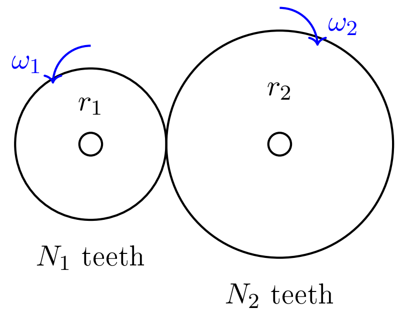

Figure 8.9: Gear pair.

Gear Ratio:

$$n = \frac{N_2}{N_1} = \frac{r_2}{r_1} = \frac{\omega_1}{\omega_2} = \frac{\tau_2}{\tau_1} \tag{8.29}$$

Reflected Inertia: When reflecting load inertia J_L to the motor side:

$$J_{\text{reflected}} = \frac{J_L}{n^2} \tag{8.30}$$

Reflected Damping:

$$B_{\text{reflected}} = \frac{B_L}{n^2} \tag{8.31}$$

Gear Ratio Selection

The gear ratio affects both speed and torque:

- High gear ratio ($n > 1$): Reduces speed, increases torque, reduces reflected inertia
- Low gear ratio ($n < 1$): Increases speed, reduces torque, increases reflected inertia

Optimal gear ratio often maximizes acceleration: $n_{\text{opt}} = \sqrt{J_L/J_m}$

8.6 Multi-Body Mechanical Systems

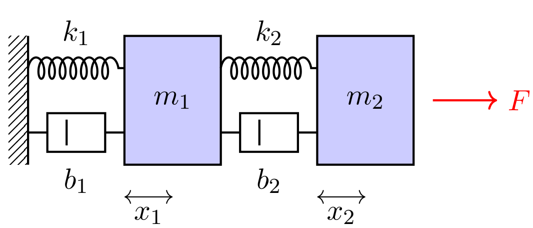

Figure 8.10: Two-mass mechanical system.

Example 1.5: Two-Mass System

Consider two masses connected by springs and dampers (see Figure 8.10).
Equations of motion:

$$m_1\ddot{x}_1 = -k_1x_1 - b_1\dot{x}_1 + k_2(x_2 - x_1) + b_2(\dot{x}_2 - \dot{x}_1) \tag{8.32}$$

$$m_2\ddot{x}_2 = F - k_2(x_2 - x_1) - b_2(\dot{x}_2 - \dot{x}_1) \tag{8.33}$$

State-space form with $\mathbf{x} = [x_1, \dot{x}_1, x_2, \dot{x}_2]^\top$:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -(k_1+k_2)/m_1 & -(b_1+b_2)/m_1 & k_2/m_1 & b_2/m_1 \\ 0 & 0 & 0 & 1 \\ k_2/m_2 & b_2/m_2 & -k_2/m_2 & -b_2/m_2 \end{bmatrix} \tag{8.34}$$

8.7 Pendulum Systems

8.7.1 Simple Pendulum

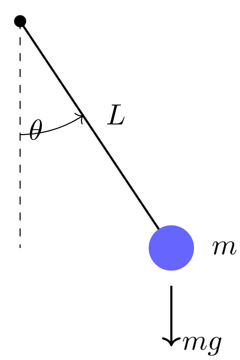

Figure 8.11: Simple pendulum.

The equation of motion (nonlinear):

$$mL^2\ddot{\theta} + mgL\sin\theta = \tau \tag{8.35}$$

For small angles ($\sin\theta \approx \theta$), the linearized equation:

$$\ddot{\theta} + \frac{g}{L}\theta = \frac{\tau}{mL^2} \tag{8.36}$$

Natural frequency: $\omega_n = \sqrt{g/L}$

8.7.2 Inverted Pendulum on Cart

The classic control benchmark problem:

Linearized equations about the upright position ($\theta = 0$):

$$(M + m)\ddot{x} + m\ell\ddot{\theta} = F \tag{8.37}$$

$$m\ell\ddot{x} + m\ell^2\ddot{\theta} - mg\ell\theta = 0 \tag{8.38}$$

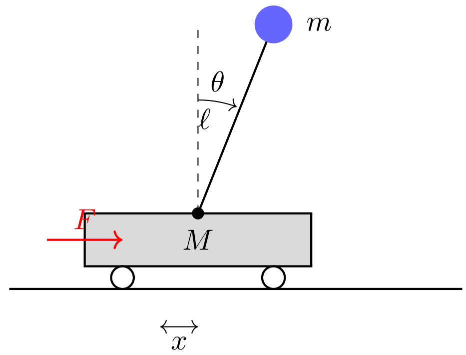

Figure 8.12: Inverted pendulum on cart.

Transfer function from force to angle:

$$\frac{\Theta(s)}{F(s)} = \frac{1}{(M\ell)s^2 - (M+m)g} \tag{8.39}$$

This system is **open-loop unstable** (pole in the right half-plane) and requires feedback control for stabilization.

Chapter 9

Electrical Systems

Electrical circuits provide clean examples of dynamic systems with well-defined mathematical models. The concepts developed here—impedance, frequency response, filtering—form the historical foundation of control theory.

Historical Context: Black's Negative Feedback Amplifier and Frequency Response Methods

The foundation of modern control theory emerged not from mechanical engineers but from electrical engineers at Bell Telephone Laboratories. In 1927, Harold Black invented the negative feedback amplifier, recognizing that feeding a portion of the output back inverted to the input could stabilize amplifier gain against component variations and temperature changes. This elegant insight—that feedback could improve performance—was initially rejected by patent offices, who considered it obvious. Yet Black's feedback amplifier enabled long-distance telephone transmission by compensating for signal loss over cables, fundamentally changing telecommunications.

The mathematical analysis of feedback systems was subsequently developed by Harry Nyquist (1932) and Hendrik Bode (1945), who created graphical methods to analyze stability and performance from frequency response measurements. The Nyquist stability criterion and Bode plots transformed control design from an empirical art into a quantitative science. These frequency response techniques, born from electrical circuits and amplifiers, became universal tools applicable to all domains—mechanical, thermal, biological, and beyond—establishing electrical systems as the historical birthplace of modern control theory.

9.1 Passive Circuit Elements

9.1.1 Resistor

Ohm's law relates voltage and current:

$$v(t) = Ri(t) \tag{9.1}$$

In the Laplace domain: $V(s) = RI(s)$
Impedance: $Z_R(s) = R$

Figure 9.1: Resistor element.

9.1.2 Capacitor

The capacitor stores charge:

$$i(t) = C\frac{\mathrm{d}v}{\mathrm{d}t} \quad \Leftrightarrow \quad v(t) = \frac{1}{C}\int i(\tau)\,\mathrm{d}\tau \tag{9.2}$$

In the Laplace domain: $I(s) = CsV(s)$ or $V(s) = \frac{1}{Cs}I(s)$
Impedance: $Z_C(s) = \frac{1}{Cs}$

Figure 9.2: Capacitor element.

9.1.3 Inductor

The inductor stores magnetic energy:

$$v(t) = L\frac{\mathrm{d}i}{\mathrm{d}t} \quad \Leftrightarrow \quad i(t) = \frac{1}{L}\int v(\tau)\,\mathrm{d}\tau \tag{9.3}$$

In the Laplace domain: $V(s) = LsI(s)$ or $I(s) = \frac{1}{Ls}V(s)$
Impedance: $Z_L(s) = Ls$

Figure 9.3: Inductor element.

9.2 RC Circuit (First-Order System)

9.2.1 Time-Domain Analysis

Applying KVL:

$$v_{\text{in}} = v_R + v_{\text{out}} = Ri + v_{\text{out}} \tag{9.4}$$

Since $i = C\frac{\mathrm{d}v_{\text{out}}}{\mathrm{d}t}$:

$$\boxed{RC\frac{\mathrm{d}v_{\text{out}}}{\mathrm{d}t} + v_{\text{out}} = v_{\text{in}}} \tag{9.5}$$

This is a first-order ODE with time constant $\tau = RC$.

9.2.2 Transfer Function

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{1}{RCs + 1} = \frac{1}{\tau s + 1} \tag{9.6}$$

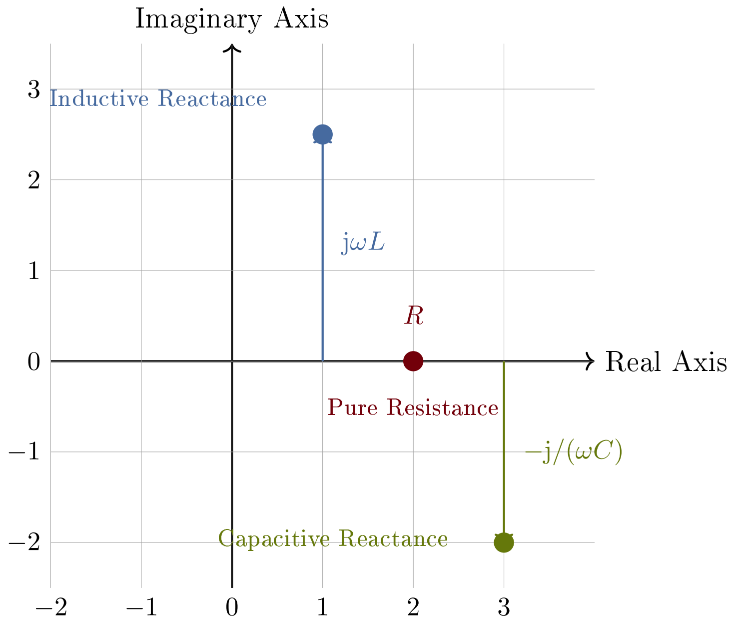

Figure 9.4: Impedance in the complex plane showing resistive, inductive, and capacitive elements.

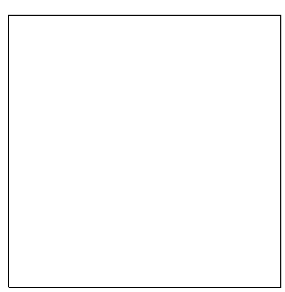

Figure 9.5: RC low-pass filter circuit.

9.2.3 Frequency Response

Substituting $s = \mathrm{j}\omega$:

$$G(\mathrm{j}\omega) = \frac{1}{1 + \mathrm{j}\omega RC} \tag{9.7}$$

Magnitude and phase:

$$|G(\mathrm{j}\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}} \tag{9.8}$$

$$\angle G(\mathrm{j}\omega) = -\arctan(\omega RC) \tag{9.9}$$

Cutoff frequency (where magnitude is $1/\sqrt{2}$ or -3 dB):

$$\omega_c = \frac{1}{RC} = \frac{1}{\tau} \tag{9.10}$$

Example 1.6: RC Circuit Step Response

An RC circuit has $R = 10\ \mathrm{k}\Omega$ and $C = 1\ \mu\mathrm{F}$. Find:

(a) Time constant

(b) Cutoff frequency

(c) Response to a 5V step input

Solution:
(a) Time constant:

$$\tau = RC = 10{,}000 \times 10^{-6} = 0.01\ \mathrm{s} = 10\ \mathrm{ms} \tag{9.11}$$

(b) Cutoff frequency:

$$f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 0.01} = 15.9\ \mathrm{Hz} \tag{9.12}$$

(c) Step response:

$$v_{\mathrm{out}}(t) = 5(1 - \mathrm{e}^{-t/0.01}) = 5(1 - \mathrm{e}^{-100t})\ \mathrm{V} \tag{9.13}$$

At $t = \tau = 10$ ms: $v_{\mathrm{out}} = 5(1 - \mathrm{e}^{-1}) = 3.16$ V (63.2% of final)

9.3 RL Circuit

For current as output:

$$L\frac{\mathrm{d}i}{\mathrm{d}t} + Ri = v_{\mathrm{in}} \tag{9.14}$$

Transfer function (voltage to current):

$$\frac{I(s)}{V_{\mathrm{in}}(s)} = \frac{1}{Ls + R} = \frac{1/R}{(L/R)s + 1} \tag{9.15}$$

Time constant: $\tau = L/R$

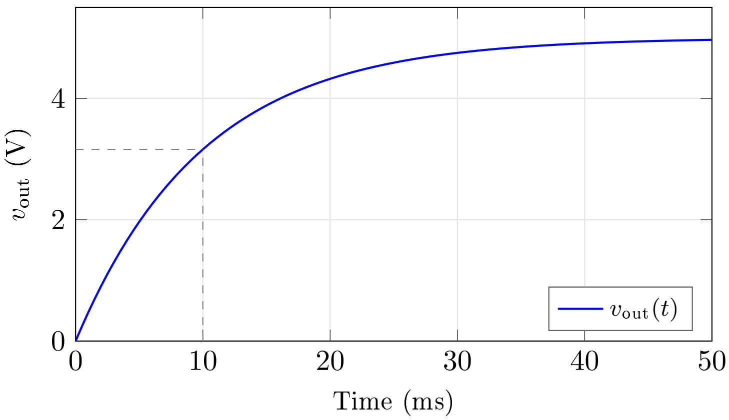

Figure 9.6: RC circuit step response.

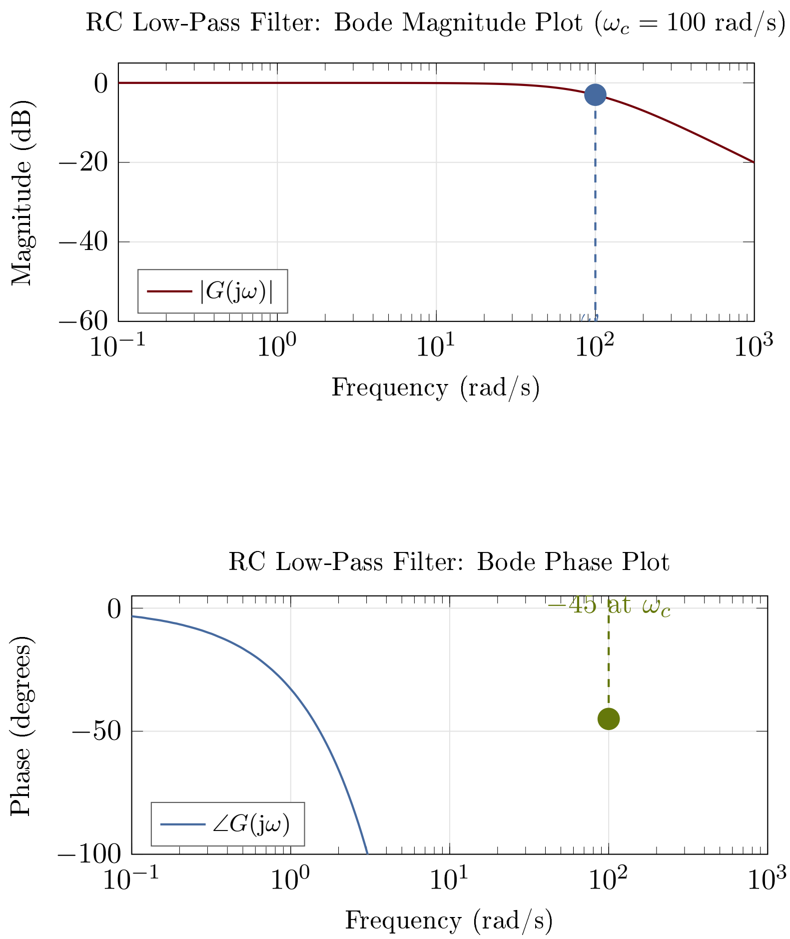

Figure 9.7: RC low-pass filter Bode plots (magnitude and phase).

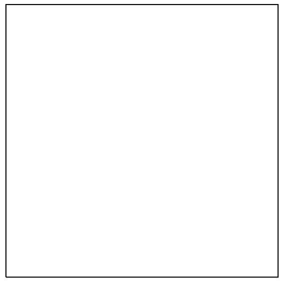

Figure 9.8: RL circuit.

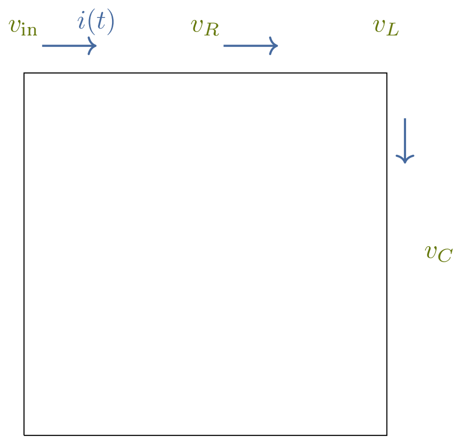

Figure 9.9: Series RLC circuit with current direction and node voltages labeled.

9.4 RLC Circuit (Second-Order System)

9.4.1 Differential Equation

Applying KVL:

$$v_{\text{in}} = v_R + v_L + v_C = Ri + L\frac{\mathrm{d}i}{\mathrm{d}t} + v_{\text{out}} \tag{9.16}$$

Since $i = C\frac{\mathrm{d}v_{\text{out}}}{\mathrm{d}t}$:

$$\boxed{LC\frac{\mathrm{d}^2v_{\text{out}}}{\mathrm{d}t^2} + RC\frac{\mathrm{d}v_{\text{out}}}{\mathrm{d}t} + v_{\text{out}} = v_{\text{in}}} \tag{9.17}$$

9.4.2 Transfer Function

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = \frac{1}{LCs^2 + RCs + 1} \tag{9.18}$$

In standard form:

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{9.19}$$

where:

$$\omega_n = \frac{1}{\sqrt{LC}} \quad \text{(resonant frequency)} \tag{9.20}$$

$$\zeta = \frac{R}{2}\sqrt{\frac{C}{L}} \quad \text{(damping ratio)} \tag{9.21}$$

9.4.3 Quality Factor

The quality factor Q measures the sharpness of resonance:

$$Q = \frac{1}{2\zeta} = \frac{1}{R}\sqrt{\frac{L}{C}} \tag{9.22}$$

High Q means sharp resonance peak and low damping.

Example 1.7: RLC Circuit Design

Design an RLC circuit with resonant frequency $f_n = 1$ kHz and $Q = 10$.
Solution:
Choose $C = 0.1$ μF. Then:

$$L = \frac{1}{(2\pi f_n)^2 C} = \frac{1}{(2\pi \times 1000)^2 \times 10^{-7}} = 0.253 \text{ H} \tag{9.23}$$

For $Q = 10$:

$$R = \frac{1}{Q}\sqrt{\frac{L}{C}} = \frac{1}{10}\sqrt{\frac{0.253}{10^{-7}}} = 159\ \Omega \tag{9.24}$$

Damping ratio: $\zeta = 1/(2Q) = 0.05$ (highly underdamped)

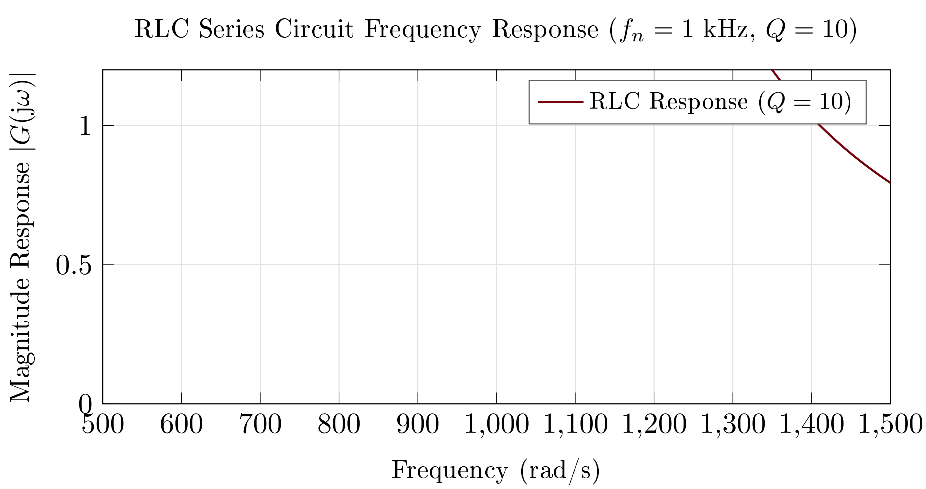

Figure 9.10: RLC series circuit frequency response showing resonance peak at ω_n and bandwidth.

9.5 Operational Amplifier Circuits

Operational amplifiers (op-amps) are fundamental building blocks for analog control systems.

9.5.1 Ideal Op-Amp Assumptions

- Infinite open-loop gain: $A \to \infty$
- Infinite input impedance: $Z_{\text{in}} \to \infty$ (no input current)
- Zero output impedance: $Z_{\text{out}} = 0$
- Infinite bandwidth

With negative feedback, these lead to the "golden rules":

1. $V_+ = V_-$ (virtual short)
2. $I_+ = I_- = 0$ (no input current)

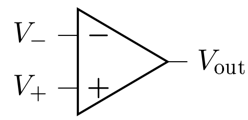

Figure 9.11: Op-amp symbol.

9.5.2 Inverting Amplifier

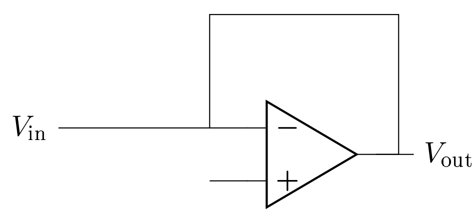

Figure 9.12: Inverting amplifier.

Transfer function:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_2}{R_1} \tag{9.25}$$

9.5.3 Non-Inverting Amplifier

Transfer function:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_2}{R_1} \tag{9.26}$$

9.5.4 Integrator

Transfer function:

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = -\frac{1}{RCs} \tag{9.27}$$

Time-domain:

$$v_{\text{out}}(t) = -\frac{1}{RC} \int_0^t v_{\text{in}}(\tau)\, \mathrm{d}\tau \tag{9.28}$$

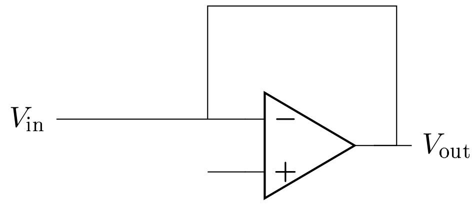

Figure 9.13: Op-amp integrator.

9.5.5 Differentiator

Transfer function:

$$G(s) = \frac{V_{\text{out}}(s)}{V_{\text{in}}(s)} = -RCs \tag{9.29}$$

Important Note

Pure differentiators amplify high-frequency noise. In practice, add a small resistor in series with C to limit high-frequency gain.

9.5.6 PID Controller Implementation

A PID controller can be implemented with op-amps:

$$G_{PID}(s) = K_p + \frac{K_i}{s} + K_d s \tag{9.30}$$

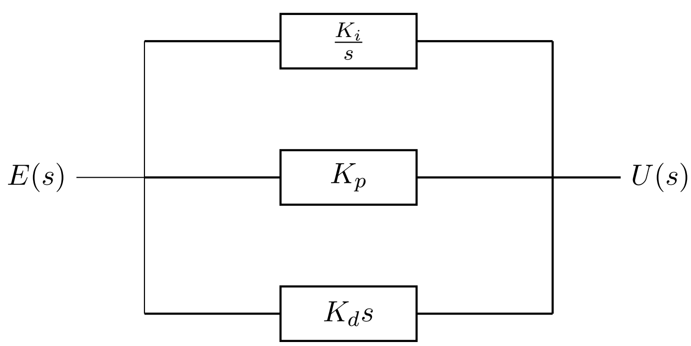

Figure 9.14: PID controller block diagram.

9.6 Active Filters

9.6.1 First-Order Low-Pass Filter

$$G(s) = \frac{K}{s/\omega_c + 1} \tag{9.31}$$

Passes low frequencies, attenuates high frequencies.

9.6.2 First-Order High-Pass Filter

$$G(s) = \frac{Ks/\omega_c}{s/\omega_c + 1} \tag{9.32}$$

Passes high frequencies, attenuates low frequencies (including DC).

9.6.3 Second-Order Butterworth Low-Pass

Maximally flat passband response:

$$G(s) = \frac{\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2} \tag{9.33}$$

Damping ratio $\zeta = 1/\sqrt{2} \approx 0.707$, giving no overshoot in step response.

9.6.4 Band-Pass Filter

$$G(s) = \frac{K(\omega_0/Q)s}{s^2 + (\omega_0/Q)s + \omega_0^2} \tag{9.34}$$

Passes frequencies near ω_0, attenuates both lower and higher frequencies.

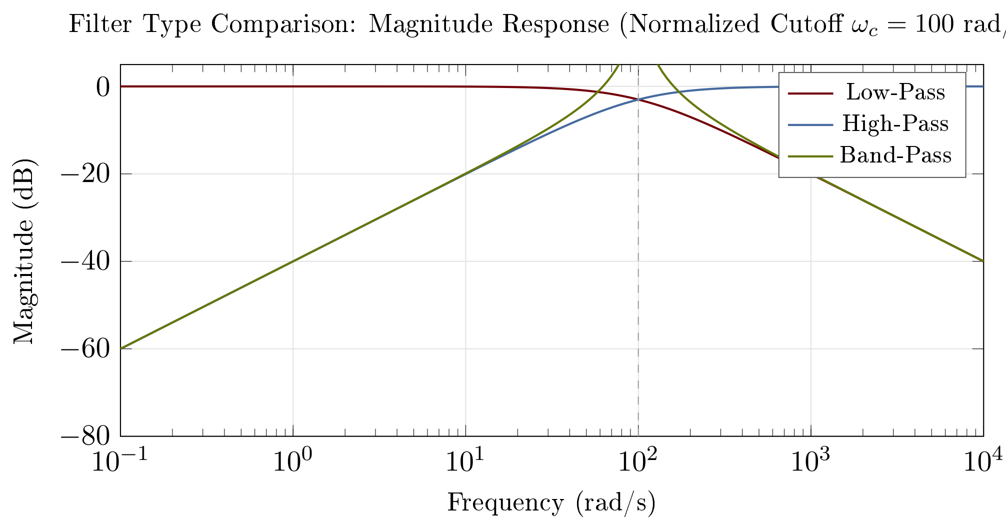

Figure 9.15: Comparison of filter types showing magnitude response of low-pass, high-pass, and band-pass filters with normalized cutoff frequency.

9.7 Power Electronics

9.7.1 Buck Converter (Step-Down)

Steady-state output (ideal):

$$V_{\text{out}} = D \cdot V_{\text{in}} \tag{9.35}$$

where D is the duty cycle ($0 \leq D \leq 1$).

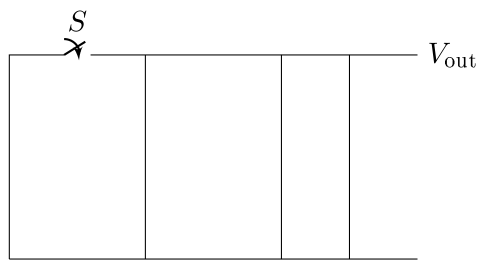

Figure 9.16: Buck converter circuit.

Small-signal transfer function (control to output):

$$\frac{\hat{v}_{\text{out}}(s)}{\hat{d}(s)} = V_{\text{in}} \cdot \frac{1}{LCs^2 + (L/R_L)s + 1} \tag{9.36}$$

Chapter 10

Electromechanical Systems

Electromechanical systems convert between electrical and mechanical energy. They are ubiquitous in control applications—from servo motors in robots to voice coils in hard drives.

10.1 DC Motor Fundamentals

The DC motor is the workhorse of motion control, converting electrical power to rotational mechanical power.

Historical Context: Faraday's Electromagnetic Induction and the Motor-Generator Principle

The DC motor's operation is rooted in Michael Faraday's 1831 discovery of electromagnetic induction—that a changing magnetic flux induces an electric current. Faraday's insight that electricity and magnetism were fundamentally related led to the realization that the inverse was also true: a current-carrying conductor in a magnetic field experiences a force. This force-current relationship, quantified as $F = BIL$ (where B is magnetic flux density and L is conductor length), became the principle underlying the DC motor. Early motors (1830s-1890s) were crude devices driven by this principle, but lacked practical commutation until the development of split-ring commutators.

The fierce competition between Thomas Edison's direct current (DC) and Nikola Tesla's alternating current (AC) systems (the "War of Currents," 1880s-1890s) hinged largely on motor capabilities. DC motors' superior controllability made them ideal for variable-speed applications and industrial drives, while AC motors' simplicity eventually won for grid applications. The DC motor's ease of control—coupled with its simple mathematical model involving back-EMF and torque constants—made it the canonical example for teaching control theory, a role it maintains to this day even as brushless motors have replaced traditional commutators in many applications.

10.1.1 Governing Equations

Electrical equation (armature circuit)

$$V_a = R_a i_a + L_a \frac{\mathrm{d}i_a}{\mathrm{d}t} + e_b \tag{10.1}$$

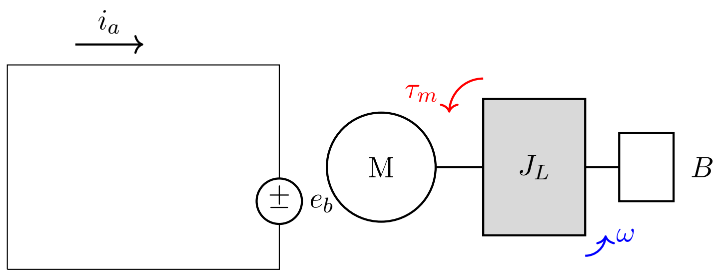

Figure 10.1: DC motor equivalent circuit and mechanical load.

Back-EMF (proportional to speed)

$$e_b = K_e\omega = K_e\dot{\theta} \tag{10.2}$$

Motor torque (proportional to current)

$$\tau_m = K_t i_a \tag{10.3}$$

Mechanical equation

$$J\ddot{\theta} = \tau_m - B\dot{\theta} - \tau_L = K_t i_a - B\omega - \tau_L \tag{10.4}$$

where:

- V_a = armature voltage (V)
- R_a = armature resistance (Ω)
- L_a = armature inductance (H)
- i_a = armature current (A)
- e_b = back-EMF (V)
- K_e = back-EMF constant (V·s/rad)
- K_t = torque constant (N·m/A)
- J = total inertia (kg·m^2)
- B = viscous friction (N·m·s/rad)
- $\omega = \dot{\theta}$ = angular velocity (rad/s)

Motor Constants

For an ideal DC motor: $K_t = K_e$ (in SI units). This follows from energy conservation—electrical power in equals mechanical power out.

10.1.2 Transfer Functions

Taking Laplace transforms:

$$V_a(s) = (R_a + L_a s)I_a(s) + K_e s\Theta(s) \tag{10.5}$$
$$Js^2\Theta(s) = K_t I_a(s) - Bs\Theta(s) - \tau_L(s) \tag{10.6}$$

Voltage to angular velocity (with $\tau_L = 0$)

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(L_a s + R_a)(Js + B) + K_t K_e} \tag{10.7}$$

Voltage to angular position

$$\frac{\Theta(s)}{V_a(s)} = \frac{K_t}{s[(L_a s + R_a)(Js + B) + K_t K_e]} \tag{10.8}$$

10.1.3 Simplified Model (Neglecting Inductance)

For many motors, the electrical time constant $\tau_e = L_a/R_a$ is much smaller than the mechanical time constant $\tau_m = J/B$. Setting $L_a \approx 0$:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t/R_a}{Js + B + K_t K_e/R_a} = \frac{K_m}{\tau_m s + 1} \tag{10.9}$$

where:

$$K_m = \frac{K_t}{R_a B + K_t K_e} = \frac{K_t}{K_t K_e + R_a B} \quad \text{(motor gain, rad/s/V)} \tag{10.10}$$

$$\tau_m = \frac{R_a J}{R_a B + K_t K_e} \quad \text{(mechanical time constant)} \tag{10.11}$$

Example 1.8: DC Motor Analysis

A DC motor has the following parameters:

- $R_a = 2\ \Omega$, $L_a = 0.5$ mH
- $K_t = K_e = 0.1$ N·m/A = 0.1 V·s/rad
- $J = 0.01$ kg·m^2, $B = 0.001$ N·m·s/rad

Find the transfer function and steady-state speed for $V_a = 24$ V.
Solution:
Electrical time constant: $\tau_e = L_a/R_a = 0.0005/2 = 0.25$ ms
Since τ_e is small, use the simplified model:

$$K_m = \frac{0.1}{2 \times 0.001 + 0.1 \times 0.1} = \frac{0.1}{0.012} = 8.33 \text{ rad/s/V} \tag{10.12}$$

$$\tau_m = \frac{2 \times 0.01}{0.012} = 1.67 \text{ s} \tag{10.13}$$

Transfer function:

$$\frac{\Omega(s)}{V_a(s)} = \frac{8.33}{1.67s + 1} \tag{10.14}$$

Steady-state speed at 24V:

$$\omega_{ss} = K_m \times V_a = 8.33 \times 24 = 200 \text{ rad/s} = 1910 \text{ rpm} \tag{10.15}$$

10.2 DC Motor with Position Control

For position control applications, we need the voltage-to-position transfer function:

$$\frac{\Theta(s)}{V_a(s)} = \frac{K_m/s}{\tau_m s + 1} = \frac{K_m}{s(\tau_m s + 1)} \tag{10.16}$$

This is a Type 1 system (one free integrator), which can track step position commands with zero steady-state error.

10.2.1 State-Space Model

Choosing states $\mathbf{x} = [\theta, \omega, i_a]^\top$:

$$\begin{bmatrix} \dot{\theta} \\ \dot{\omega} \\ \dot{i}_a \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -B/J & K_t/J \\ 0 & -K_e/L_a & -R_a/L_a \end{bmatrix} \begin{bmatrix} \theta \\ \omega \\ i_a \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/L_a \end{bmatrix} V_a \tag{10.17}$$

Output (position): $y = \theta = [1, 0, 0]\mathbf{x}$

10.3 Brushless DC Motors (BLDC)

Brushless DC motors use electronic commutation instead of mechanical brushes. The control-relevant model is similar to the brushed DC motor:

$$\frac{\Omega(s)}{V_q(s)} = \frac{K_t/R}{(L/R)Js^2 + Js + K_t K_e/R} \tag{10.18}$$

where V_q is the q-axis voltage in the field-oriented control frame.

10.4 Stepper Motors

Stepper motors move in discrete angular steps, typically 1.8° (200 steps/revolution).

Open-loop position (idealized)

$$\theta = N_{\text{steps}} \times \theta_{\text{step}} \tag{10.19}$$

Dynamic model (simplified single phase)

$$J\ddot{\theta} + B\dot{\theta} + K_d \sin(N_r\theta) = K_t i \tag{10.20}$$

where N_r is the number of rotor teeth and K_d is the detent torque constant.

10.5 Voice Coil Actuators

Voice coils provide linear motion and are used in hard drives, loudspeakers, and precision positioning.

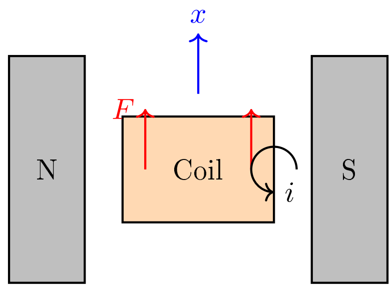

Figure 10.2: Voice coil actuator schematic.

Governing equations

$$V = Ri + L\frac{\mathrm{d}i}{\mathrm{d}t} + K_e\dot{x} \tag{10.21}$$

$$F = K_f i = m\ddot{x} + b\dot{x} \tag{10.22}$$

Transfer function (voltage to position):

$$\frac{X(s)}{V(s)} = \frac{K_f}{s[(Ls+R)(ms+b)+K_fK_e]} \tag{10.23}$$

10.6 Solenoids

Solenoids provide on/off linear actuation. The force depends nonlinearly on position:

$$F = \frac{1}{2}i^2\frac{\mathrm{d}L(x)}{\mathrm{d}x} \tag{10.24}$$

where $L(x)$ is the position-dependent inductance.
Linearized about operating point (x_0, i_0):

$$\Delta F = K_i\Delta i + K_x\Delta x \tag{10.25}$$

where K_i and K_x are partial derivatives of force with respect to current and position.

10.7 Piezoelectric Actuators

Piezoelectric actuators provide very small displacements (nm to μm) with high force and fast response.

Constitutive equations

$$x = d_{33}V + s_{33}F/A \tag{10.26}$$

$$Q = CV + d_{33}F \tag{10.27}$$

where:

- d_{33} = piezoelectric coefficient (m/V or C/N)

- s_{33} = elastic compliance (m^2/N)
- C = capacitance (F)

Transfer function (simplified, free displacement):

$$\frac{X(s)}{V(s)} = \frac{d_{33}\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{10.28}$$

10.8 Generators and Regeneration

When a motor is driven faster than its no-load speed, or when decelerating a load, it acts as a generator:

$$P_{\text{regen}} = e_b \cdot i_a = K_e\omega \cdot i_a \tag{10.29}$$

This regenerated power can be:

- Dissipated in braking resistors
- Returned to the power supply (regenerative braking)
- Stored in capacitors or batteries

10.9 Motor Selection Example

Example 1.9: Robot Joint Motor Selection

A robot joint must:

- Move a load with inertia $J_L = 0.05$ kg·m^2
- Achieve maximum velocity $\omega_{\text{max}} = 10$ rad/s
- Achieve maximum acceleration $\alpha_{\text{max}} = 50$ rad/s^2

Select an appropriate motor and gear ratio.
Solution:
Required torque at load:

$$\tau_L = J_L\alpha_{\text{max}} = 0.05 \times 50 = 2.5 \text{ N·m} \tag{10.30}$$

Consider a motor with:

- Rated torque: $\tau_m = 0.3$ N·m
- Rated speed: 3000 rpm = 314 rad/s
- Rotor inertia: $J_m = 5 \times 10^{-5}$ kg·m^2

Required gear ratio (from torque):

$$n \geq \frac{\tau_L}{\tau_m} = \frac{2.5}{0.3} = 8.3 \tag{10.31}$$

Check speed (with $n = 10$):

$$\omega_m = n \cdot \omega_L = 10 \times 10 = 100 \text{ rad/s} < 314 \text{ rad/s} \quad \checkmark \tag{10.32}$$

Reflected inertia check:

$$J_{\text{reflected}} = \frac{J_L}{n^2} = \frac{0.05}{100} = 5 \times 10^{-4} \text{ kg·m}^2 \tag{10.33}$$

Total inertia at motor: $J_{\text{total}} = J_m + J_{\text{reflected}} = 5.5 \times 10^{-4}$ kg·m^2
Motor acceleration:

$$\alpha_m = \frac{\tau_m}{J_{\text{total}}} = \frac{0.3}{5.5 \times 10^{-4}} = 545 \text{ rad/s}^2 \tag{10.34}$$

Load acceleration achieved:

$$\alpha_L = \frac{\alpha_m}{n} = \frac{545}{10} = 54.5 \text{ rad/s}^2 > 50 \text{ rad/s}^2 \quad \checkmark \tag{10.35}$$

The motor with 10:1 gear ratio meets all requirements.

Chapter 11

Thermal and Fluid Systems

Thermal and fluid systems are governed by energy and mass conservation. These systems often exhibit first-order dynamics with relatively slow time constants, making them excellent candidates for feedback control.

11.1 Thermal Systems Fundamentals

11.1.1 Heat Transfer Mechanisms

Conduction (through solids)

$$\dot{Q} = \frac{kA}{L}(T_1 - T_2) = \frac{T_1 - T_2}{R_{th}} \tag{11.1}$$

where $R_{th} = L/(kA)$ is the thermal resistance.

Convection (between solid and fluid)

$$\dot{Q} = hA(T_s - T_\infty) \tag{11.2}$$

where h is the convection coefficient ($\mathrm{W/m^2K}$).

Radiation

$$\dot{Q} = \epsilon\sigma A(T_1^4 - T_2^4) \tag{11.3}$$

For small temperature differences, linearized: $\dot{Q} \approx h_r A(T_1 - T_2)$ where $h_r = 4\epsilon\sigma T_{\text{avg}}^3$.

11.1.2 Thermal Capacitance

The energy stored in a body:

$$E = mC_pT = \rho V C_p T \tag{11.4}$$

Rate of temperature change:

$$mC_p\frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} \tag{11.5}$$

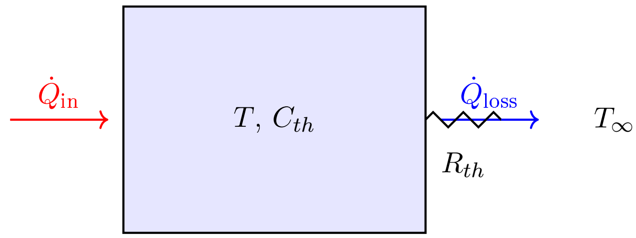

Figure 11.1: Single-capacitance thermal system.

11.2 Single-Capacitance Thermal System

Energy balance:

$$C_{th}\frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\text{in}} - \frac{T - T_\infty}{R_{th}} \tag{11.6}$$

where $C_{th} = mC_p$ is the thermal capacitance.

Rearranging:

$$\boxed{R_{th}C_{th}\frac{\mathrm{d}T}{\mathrm{d}t} + T = R_{th}\dot{Q}_{\text{in}} + T_\infty} \tag{11.7}$$

11.2.1 Transfer Function

Defining $\theta = T - T_\infty$ (temperature above ambient):

$$G(s) = \frac{\Theta(s)}{\dot{Q}_{\text{in}}(s)} = \frac{R_{th}}{\tau s + 1} \tag{11.8}$$

where $\tau = R_{th}C_{th}$ is the thermal time constant.

Example 1.10: Room Heating

A room has thermal capacitance $C_{th} = 500$ kJ/K and thermal resistance to outdoors $R_{th} = 0.01$ K/W. Find:

(a) The thermal time constant

(b) The heater power needed to maintain 20°C when outdoor temperature is 0°C

(c) The room temperature 1 hour after turning off the heater

Solution:

(a) Time constant:

$$\tau = R_{th}C_{th} = 0.01 \times 500{,}000 = 5000 \text{ s} = 83.3 \text{ min} \tag{11.9}$$

(b) Steady-state heat balance:

$$\dot{Q}_{\text{in}} = \frac{T - T_\infty}{R_{th}} = \frac{20 - 0}{0.01} = 2000 \text{ W} = 2 \text{ kW} \tag{11.10}$$

(c) After heater off (cooling from 20°C):

$$T(t) = T_\infty + (T_0 - T_\infty)\mathrm{e}^{-t/\tau} = 0 + 20\mathrm{e}^{-3600/5000} = 9.7\text{C} \tag{11.11}$$

11.3 Multi-Capacitance Thermal Systems

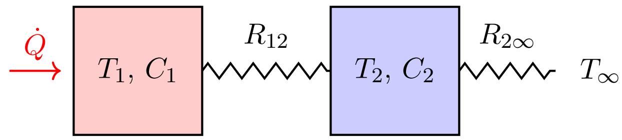

Figure 11.2: Two-capacitance thermal system.

Energy balances:

$$C_1 \frac{\mathrm{d}T_1}{\mathrm{d}t} = \dot{Q} - \frac{T_1 - T_2}{R_{12}} \tag{11.12}$$

$$C_2 \frac{\mathrm{d}T_2}{\mathrm{d}t} = \frac{T_1 - T_2}{R_{12}} - \frac{T_2 - T_\infty}{R_{2\infty}} \tag{11.13}$$

This yields a second-order system with two thermal time constants.

11.4 Fluid Systems: Tank Level

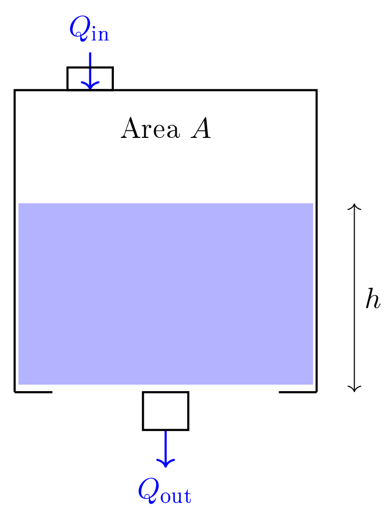

Figure 11.3: Single tank level system.

11.4.1 Mass Balance

For incompressible fluid:

$$A \frac{\mathrm{d}h}{\mathrm{d}t} = Q_{\text{in}} - Q_{\text{out}} \tag{11.14}$$

11.4.2 Outflow Models

Linear resistance (laminar flow through valve)

$$Q_{\text{out}} = \frac{h}{R_f} \tag{11.15}$$

where R_f is the fluid resistance.

Nonlinear resistance (turbulent flow, orifice)

$$Q_{\text{out}} = C_d A_o \sqrt{2gh} \tag{11.16}$$

This is nonlinear but can be linearized about an operating point.

11.4.3 Linear Tank Model

With linear resistance:

$$A\frac{\mathrm{d}h}{\mathrm{d}t} + \frac{h}{R_f} = Q_{\text{in}} \tag{11.17}$$

Transfer function:

$$G(s) = \frac{H(s)}{Q_{\text{in}}(s)} = \frac{R_f}{\tau s + 1} \tag{11.18}$$

where $\tau = AR_f$ is the hydraulic time constant.

Example 1.11: Water Tank

A cylindrical tank has diameter 2 m and drains through a valve with resistance $R_f = 100$ s/m^2. Find:

(a) The time constant

(b) Steady-state level for inflow of 0.01 m^3/s

(c) Time to reach 90% of steady-state from empty

Solution:
Tank area: $A = \pi(1)^2 = 3.14$ m^2
(a) Time constant:

$$\tau = AR_f = 3.14 \times 100 = 314 \text{ s} = 5.2 \text{ min} \tag{11.19}$$

(b) Steady-state level:

$$h_{ss} = R_f \times Q_{\text{in}} = 100 \times 0.01 = 1 \text{ m} \tag{11.20}$$

(c) Time to 90%:

$$0.9 = 1 - \mathrm{e}^{-t/\tau} \implies t = -\tau \ln(0.1) = 314 \times 2.3 = 723 \text{ s} = 12 \text{ min} \tag{11.21}$$

11.5 Interacting Tank Systems

Mass balances:

$$A_1\frac{\mathrm{d}h_1}{\mathrm{d}t} = Q_{\text{in}} - \frac{h_1 - h_2}{R_1} \tag{11.22}$$

$$A_2\frac{\mathrm{d}h_2}{\mathrm{d}t} = \frac{h_1 - h_2}{R_1} - \frac{h_2}{R_2} \tag{11.23}$$

This is a second-order system. The interaction between tanks creates coupling that affects the system dynamics.

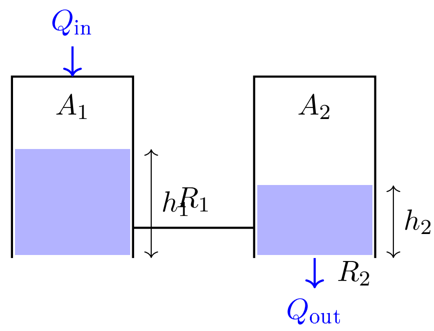

Figure 11.4: Two interacting tanks.

11.6 Hydraulic Actuators

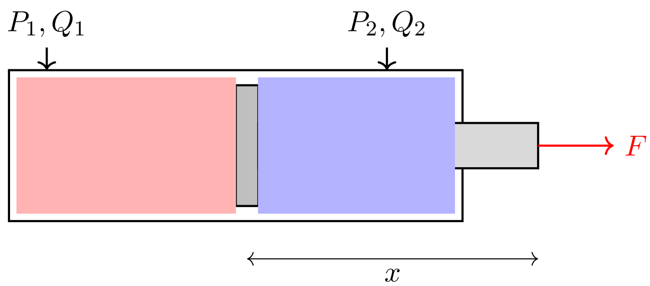

Figure 11.5: Double-acting hydraulic cylinder.

11.6.1 Flow Continuity

For incompressible fluid:

$$Q_1 = A_1\dot{x} + \frac{V_1}{\beta}\dot{P}_1 \tag{11.24}$$

where β is the bulk modulus.

11.6.2 Force Balance

$$M\ddot{x} = P_1A_1 - P_2A_2 - F_{\text{load}} - B\dot{x} \tag{11.25}$$

11.6.3 Servo Valve Model

Flow through a servo valve:

$$Q = C_d w x_v \sqrt{\frac{2(P_s - P_L)}{\rho}} \tag{11.26}$$

where x_v is the valve spool position.
Linearized about operating point:

$$Q = K_q x_v - K_c P_L \tag{11.27}$$

11.7 Pneumatic Systems

Pneumatic systems use compressible gas (usually air), adding complexity:

Ideal gas law

$$PV = mRT \tag{11.28}$$

Mass flow through orifice

For subsonic flow ($P_2/P_1 > 0.528$):

$$\dot{m} = C_d A P_1 \sqrt{\frac{2}{RT_1}} \sqrt{\frac{\gamma}{\gamma - 1}\left[\left(\frac{P_2}{P_1}\right)^{2/\gamma} - \left(\frac{P_2}{P_1}\right)^{(\gamma+1)/\gamma}\right]} \tag{11.29}$$

For choked flow ($P_2/P_1 \leq 0.528$):

$$\dot{m} = C_d A P_1 \sqrt{\frac{\gamma}{RT_1}} \left(\frac{2}{\gamma + 1}\right)^{(\gamma+1)/(2(\gamma-1))} \tag{11.30}$$

11.7.1 Pneumatic Cylinder

Pressure dynamics in chamber:

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \frac{\gamma RT}{V}(\dot{m}_{\text{in}} - \dot{m}_{\text{out}}) - \frac{\gamma P}{V}\dot{V} \tag{11.31}$$

11.8 Heat Exchangers

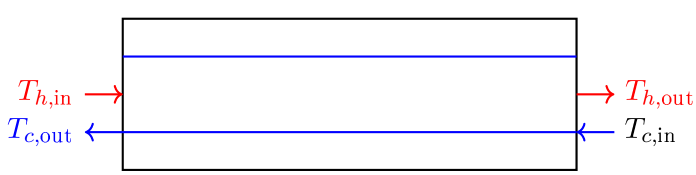

Figure 11.6: Counter-flow heat exchanger.

Simplified lumped model:

$$C_h \frac{\mathrm{d}T_h}{\mathrm{d}t} = \dot{m}_h c_{p,h}(T_{h,\text{in}} - T_h) - UA(T_h - T_c) \tag{11.32}$$

$$C_c \frac{\mathrm{d}T_c}{\mathrm{d}t} = \dot{m}_c c_{p,c}(T_{c,\text{in}} - T_c) + UA(T_h - T_c) \tag{11.33}$$

where UA is the overall heat transfer coefficient times area.

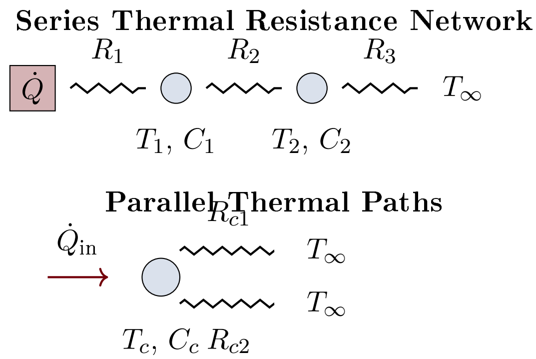

Figure 11.7: Thermal resistance networks showing series (top) and parallel (bottom) heat flow paths with thermal capacitances at nodes.

11.9 Thermal Resistance Networks

11.9.1 Network Analysis

For **series resistances**, the total thermal resistance equals the sum:

$$R_{\text{total}} = R_1 + R_2 + R_3 + \cdots \tag{11.34}$$

The heat flow through all resistances is identical (same current). The temperature difference across each resistance is proportional to its value.

For **parallel resistances**, the equivalent resistance is:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_{c1}} + \frac{1}{R_{c2}} + \cdots \tag{11.35}$$

Each path dissipates heat proportionally to the inverse of its resistance. Parallel paths reduce the effective resistance, increasing heat flow.

11.10 Two-Tank Fluid System Dynamics

This configuration creates a second-order system where tank interactions cause non-minimum-phase behavior. The inter-tank resistance R_1 introduces coupling between the tanks.

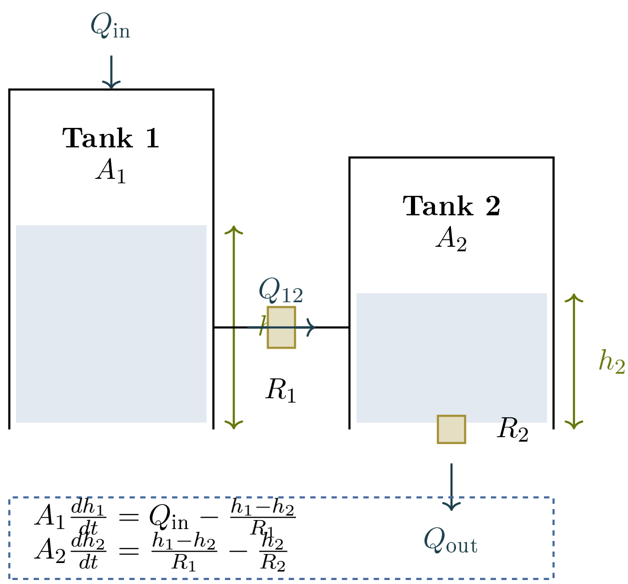

Figure 11.8: Two-tank fluid system showing mass conservation, inter-tank flow through resistance R_1, and outflow through R_2. Shaded regions indicate liquid level.

Chapter 12

Chemical Systems

Chemical processes involve mass and energy balances coupled with reaction kinetics. These systems often exhibit nonlinear behavior, time delays, and complex interactions, making them challenging but important control applications.

12.1 Continuous Stirred-Tank Reactor (CSTR)

The CSTR is a fundamental unit in chemical engineering and a classic control problem.

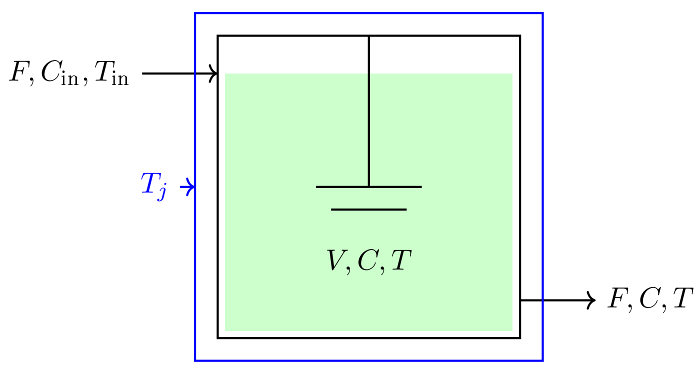

Figure 12.1: Continuous stirred-tank reactor with cooling jacket.

12.1.1 Assumptions

- Perfect mixing (uniform concentration and temperature)
- Constant volume V and density ρ
- Constant flow rate F (in = out)
- Single first-order irreversible reaction: $A \to B$

12.1.2 Component Mass Balance

$$V\frac{\mathrm{d}C_A}{\mathrm{d}t} = F(C_{A,\text{in}} - C_A) - Vk(T)C_A \tag{12.1}$$

where $k(T)$ is the temperature-dependent reaction rate constant.

12.1.3 Arrhenius Equation

$$k(T) = k_0 \exp\left(-\frac{E_a}{RT}\right) \tag{12.2}$$

where:

- k_0 = pre-exponential factor (1/s)
- E_a = activation energy (J/mol)
- R = universal gas constant (8.314 J/mol·K)

12.1.4 Energy Balance

$$\rho V C_p \frac{\mathrm{d}T}{\mathrm{d}t} = F\rho C_p(T_{\text{in}} - T) + (-\Delta H_r)Vk(T)C_A - UA(T - T_j) \tag{12.3}$$

where:

- ΔH_r = heat of reaction (J/mol, negative for exothermic)
- UA = heat transfer coefficient times area (W/K)
- T_j = jacket temperature (K)

12.1.5 Linearized Model

Linearizing about steady-state (C_{A0}, T_0):

$$\frac{\mathrm{d}\Delta C_A}{\mathrm{d}t} = a_{11}\Delta C_A + a_{12}\Delta T + b_1 \Delta C_{A,\text{in}} \tag{12.4}$$

$$\frac{\mathrm{d}\Delta T}{\mathrm{d}t} = a_{21}\Delta C_A + a_{22}\Delta T + b_2 \Delta T_j \tag{12.5}$$

State-space form:

$$\begin{bmatrix} \Delta \dot{C}_A \\ \Delta \dot{T} \end{bmatrix} = \mathbf{A} \begin{bmatrix} \Delta C_A \\ \Delta T \end{bmatrix} + \mathbf{B} \begin{bmatrix} \Delta C_{A,\text{in}} \\ \Delta T_j \end{bmatrix} \tag{12.6}$$

Example 1.12: CSTR Linearization

A CSTR operates at steady-state with:

- $V = 1\ \mathrm{m}^3$, $F = 0.1\ \mathrm{m}^3/\mathrm{min}$
- $C_{A0} = 2\ \mathrm{mol/L}$, $T_0 = 350$ K
- $k_0 = 10^{10}\ \mathrm{min}^{-1}$, $E_a/R = 8750$ K

The linearized **A** matrix elements are:

$$a_{11} = -\frac{F}{V} - k(T_0) = -0.1 - k_0 \mathrm{e}^{-8750/350} = -0.1 - 1.2 = -1.3\ \mathrm{min}^{-1} \tag{12.7}$$

$$a_{12} = -C_{A0} \left.\frac{\mathrm{d}k}{\mathrm{d}T}\right|_{T_0} = -C_{A0} k(T_0) \frac{E_a}{RT_0^2} = -0.17\ \mathrm{mol/L{\cdot}K{\cdot}min} \tag{12.8}$$

The negative a_{12} indicates that increased temperature increases reaction rate, consuming more C_A.

12.2 Mixing Tank

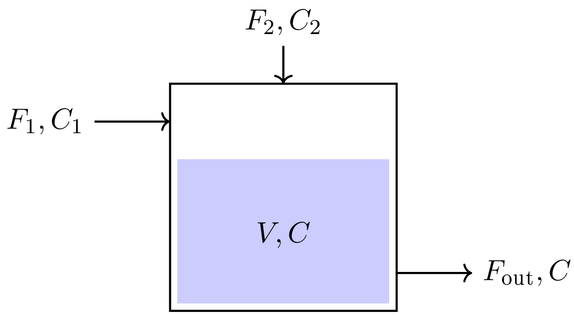

Figure 12.2: Mixing tank with two inlets.

12.2.1 Mass Balance (Total)

$$\frac{\mathrm{d}(\rho V)}{\mathrm{d}t} = \rho_1 F_1 + \rho_2 F_2 - \rho F_{\text{out}} \tag{12.9}$$

For constant density and volume: $F_{\text{out}} = F_1 + F_2$

12.2.2 Component Balance

$$V \frac{\mathrm{d}C}{\mathrm{d}t} = F_1 C_1 + F_2 C_2 - (F_1 + F_2)C \tag{12.10}$$

Transfer function (for constant F_1, F_2):

$$\frac{C(s)}{C_1(s)} = \frac{F_1/V}{s + (F_1 + F_2)/V} = \frac{F_1/(F_1 + F_2)}{\tau s + 1} \tag{12.11}$$

where $\tau = V/(F_1 + F_2)$ is the residence time.

12.3 pH Neutralization

pH control is notoriously difficult due to the highly nonlinear relationship between acid/base concentration and pH.

$$\text{pH} = -\log_{10}[H^+] \tag{12.12}$$

For a strong acid-strong base system:

$$[H^+] = \frac{C_a - C_b + \sqrt{(C_a - C_b)^2 + 4K_w}}{2} \tag{12.13}$$

where $K_w = 10^{-14}$ at 25°C.

Titration curve

The pH changes dramatically near the equivalence point, creating a high-gain nonlinearity.

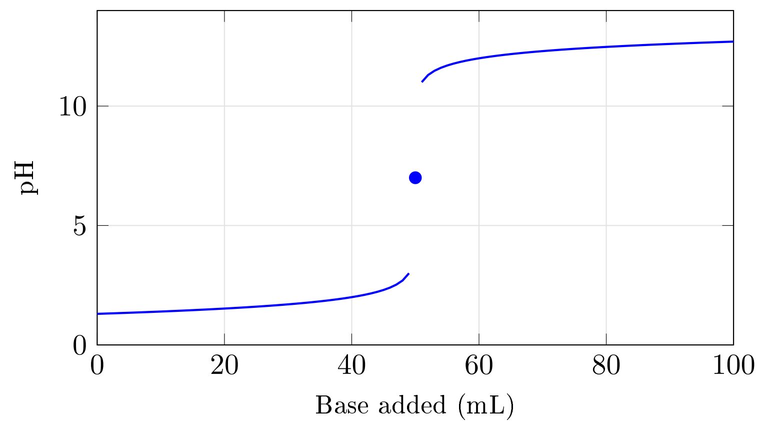

Figure 12.3: Typical pH titration curve showing high gain near equivalence.

12.4 Distillation Column

Distillation separates mixtures based on volatility differences.

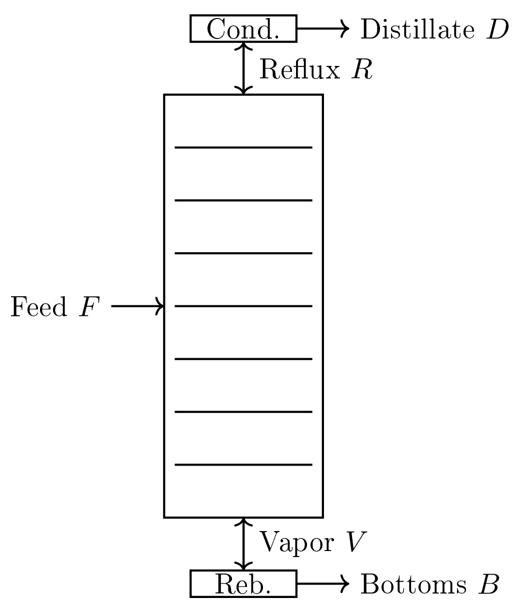

Figure 12.4: Distillation column schematic.

12.4.1 Simplified Model

For a binary mixture, the composition dynamics on tray n:

$$M_n \frac{\mathrm{d}x_n}{\mathrm{d}t} = L_{n-1}x_{n-1} + V_{n+1}y_{n+1} - L_n x_n - V_n y_n \tag{12.14}$$

where:

- M_n = liquid holdup on tray
- L_n = liquid flow rate

- V_n = vapor flow rate
- x_n = liquid composition (mole fraction)
- y_n = vapor composition

Vapor-liquid equilibrium:

$$y_n = \frac{\alpha x_n}{1 + (\alpha - 1)x_n} \tag{12.15}$$

where α is the relative volatility.

12.5 Batch Reactor

Unlike continuous reactors, batch reactors have no flow during reaction:

$$\frac{\mathrm{d}C_A}{\mathrm{d}t} = -k(T)C_A \tag{12.16}$$

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{(-\Delta H_r)k(T)C_A}{\rho C_p} - \frac{UA(T - T_j)}{\rho V C_p} \tag{12.17}$$

The concentration decreases exponentially (for first-order kinetics):

$$C_A(t) = C_{A0} \exp\left(-\int_0^t k(T(\tau))\,\mathrm{d}\tau\right) \tag{12.18}$$

12.6 Reaction Kinetics

12.6.1 Zero-Order Reaction

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -k \tag{12.19}$$

Solution: $C(t) = C_0 - kt$

12.6.2 First-Order Reaction

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -kC \tag{12.20}$$

Solution: $C(t) = C_0 \mathrm{e}^{-kt}$
Half-life: $t_{1/2} = \ln(2)/k$

12.6.3 Second-Order Reaction

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -kC^2 \tag{12.21}$$

Solution: $\frac{1}{C(t)} = \frac{1}{C_0} + kt$

12.6.4 Enzyme Kinetics (Michaelis-Menten)

$$\frac{\mathrm{d}C}{\mathrm{d}t} = -\frac{V_{\max}C}{K_m + C} \tag{12.22}$$

where $V_{\max}$ is maximum rate and K_m is the Michaelis constant.

12.7 Process Dead Time

Many chemical processes exhibit significant time delays (dead time, transport lag):

$$y(t) = u(t - \theta) \tag{12.23}$$

Transfer function:

$$G(s) = \mathrm{e}^{-\theta s} \tag{12.24}$$

Dead time makes control more challenging. Common approximations:

First-order Pade

$$\mathrm{e}^{-\theta s} \approx \frac{1 - \theta s/2}{1 + \theta s/2} \tag{12.25}$$

Second-order Pade

$$\mathrm{e}^{-\theta s} \approx \frac{1 - \theta s/2 + (\theta s)^2/12}{1 + \theta s/2 + (\theta s)^2/12} \tag{12.26}$$

Example 1.13: First-Order Plus Dead Time (FOPDT)

Many chemical processes can be approximated as:

$$G(s) = \frac{K\mathrm{e}^{-\theta s}}{\tau s + 1} \tag{12.27}$$

For a heat exchanger with $K = 2$ °C/%, $\tau = 60$ s, $\theta = 10$ s:

$$G(s) = \frac{2\mathrm{e}^{-10s}}{60s + 1} \tag{12.28}$$

The dead-time-to-time-constant ratio $\theta/\tau = 10/60 = 0.17$ indicates this system is relatively easy to control.

12.8 CSTR Schematic and Inlet/Outlet Configuration

The inlet provides fresh reactant at concentration $C_{A,\text{in}}$ and temperature T_{in}, while the outlet removes mixture at the reactor conditions (C_A, T) since perfect mixing is assumed. The cooling jacket maintains temperature control through T_j.

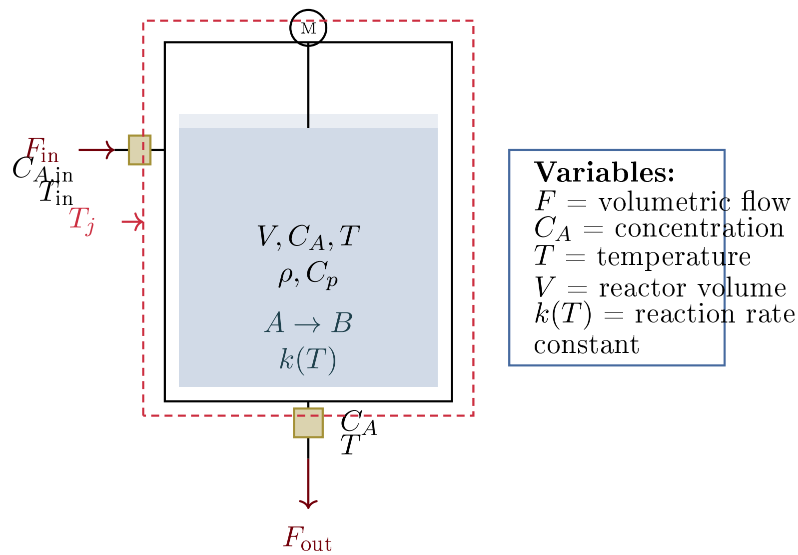

Figure 12.5: Continuous stirred-tank reactor (CSTR) with detailed inlet/outlet ports, cooling jacket, and labeled flows showing reactant concentration and temperature at inlet and outlet.

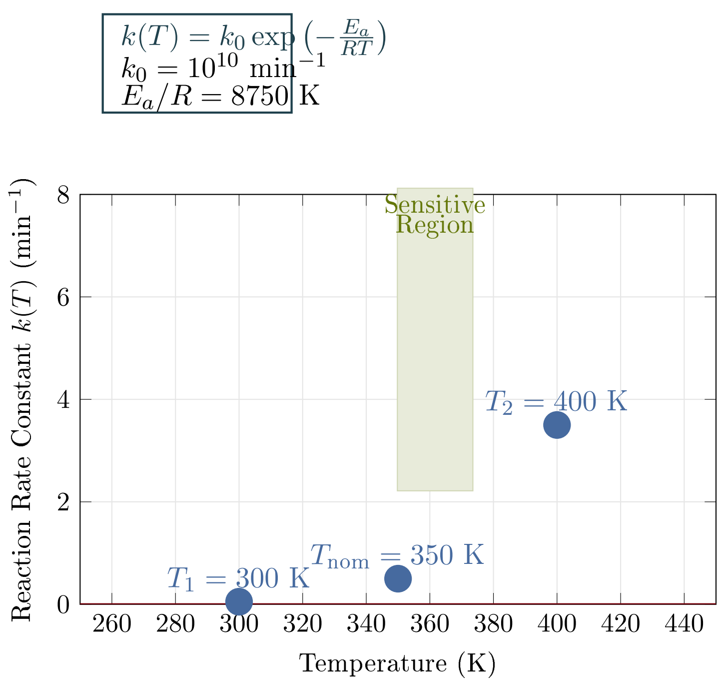

Figure 12.6: Arrhenius equation showing exponential increase in reaction rate with temperature. The nonlinear relationship creates control challenges, with the system becoming increasingly sensitive to temperature in the operating range.

12.9 Reaction Rate vs. Temperature: Arrhenius Behavior

The exponential dependence means small temperature changes can dramatically affect reaction rate. For a 50 K increase from 350 K to 400 K, the reaction rate increases approximately 7-fold. This nonlinear behavior is fundamental to exothermic reaction dynamics and runaway concerns.

Chapter 13

Aerospace Systems

Aerospace systems present unique control challenges: operation in three dimensions, coupling between axes, varying dynamics with flight conditions, and stringent safety requirements.

13.1 Aircraft Longitudinal Dynamics

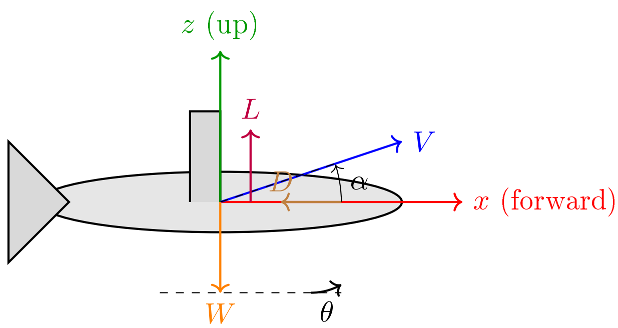

Figure 13.1: Aircraft longitudinal plane showing forces and angles.

13.1.1 State Variables

- u = forward velocity perturbation (m/s)
- w = vertical velocity perturbation (m/s)
- q = pitch rate (rad/s)
- θ = pitch angle (rad)

13.1.2 Linearized Equations of Motion

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & X_w & 0 & -g \\ Z_u & Z_w & U_0 & 0 \\ M_u & M_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} X_{\delta_e} \\ Z_{\delta_e} \\ M_{\delta_e} \\ 0 \end{bmatrix} \delta_e \tag{13.1}$$

where δ_e is elevator deflection and X_u, Z_w, etc. are stability derivatives.

13.1.3 Characteristic Modes

The longitudinal dynamics typically have two modes:

Short Period Mode

- Fast oscillation (1-3 seconds period)
- Primarily α and q motion
- Well-damped ($\zeta \approx 0.3 - 0.7$)

Approximate transfer function:

$$\frac{\alpha(s)}{\delta_e(s)} \approx \frac{Z_{\delta_e}/U_0}{s^2 - (M_q + M_{\dot{\alpha}} + Z_w/U_0)s + (Z_w M_q/U_0 - M_w)} \tag{13.2}$$

Phugoid Mode

- Slow oscillation (30-100 seconds period)
- Exchange between kinetic and potential energy
- Lightly damped ($\zeta \approx 0.04 - 0.1$)

Approximate frequency and damping:

$$\omega_p \approx \sqrt{2}\frac{g}{U_0} \tag{13.3}$$

$$\zeta_p \approx \frac{1}{\sqrt{2}}\frac{C_D}{C_L} \tag{13.4}$$

Example 1.14: Boeing 747 Longitudinal Modes

At cruise ($U_0 = 250$ m/s, altitude 10 km):
Short period:

- $\omega_{sp} = 0.9$ rad/s, $\zeta_{sp} = 0.35$
- Period: $T_{sp} = 2\pi/\omega_{sp} = 7$ s

Phugoid:

- $\omega_p = 0.06$ rad/s, $\zeta_p = 0.05$
- Period: $T_p = 2\pi/\omega_p = 105$ s

13.2 Aircraft Body Axes and Rotational Dynamics

13.3 Aircraft Lateral-Directional Dynamics

13.3.1 State Variables

- β = sideslip angle (rad)

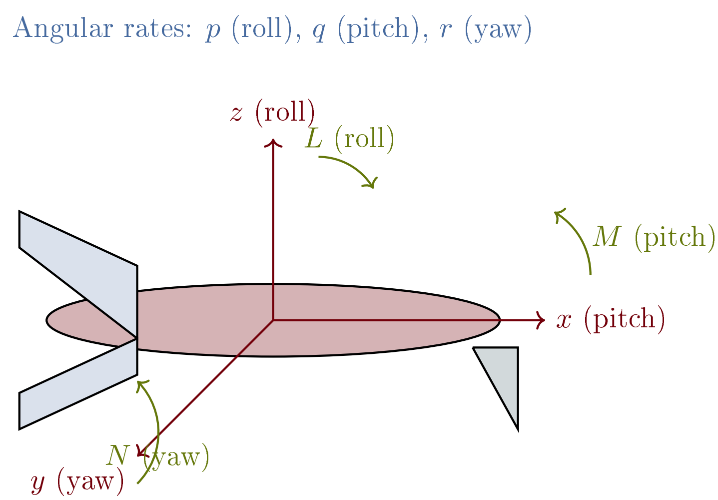

Figure 13.2: Aircraft body-fixed coordinate axes showing pitch, roll, and yaw moments with associated angular velocities.

- p = roll rate (rad/s)
- r = yaw rate (rad/s)
- ϕ = bank angle (rad)

13.3.2 Characteristic Modes

Dutch Roll

- Coupled yaw-roll oscillation
- Period: 3-10 seconds
- Damping varies with aircraft design

Roll Mode

- Pure rolling motion
- First-order, typically fast (time constant 0.5-2 s)

Spiral Mode

- Slow divergence or convergence
- Often slightly unstable (acceptable if slow)

13.4 Rocket Dynamics

13.4.1 Pitch Dynamics (Simplified)

For a rocket with thrust vector control:

$$I_{yy}\ddot{\theta} = T \cdot \ell_T \cdot \delta \tag{13.5}$$

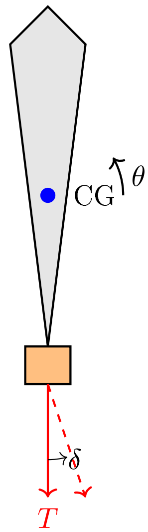

Figure 13.3: Rocket with thrust vector control.

where:

- I_{yy} = pitch moment of inertia
- T = thrust
- ℓ_T = distance from CG to gimbal point
- δ = gimbal angle

Transfer function:

$$\frac{\Theta(s)}{\delta(s)} = \frac{T\ell_T/I_{yy}}{s^2} = \frac{K_r}{s^2} \tag{13.6}$$

This is a double integrator—marginally stable and requires feedback for control.

13.4.2 Aerodynamic Effects

With aerodynamic forces, additional terms appear:

$$I_{yy}\ddot{\theta} = T\ell_T\delta + M_\alpha\alpha + M_q q \tag{13.7}$$

For a statically unstable rocket ($M_\alpha > 0$), the system has a pole in the right half-plane.

13.5 Satellite Attitude Dynamics

13.5.1 Euler's Equations

$$I_{xx}\dot{\omega}_x + (I_{zz} - I_{yy})\omega_y\omega_z = \tau_x \tag{13.8}$$

$$I_{yy}\dot{\omega}_y + (I_{xx} - I_{zz})\omega_z\omega_x = \tau_y \tag{13.9}$$

$$I_{zz}\dot{\omega}_z + (I_{yy} - I_{xx})\omega_x\omega_y = \tau_z \tag{13.10}$$

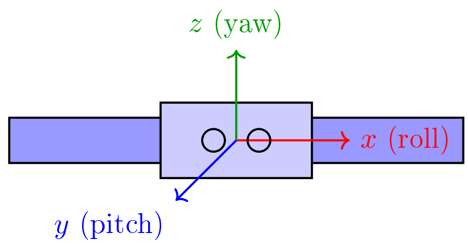

Figure 13.4: Satellite with body-fixed axes.

13.5.2 Linearized Single-Axis Model

For small angles about one axis:

$$I\ddot{\theta} = \tau \tag{13.11}$$

Transfer function:

$$\frac{\Theta(s)}{\tau(s)} = \frac{1}{Is^2} \tag{13.12}$$

13.5.3 Reaction Wheel Dynamics

A reaction wheel applies torque by changing its angular momentum:

$$\tau_{\text{satellite}} = -I_w\dot{\omega}_w \tag{13.13}$$

Combined system:

$$I_s\ddot{\theta} = -I_w\dot{\omega}_w = I_w\ddot{\theta}_w \tag{13.14}$$

13.6 Quadrotor Dynamics

13.6.1 Control Inputs

$$\text{Total thrust:} \quad T = F_1 + F_2 + F_3 + F_4 \tag{13.15}$$

$$\text{Roll torque:} \quad \tau_\phi = \ell(F_1 - F_2) \tag{13.16}$$

$$\text{Pitch torque:} \quad \tau_\theta = \ell(F_3 - F_4) \tag{13.17}$$

$$\text{Yaw torque:} \quad \tau_\psi = k_\tau(F_1 + F_2 - F_3 - F_4) \tag{13.18}$$

13.6.2 Equations of Motion

Translational (in body frame):

$$m\ddot{x} = -mg\sin\theta \tag{13.19}$$

$$m\ddot{y} = mg\cos\theta\sin\phi \tag{13.20}$$

$$m\ddot{z} = mg\cos\theta\cos\phi - T \tag{13.21}$$

Rotational:

$$I_{xx}\ddot{\phi} = \tau_\phi \tag{13.22}$$

$$I_{yy}\ddot{\theta} = \tau_\theta \tag{13.23}$$

$$I_{zz}\ddot{\psi} = \tau_\psi \tag{13.24}$$

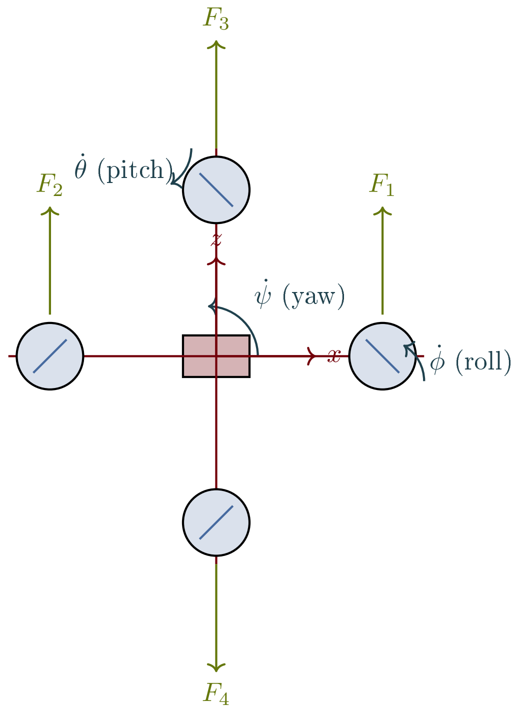

Figure 13.5: Quadrotor free body diagram showing thrust forces at each rotor and angular velocities for roll, pitch, and yaw control.

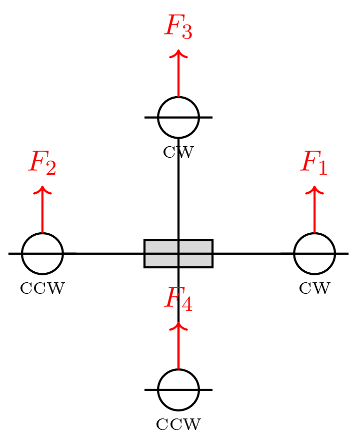

Figure 13.6: Quadrotor configuration showing rotor positions and control mixing.

13.6.3 Linearized Model (Hover)

Near hover ($\phi, \theta, \psi \approx 0$):

$$\ddot{\phi} = \frac{\ell}{I_{xx}}(F_1 - F_2), \quad \ddot{\theta} = \frac{\ell}{I_{yy}}(F_3 - F_4), \quad \ddot{z} = g - \frac{T}{m} \tag{13.25}$$

13.7 Spacecraft Orbital Mechanics

13.7.1 Hill-Clohessy-Wiltshire Equations

Relative motion near a circular reference orbit:

$$\ddot{x} - 2n\dot{y} - 3n^2 x = a_x \tag{13.26}$$

$$\ddot{y} + 2n\dot{x} = a_y \tag{13.27}$$

$$\ddot{z} + n^2 z = a_z \tag{13.28}$$

where $n = \sqrt{\mu/a^3}$ is the orbital rate.
The out-of-plane motion (z) is decoupled and is a simple harmonic oscillator.
The in-plane motion (x, y) is coupled, with the solution:

$$x(t) = (4 - 3\cos nt)x_0 + \frac{\sin nt}{n}\dot{x}_0 + \frac{2(1 - \cos nt)}{n}\dot{y}_0 \tag{13.29}$$

Chapter 14

Biomedical Systems

Biomedical control systems interface with the human body, requiring special consideration for safety, variability between patients, and the inherent complexity of biological processes.

14.1 Pharmacokinetics: Drug Distribution

Pharmacokinetics describes how drugs move through the body.

14.1.1 One-Compartment Model

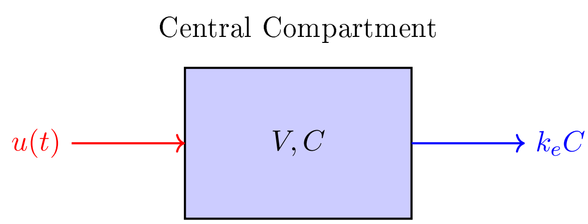

Figure 14.1: One-compartment pharmacokinetic model.

Mass balance:

$$V \frac{\mathrm{d}C}{\mathrm{d}t} = u(t) - k_e V C \tag{14.1}$$

where:

- V = volume of distribution (L)
- C = drug concentration (mg/L)
- $u(t)$ = infusion rate (mg/min)
- k_e = elimination rate constant (1/min)

Transfer function:

$$\frac{C(s)}{U(s)} = \frac{1/V}{s + k_e} \tag{14.2}$$

Half-life: $t_{1/2} = \ln(2)/k_e$

Example 1.15: IV Drug Infusion

A drug has $V = 50$ L and $k_e = 0.1$ min^{-1}. Find:

(a) The half-life

(b) The infusion rate needed for steady-state concentration of 10 mg/L

(c) Time to reach 90% of steady-state

Solution:
(a) Half-life: $t_{1/2} = \ln(2)/0.1 = 6.93$ min
(b) At steady-state: $u_{ss} = k_e V C_{ss} = 0.1 \times 50 \times 10 = 50$ mg/min
(c) Time constant: $\tau = 1/k_e = 10$ min
Time to 90%: $t = -\tau \ln(0.1) = 10 \times 2.3 = 23$ min

14.1.2 Two-Compartment Model

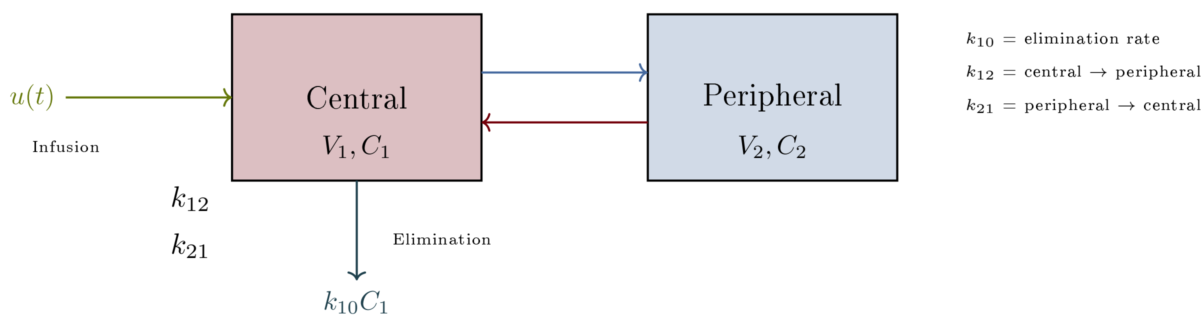

Figure 14.2: Two-compartment pharmacokinetic model diagram showing drug infusion, inter-compartment transfer, and elimination with rate constants labeled.

Mass balances:

$$V_1 \frac{dC_1}{dt} = u(t) - (k_{10} + k_{12})V_1 C_1 + k_{21} V_2 C_2 \tag{14.3}$$
$$V_2 \frac{dC_2}{dt} = k_{12} V_1 C_1 - k_{21} V_2 C_2 \tag{14.4}$$

This yields a second-order system with two exponential phases:

- **Distribution phase** (α): rapid initial decline
- **Elimination phase** (β): slower terminal decline

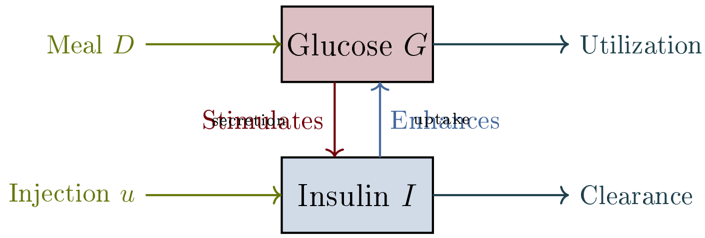

Figure 14.3: Glucose-insulin feedback loop block diagram showing meal and infusion inputs, utilization/clearance outputs, and the bidirectional regulatory feedback.

14.2 Glucose-Insulin Dynamics

14.2.1 Minimal Model (Bergman)

$$\frac{\mathrm{d}G}{\mathrm{d}t} = -p_1 G - X(G + G_b) + D(t) \tag{14.5}$$

$$\frac{\mathrm{d}X}{\mathrm{d}t} = -p_2 X + p_3(I - I_b) \tag{14.6}$$

$$\frac{\mathrm{d}I}{\mathrm{d}t} = -n(I - I_b) + \gamma(G - h)^+ + u(t) \tag{14.7}$$

where:

- G = glucose concentration (mg/dL)
- I = insulin concentration (μU/mL)
- X = insulin action (remote compartment)
- $D(t)$ = meal disturbance
- $u(t)$ = insulin infusion (for artificial pancreas)

14.2.2 Linearized Model for Control Design

Near operating point (G_0, I_0):

$$\begin{bmatrix} \Delta\dot{G} \\ \Delta\dot{I} \end{bmatrix} = \begin{bmatrix} -p_1 - X_0 & -K_G \\ K_I & -n \end{bmatrix} \begin{bmatrix} \Delta G \\ \Delta I \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Delta D + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Delta u \tag{14.8}$$

14.3 Cardiovascular System

14.3.1 Windkessel Model

The arterial system can be modeled as an electrical analog:

Governing equation:

$$C\frac{\mathrm{d}P}{\mathrm{d}t} + \frac{P}{R} = Q(t) \tag{14.9}$$

where:

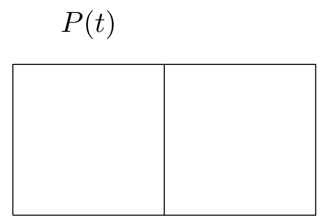

Figure 14.4: Two-element Windkessel model.

- P = arterial pressure (mmHg)
- Q = cardiac output (mL/s)
- C = arterial compliance (mL/mmHg)
- R = systemic vascular resistance (mmHg·s/mL)

14.3.2 Blood Pressure Control

The baroreflex regulates blood pressure:

$$\frac{\mathrm{d}P}{\mathrm{d}t} = -\frac{1}{\tau_P}(P - P_{\mathrm{set}}) + K_u u(t) \tag{14.10}$$

where $u(t)$ represents vasoactive drug infusion.

14.4 Respiratory System

14.4.1 Mechanical Ventilation Model

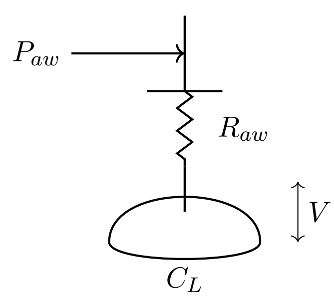

Figure 14.5: Single-compartment lung model.

Equation of motion:

$$P_{aw} = R_{aw}\dot{V} + \frac{V}{C_L} + P_{\mathrm{PEEP}} \tag{14.11}$$

where:

- P_{aw} = airway pressure (cmH_2O)
- R_{aw} = airway resistance (cmH_2O·s/L)

- C_L = lung compliance (L/cmH_2O)
- V = lung volume (L)
- P_{PEEP} = positive end-expiratory pressure

Transfer function:

$$\frac{V(s)}{P_{aw}(s)} = \frac{C_L}{R_{aw}C_L s + 1} \tag{14.12}$$

14.5 Anesthesia Delivery

14.5.1 Propofol Pharmacokinetics

Three-compartment mammillary model:

$$V_1\dot{C}_1 = u - k_{10}V_1C_1 - k_{12}V_1C_1 + k_{21}V_2C_2 - k_{13}V_1C_1 + k_{31}V_3C_3 \tag{14.13}$$

$$V_2\dot{C}_2 = k_{12}V_1C_1 - k_{21}V_2C_2 \tag{14.14}$$

$$V_3\dot{C}_3 = k_{13}V_1C_1 - k_{31}V_3C_3 \tag{14.15}$$

14.5.2 Pharmacodynamic Effect

The effect-site concentration C_e relates to the measured output (e.g., BIS index):

$$\frac{\mathrm{d}C_e}{\mathrm{d}t} = k_{e0}(C_1 - C_e) \tag{14.16}$$

Hill equation for effect:

$$\text{Effect} = E_0 - E_{\max}\frac{C_e^\gamma}{C_e^\gamma + EC_{50}^\gamma} \tag{14.17}$$

14.6 Medical Device Examples

14.6.1 Infusion Pump

Transfer function (actuator dynamics):

$$\frac{Q(s)}{V_{\text{cmd}}(s)} = \frac{K_p}{\tau_p s + 1} \tag{14.18}$$

where Q is flow rate and V_{cmd} is the command voltage.

14.6.2 Pacemaker Rate Response

Activity-based pacing rate:

$$\tau\frac{\mathrm{d}R}{\mathrm{d}t} + R = R_{\text{base}} + K \cdot A(t) \tag{14.19}$$

where $A(t)$ is the activity sensor signal and R is the pacing rate.

14.6.3 Cochlear Implant Signal Processing

Filter bank model:

$$H_k(s) = \frac{\omega_k^2}{s^2 + 2\zeta\omega_k s + \omega_k^2} \tag{14.20}$$

for frequency band k centered at ω_k.

14.7 Neural Systems

14.7.1 Hodgkin-Huxley Neuron Model

$$C_m \frac{\mathrm{d}V}{\mathrm{d}t} = I_{\text{ext}} - g_{Na} m^3 h (V - E_{Na}) - g_K n^4 (V - E_K) - g_L (V - E_L) \tag{14.21}$$

with gating variables m, h, n following first-order kinetics.

14.7.2 Simplified Integrate-and-Fire Model

$$\tau_m \frac{\mathrm{d}V}{\mathrm{d}t} = -(V - V_{\text{rest}}) + R_m I_{\text{ext}} \tag{14.22}$$

When V reaches threshold V_{th}, the neuron fires and resets.

Chapter 15

Industrial and Computer Systems

This section covers systems from manufacturing, process control, and computer science—domains where control theory meets discrete events, queueing, and information flow.

15.1 Industrial Process Control

15.1.1 Conveyor Belt System

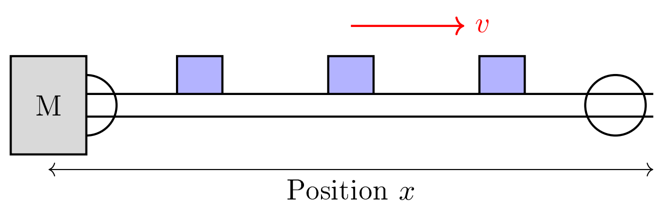

Figure 15.1: Conveyor belt system with motor drive.

Motor-driven conveyor dynamics:

$$J_{\mathrm{eq}}\dot{\omega} + B\omega = K_t i - \tau_L \tag{15.1}$$

where J_{eq} includes reflected inertia of belt and load.
Belt velocity: $v = r\omega$ where r is the drive roller radius.
Transfer function (speed control):

$$\frac{V(s)}{I(s)} = \frac{rK_t}{J_{\mathrm{eq}}s + B} \tag{15.2}$$

15.1.2 Temperature Control Zone

Industrial ovens often have multiple zones:
Each zone:

$$C_i \frac{\mathrm{d}T_i}{\mathrm{d}t} = Q_i - \frac{T_i - T_{i-1}}{R_{i-1,i}} - \frac{T_i - T_{i+1}}{R_{i,i+1}} - \frac{T_i - T_\infty}{R_{i,\infty}} \tag{15.3}$$

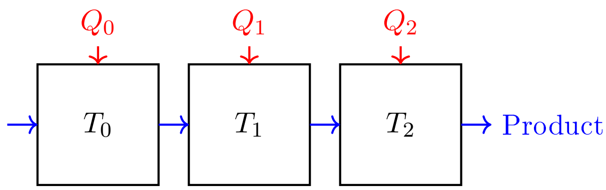

Figure 15.2: Multi-zone temperature control.

15.2 Robotic Manipulators

15.2.1 Single-Link Robot Arm

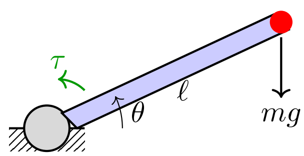

Figure 15.3: Single-link robot arm.

Equation of motion:

$$J\ddot{\theta} + B\dot{\theta} + mg\ell\cos\theta = \tau \tag{15.4}$$

Linearized about horizontal ($\theta_0 = 0$):

$$J\ddot{\theta} + B\dot{\theta} + mg\ell = \tau \tag{15.5}$$

Note: Gravity creates a constant disturbance torque.

15.2.2 Two-Link Planar Arm

The dynamics become coupled and nonlinear:

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau} \tag{15.6}$$

where:

- $\mathbf{M}(\mathbf{q})$ = mass matrix (configuration-dependent)
- $\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})$ = Coriolis and centrifugal terms
- $\mathbf{g}(\mathbf{q})$ = gravity vector
- $\boldsymbol{\tau}$ = joint torques

For a two-link arm:

$$M_{11} = m_1\ell_{c1}^2 + m_2(\ell_1^2 + \ell_{c2}^2 + 2\ell_1\ell_{c2}\cos\theta_2) + I_1 + I_2 \tag{15.7}$$

$$M_{12} = m_2(\ell_{c2}^2 + \ell_1\ell_{c2}\cos\theta_2) + I_2 \tag{15.8}$$

$$M_{22} = m_2\ell_{c2}^2 + I_2 \tag{15.9}$$

15.3 CNC Machine Tools

15.3.1 Axis Drive Model

Each axis of a CNC machine:

$$J_{\text{axis}}\ddot{x} + B\dot{x} + F_c\,\text{sgn}(\dot{x}) = F_{\text{motor}} - F_{\text{cut}} \tag{15.10}$$

where F_c is Coulomb friction and F_{cut} is cutting force disturbance.

15.3.2 Servo System

Position loop transfer function:

$$\frac{X(s)}{X_{\text{cmd}}(s)} = \frac{K_pK_v/s}{1 + K_pK_v/s + s^2/\omega_n^2} \approx \frac{K_p}{\tau_p s + 1} \tag{15.11}$$

for well-tuned systems where $K_v = K_p/\tau_p$.

15.3.3 CNC Machine Control Loop

A complete CNC control architecture uses nested feedback loops for position tracking with real-time disturbance rejection:

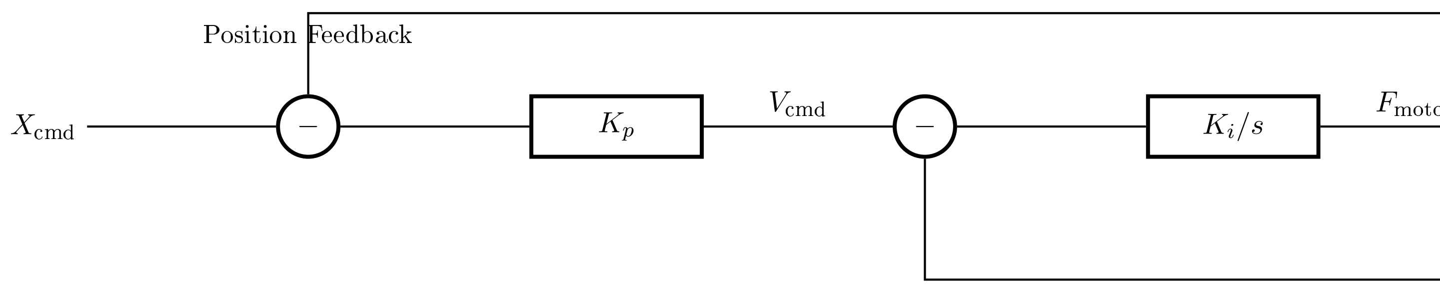

Figure 15.4: CNC machine position control loop with nested feedback and disturbance rejection. The proportional controller commands a velocity set-point, while the integrating controller regulates actual velocity feedback to track the commanded profile.

15.4 Power Systems

15.4.1 Synchronous Generator

Swing equation:

$$\frac{2H}{\omega_s}\frac{\mathrm{d}^2\delta}{\mathrm{d}t^2} + D\frac{\mathrm{d}\delta}{\mathrm{d}t} = P_m - P_e \tag{15.12}$$

where:

- H = inertia constant (seconds)
- δ = rotor angle (electrical radians)
- ω_s = synchronous speed

- P_m = mechanical power
- P_e = electrical power

Linearized about operating point:

$$\frac{\Delta\delta(s)}{\Delta P_m(s)} = \frac{\omega_s/2H}{s^2 + (D\omega_s/2H)s + K_s\omega_s/2H} \tag{15.13}$$

where $K_s = \partial P_e/\partial\delta$ is the synchronizing coefficient.

15.4.2 Automatic Generation Control (AGC)

Area control error:

$$\mathrm{ACE} = \Delta P_{\mathrm{tie}} + B\Delta f \tag{15.14}$$

where B is the frequency bias setting.

15.5 Queueing Systems

15.5.1 M/M/1 Queue

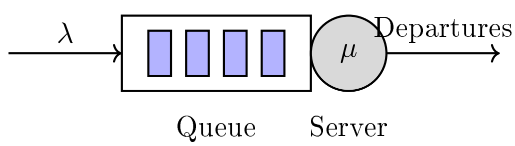

Figure 15.5: M/M/1 queue model.

Steady-state:

$$\text{Utilization:} \quad \rho = \lambda/\mu \tag{15.15}$$

$$\text{Mean queue length:} \quad L = \frac{\rho}{1-\rho} \tag{15.16}$$

$$\text{Mean waiting time:} \quad W = \frac{1}{\mu - \lambda} \tag{15.17}$$

15.5.2 Fluid Flow Model

For control design, queues can be approximated as fluid:

$$\frac{\mathrm{d}q}{\mathrm{d}t} = \lambda(t) - \mu(t) \tag{15.18}$$

where q is queue length, λ is arrival rate, and μ is service rate.
Transfer function:

$$\frac{Q(s)}{\Lambda(s) - M(s)} = \frac{1}{s} \tag{15.19}$$

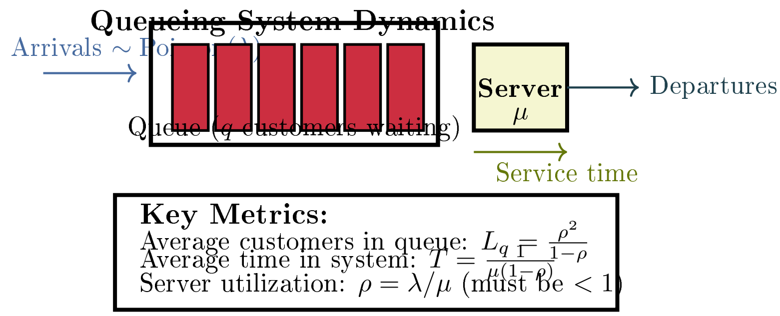

Figure 15.6: Queueing system flow and performance metrics. Customers arrive according to a Poisson process with rate λ, wait in queue, and receive service with exponential service time (mean $1/\mu$). System stability requires $\rho < 1$.

15.6 Computer Systems

15.6.1 CPU Load Control

CPU utilization dynamics:

$$\tau \frac{\mathrm{d}U}{\mathrm{d}t} + U = K_u \cdot R \tag{15.20}$$

where U is utilization, R is admission rate, and τ depends on job service time.

15.6.2 Network Congestion Control

TCP congestion window dynamics (simplified):

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{1}{RTT} - \frac{W}{2} \cdot p \tag{15.21}$$

where W is congestion window, RTT is round-trip time, and p is packet loss probability.

15.6.3 Memory Management

Page fault rate model:

$$f = f_0 \left(\frac{W_s}{M}\right)^{\alpha} \tag{15.22}$$

where M is allocated memory, W_s is working set size, and α is a workload parameter.

15.7 Data Center Thermal Management

Thermal dynamics:

$$C_{\text{room}} \frac{\mathrm{d}T}{\mathrm{d}t} = Q_{\text{IT}} - \dot{m} c_p (T - T_{\text{supply}}) \tag{15.23}$$

where $\dot{m}$ is air mass flow rate (controlled by CRAC fans).

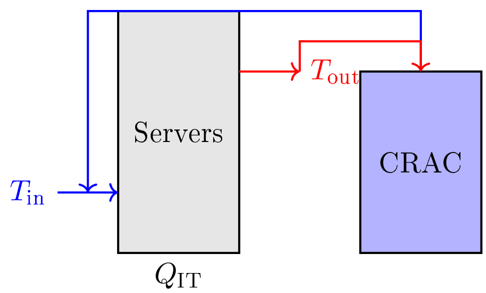

Figure 15.7: Data center thermal model.

15.8 Building Automation

15.8.1 HVAC Zone Model

$$C_z \frac{\mathrm{d}T_z}{\mathrm{d}t} = Q_{\text{HVAC}} + Q_{\text{internal}} + Q_{\text{solar}} - UA(T_z - T_{\text{out}}) \tag{15.24}$$

Transfer function from HVAC power to zone temperature:

$$\frac{T_z(s)}{Q_{\text{HVAC}}(s)} = \frac{1/UA}{\tau s + 1} \tag{15.25}$$

where $\tau = C_z/UA$.

15.8.2 Lighting Control

Dimming dynamics:

$$\tau_L \frac{\mathrm{d}L}{\mathrm{d}t} + L = K_L \cdot d \tag{15.26}$$

where L is light level, d is dimmer setting, and τ_L is lamp response time.

15.9 Traffic Flow

15.9.1 Cell Transmission Model

For a road segment:

$$\frac{\mathrm{d}n_i}{\mathrm{d}t} = q_{i-1} - q_i \tag{15.27}$$

where n_i is vehicle count in cell i and q_i is flow rate.

Flow-density relationship:

$$q = v_f n \left(1 - \frac{n}{n_{\max}}\right) \tag{15.28}$$

15.9.2 Traffic Signal Timing

Green time allocation affects queue discharge:

$$\frac{\mathrm{d}q}{\mathrm{d}t} = \lambda - s \cdot g(t) \tag{15.29}$$

where s is saturation flow rate and $g(t) \in \{0, 1\}$ indicates green signal.

Chapter 16

Problem Set

This chapter consolidates all exercises from the modeling examples chapters. Problems are organized by domain to facilitate focused practice.

Exercise 16.1. A spring-mass system has $m = 2$ kg and $k = 200$ N/m. Find:

(a) The natural frequency in rad/s and Hz

(b) The transfer function $X(s)/F(s)$

(c) The response to a 10 N step force

Exercise 16.2. Add a damper with $b = 8$ N·s/m to the system in Exercise 1.

(a) Calculate the damping ratio

(b) Find the damped natural frequency

(c) Determine if the system is underdamped, critically damped, or overdamped

Exercise 16.3. A motor with $J_m = 0.005$ kg·m^2 drives a load with $J_L = 0.5$ kg·m^2 through a 10:1 gear ratio.

(a) Calculate the reflected load inertia at the motor

(b) Find the total effective inertia seen by the motor

(c) Derive the transfer function from motor torque to load angle

Exercise 16.4. For the quarter-car model, derive the transfer function from road velocity $\dot{X}_r(s)$ to body acceleration $s^2 X_s(s)$. This represents the acceleration felt by passengers.

Exercise 16.5. Linearize the simple pendulum equation about $\theta_0 = \pi/6$ (30 degrees). Find the equilibrium torque and the linearized dynamics.

Exercise 16.6. A tuned mass damper for building vibration control has the following parameters:

- Building mass: $M = 1000$ kg
- Damper mass: $m = 20$ kg
- Damper spring constant: $k = 5000$ N/m

- Damper coefficient: $b = 500$ N·s/m

(a) Calculate the natural frequency of the tuned damper

(b) If the building is excited at 2 Hz, find the phase lag of the damper response

(c) Design the spring constant to match the building frequency of 1.5 Hz

Exercise 16.7. Consider a flexible robot arm modeled as a two-inertia system: motor inertia $J_m = 0.01$ kg·m^2, link inertia $J_L = 0.5$ kg·m^2, shaft torsional stiffness $\kappa = 50$ N·m/rad, and bearing friction $B = 0.1$ N·m·s/rad.

(a) Derive the transfer function from motor torque to link angle

(b) Find the natural frequencies (radians and Hz)

(c) Explain why this system exhibits vibration and how to suppress it

Exercise 16.8. Concept: Compare the responses of three second-order mechanical systems with the same natural frequency ($\omega_n = 10$ rad/s) but different damping ratios: $\zeta_1 = 0.3$, $\zeta_2 = 0.7$, $\zeta_3 = 1.0$.

(a) For each, describe the qualitative step response: overshoot, oscillations, settling time

(b) Rank them by settling time (2% criterion)

(c) Which would you choose for a robotics application and why?

Exercise 16.9. Design: A vibration isolator for precision equipment must satisfy:

- Support mass: $m = 50$ kg
- Isolation efficiency $> 80\%$ at 10 Hz (reduce vibration amplitude by factor of 5)
- Static deflection < 10 cm at rated load

(a) Calculate the required spring constant

(b) Find the natural frequency

(c) Verify the isolation efficiency at 10 Hz (calculate transmissibility)

(d) Specify a damping ratio for practical implementation

Exercise 16.10. Concept: The inverted pendulum on a cart is fundamentally unstable. Explain:

(a) Why feedback control is necessary (discuss the pole locations)

(b) What happens to the system response without control when perturbed

(c) Why a cart-based solution is simpler than a balance-wheel solution

Exercise 16.11. For an RC circuit with $R = 47$ kΩ and $C = 0.22$ μF:

(a) Find the time constant and cutoff frequency

(b) Sketch the Bode plot (magnitude and phase)

(c) Find the output voltage 5 ms after a 10V step input

Exercise 16.12. Design an RLC band-pass filter with center frequency 10 kHz and bandwidth 1 kHz. Calculate R, L, C, and Q.

Exercise 16.13. An op-amp integrator has $R = 100$ kΩ and $C = 1$ μF. If the input is $v_{\text{in}}(t) = 2\sin(100t)$ V:

(a) Find the transfer function

(b) Calculate the steady-state output amplitude and phase

(c) Write the time-domain output expression

Exercise 16.14. For the series RLC circuit, find the current $i(t)$ response to a unit step voltage input when $R = 100$ Ω, $L = 0.1$ H, and $C = 100$ μF. Is the circuit underdamped, critically damped, or overdamped?

Exercise 16.15. Design a second-order Sallen-Key low-pass filter with cutoff frequency 1 kHz and Butterworth response. Specify all component values.

Exercise 16.16. Concept: Compare the frequency response characteristics of three filters:

(a) A first-order low-pass filter with $\omega_c = 1000$ rad/s

(b) A second-order Butterworth low-pass filter with $\omega_c = 1000$ rad/s

(c) A first-order high-pass filter with $\omega_c = 1000$ rad/s

For each, sketch the Bode magnitude plot and describe:

(a) The slope (dB/decade) in the stopband

(b) The phase shift at the cutoff frequency

(c) Which frequencies are attenuated and which are passed

Exercise 16.17. A boost converter steps up a 12V input to 48V at a switching frequency of 200 kHz. Assume efficiency $\eta = 0.92$.

(a) Calculate the required duty cycle

(b) If the load draws 5A at 48V, find the input current

(c) Design the filter inductance to limit current ripple to $\pm 5\%$

Exercise 16.18. Design: An active low-pass filter must attenuate 60 Hz power line noise by at least 40 dB while preserving signals below 50 Hz.

(a) Determine the minimum filter order required

(b) Choose between Butterworth, Chebyshev, and Bessel responses and justify your choice

(c) If using a second-order topology, calculate the component values for unity-gain implementation

(d) Estimate the phase lag at 20 Hz

Exercise 16.19. Computational: For the series RLC circuit with $R = 10\ \Omega$, $L = 1$ H, and $C = 1$ μF:

(a) Find the resonant frequency

(b) Calculate the quality factor Q

(c) Find the magnitude of impedance at resonance, at $\omega = \omega_n/2$, and at $2\omega_n$

(d) If driven by $v_{\text{in}}(t) = 100\sin(\omega_n t)$ V, find the steady-state current amplitude at resonance

Exercise 16.20. Concept: The PID controller can be implemented using three op-amp stages (one proportional, one integrator, one differentiator).

(a) Draw a block diagram showing how three op-amp circuits combine to form a PID controller

(b) Explain the role of each stage in controlling a second-order system

(c) Discuss practical issues with the differentiator stage (e.g., noise) and propose solutions

Exercise 16.21. A DC motor has $R_a = 1\ \Omega$, $K_t = K_e = 0.05$ N·m/A, $J = 2 \times 10^{-4}$ kg·m^2, and $B = 10^{-4}$ N·m·s/rad. Neglect inductance.

(a) Derive the transfer function from voltage to velocity

(b) Find the steady-state speed for $V_a = 12$ V with no load

(c) Find the stall torque (torque at zero speed) for $V_a = 12$ V

Exercise 16.22. For the motor in Exercise 1, a 5:1 gear reducer is added, and the load inertia is $J_L = 0.01$ kg·m^2 with load damping $B_L = 0.01$ N·m·s/rad.

(a) Find the reflected load inertia and damping at the motor

(b) Derive the new transfer function from voltage to load velocity

(c) Compare the new time constant to the original

Exercise 16.23. A voice coil actuator has $R = 5\ \Omega$, $L = 1$ mH, $K_f = K_e = 10$ N/A, $m = 0.1$ kg, and $b = 2$ N·s/m.

(a) Derive the transfer function from voltage to position

(b) Find all poles and determine if the system is stable

(c) Calculate the DC gain

Exercise 16.24. A piezoelectric actuator has $d_{33} = 500$ pm/V, resonant frequency 30 kHz, and damping ratio 0.02.

(a) Write the transfer function

(b) Find the static displacement for 100V input

(c) Find the displacement amplitude at 20 kHz for 100V sinusoidal input

Exercise 16.25. Design: A servo motor must position a robotic arm joint with:

- Load inertia: $J_L = 0.02$ kg·m^2
- Load damping: $B_L = 0.05$ N·m·s/rad
- Desired bandwidth: 20 rad/s
- Maximum allowed steady-state error: 0.01 rad

(a) Select a motor with appropriate torque and speed ratings

(b) Choose a gear ratio to match the bandwidth requirement

(c) Design a proportional feedback controller to meet the steady-state error specification

(d) Estimate the settling time (2% criterion)

Exercise 16.26. Concept: Explain the role of back-EMF in DC motor speed regulation.

(a) Derive how back-EMF creates a speed-dependent resistance to applied voltage

(b) Show that steady-state speed increases linearly with applied voltage

(c) Explain why increased load torque causes the motor to slow down

(d) Discuss how feedback control can maintain constant speed despite load variations

Exercise 16.27. A stepper motor with 200 steps/revolution (1.8° per step) drives a load with moment of inertia $J = 10^{-4}$ kg·m^2 and friction damping $B = 10^{-5}$ N·m·s/rad.

(a) Calculate the motor's step angle in radians

(b) If stepped at 1000 Hz, what is the steady-state rotational velocity?

(c) Calculate the acceleration at a 1000 rad/s^2 acceleration specification

(d) Explain the hunting and resonance issues in stepper motor control

Exercise 16.28. Computational: For a brushless DC motor (BLDC) with field-oriented control:

(a) Derive the relationship between q-axis voltage and motor torque

(b) Show that the motor behaves like a first-order system in the current loop

(c) Calculate the current loop bandwidth for $L = 5$ mH, $R = 1\ \Omega$, and proportional gain $K_p = 10$

(d) Find the settling time for a 10A step reference current

Exercise 16.29. Concept: Compare direct drive motors versus geared motors for a positioning application.

(a) List advantages and disadvantages of each approach

(b) Show how gear ratio affects reflected inertia and bandwidth

(c) Explain the tradeoff between speed and torque

(d) In what applications would you choose each approach?

Exercise 16.30. A metal block ($m = 5$ kg, $C_p = 500$ J/kg·K) is heated by a 100 W heater and loses heat to 25°C ambient through convection ($hA = 5$ W/K).

(a) Write the differential equation for block temperature

(b) Find the thermal time constant

(c) Find the steady-state temperature

(d) How long to reach 90% of steady-state from ambient?

Exercise 16.31. Thermal Resistance Network: A system has two thermal resistances in series: $R_1 = 0.5$ K/W and $R_2 = 0.8$ K/W, with a heat source $\dot{Q} = 500$ W and ambient temperature $T_\infty = 20$°C. The total thermal capacitance is $C_{th} = 10$ kJ/K.

(a) Find the total thermal resistance

(b) Find the steady-state temperature rise above ambient

(c) If the system starts at ambient, find the temperature after 100 seconds

Exercise 16.32. Two tanks in series each have $A = 1$ m^2. The first tank drains into the second through resistance $R_1 = 50$ s/m^2, and the second drains to atmosphere through $R_2 = 100$ s/m^2.

(a) Write the state-space model

(b) Find the transfer function from Q_{in} to h_2

(c) Find the system poles and time constants

(d) Explain how the inter-tank interaction affects the response

Exercise 16.33. Tank Level Dynamics: A conical tank has height-dependent area: $A(h) = \pi(rh/H)^2$ where $r = 0.5$ m is base radius and $H = 2$ m is total height. The outlet has resistance $R_f = 80$ s/m^2.

(a) Write the nonlinear ODE for level h

(b) For $h = 1$ m at steady-state with constant inflow, find Q_{in}

(c) Linearize the model about this operating point

Exercise 16.34. A hydraulic servo system has a 10 cm diameter piston, 1000 kg load mass, and bulk modulus $\beta = 1.5 \times 10^9$ Pa. The chamber volume is 0.5 L.

(a) Calculate the hydraulic natural frequency

(b) If damping ratio is 0.3, find the damping coefficient

(c) Sketch the expected step response

Exercise 16.35. Heat Exchanger Thermal Dynamics: A counter-flow heat exchanger receives hot fluid at 80°C (inlet) with flow rate 0.5 kg/s, and cold fluid at 20°C with flow rate 0.3 kg/s. The heat transfer coefficient is $UA = 2$ kW/K, and thermal capacitances are $C_h = C_c = 5$ kJ/K.

(a) Write the energy balance equations for both sides

(b) Find the steady-state outlet temperatures

(c) Determine the time constants of the hot and cold sides

Exercise 16.36. A hot water tank ($V = 200$ L) receives water at 15°C at rate 0.1 L/s and has a 3 kW heater. Heat loss to ambient is $UA = 10$ W/K. Find the steady-state temperature and time constant. What is the response time to reach 95% of steady-state?

Exercise 16.37. A CSTR with volume 500 L and flow rate 50 L/min processes a first-order reaction with $k = 0.1$ min^{-1}. Feed concentration is 2 mol/L.

(a) Find the steady-state exit concentration

(b) Find the residence time

(c) Write the transfer function from C_{in} to C

(d) Find the time constant

Exercise 16.38. A mixing tank blends two streams: Stream 1 at 100 L/min with 10% salt, and Stream 2 at 50 L/min with 2% salt. Tank volume is 1000 L.

(a) Find the steady-state exit concentration

(b) If Stream 1 concentration suddenly changes to 15%, find the new steady-state and the time to reach 90% of the change

Exercise 16.39. An exothermic batch reactor starts at 25°C with 1000 mol of reactant. Heat of reaction is -50 kJ/mol, heat capacity is 4 kJ/kg·K, mass is 1000 kg, and reaction rate constant is $k = 0.01$ min^{-1} at 25°C with $E_a/R = 5000$ K.

(a) Write the coupled ODEs for concentration and temperature (adiabatic case)

(b) Estimate the adiabatic temperature rise if reaction goes to completion

Exercise 16.40. CSTR Reaction Kinetics: A CSTR operates with a second-order reaction $2A \to B$ having rate law $r = kC_A^2$ with $k = 0.1$ L/(mol·min). The reactor volume is 100 L and inlet concentration is 1 mol/L.

(a) For inlet flow 10 L/min, calculate the steady-state concentration

(b) If temperature increases causing k to double, find the new steady-state

(c) Compare reaction rates at both operating points

Exercise 16.41. CSTR Temperature Control: An exothermic reaction in a CSTR generates heat at rate $\dot{Q}_{\text{gen}} = (-\Delta H_r)k(T)C_A \cdot V$ where $-\Delta H_r = 50$ kJ/mol, $k(T)$ follows Arrhenius with $E_a/R = 8750$ K. The cooling jacket removes heat at $\dot{Q}_{\text{cool}} = UA(T - T_j)$ with $UA = 5$ kW/K.

(a) Write the energy balance ODE

(b) For $T_j = 350$ K (set at nominal operating point), find the steady-state temperature

(c) If T_j suddenly drops to 300 K, predict the qualitative response

Exercise 16.42. A process has FOPDT model with $K = 1.5$, $\tau = 120$ s, and $\theta = 30$ s.

(a) Write the transfer function

(b) Calculate θ/τ and comment on controllability

(c) Use first-order Padé approximation and find the system poles

Exercise 16.43. pH Control Challenge: A strong acid-strong base neutralization uses $[H^+] = \frac{C_a - C_b + \sqrt{(C_a - C_b)^2 + 4K_w}}{2}$ where $K_w = 10^{-14}$.

(a) With acid concentration 0.1 M flowing at 100 mL/min and base 0.1 M at 95 mL/min, find pH at steady-state

(b) Calculate the sensitivity $\frac{d(\text{pH})}{dC_b}$ at this point

(c) If base flow increases to 100.5 mL/min, estimate the new pH and explain the control difficulty

Exercise 16.44. An aircraft has short-period approximation:

$$\frac{\alpha(s)}{\delta_e(s)} = \frac{-2}{s^2 + 1.5s + 4} \tag{16.1}$$

(a) Find the natural frequency and damping ratio

(b) Is this response acceptable for a fighter aircraft?

(c) Sketch the step response

Exercise 16.45. A rocket has pitch moment of inertia $I_{yy} = 10^6$ kg·m^2, thrust $T = 10^6$ N, and gimbal arm $\ell_T = 20$ m.

(a) Find the transfer function from gimbal angle to pitch rate

(b) Design a proportional controller to achieve 1 rad/s^2 pitch acceleration per degree of gimbal

Exercise 16.46. A satellite reaction wheel has inertia $I_w = 0.01$ kg·m^2 and the satellite has inertia $I_s = 100$ kg·m^2 about one axis.

(a) If the wheel accelerates at 100 rad/s^2, find the satellite angular acceleration

(b) Find the maximum satellite rate if the wheel is limited to 6000 rpm

Exercise 16.47. A quadrotor has mass 1.5 kg, arm length 0.25 m, and moment of inertia $I_{xx} = I_{yy} = 0.02$ kg·m^2.

(a) Find the hover thrust per motor

(b) Derive the transfer function from motor force differential to roll angle

(c) If motor response is modeled as first-order with $\tau = 0.05$ s, find the complete roll transfer function

Exercise 16.48. A satellite has inertia $I = 50$ kg·m^2 about the pitch axis. Two reaction wheels are available with inertias $I_w = 0.5$ kg·m^2 each, located 90apart about the body axes.

(a) Write the coupled equations for pitch-yaw control using both wheels

(b) If the pitch wheel accelerates at 200 rad/s^2, find the satellite pitch angular acceleration

(c) Design a control law to achieve zero final angular rate when the pitch wheel reaches 3000 rpm

Exercise 16.49. An aircraft has longitudinal dynamics with short-period natural frequency $\omega_n = 1.2$ rad/s and damping $\zeta = 0.45$. The phugoid has $\omega_p = 0.08$ rad/s and $\zeta_p = 0.06$.

(a) Sketch the pole-zero map for the longitudinal system

(b) Calculate the time constants for each mode

(c) What type of controller (P, PI, or PD) would be best for elevator control? Why?

Exercise 16.50. A rocket undergoes pitch control with gimbal actuation. The rocket parameters are: $I_{yy} = 2 \times 10^6$ kg·m^2, thrust $T = 5 \times 10^6$ N, gimbal arm $\ell = 10$ m, and aerodynamic damping coefficient $M_q = -3 \times 10^5$ N·m·s/rad.

(a) Derive the transfer function from gimbal angle δ to pitch rate q

(b) Find the system poles and determine stability

(c) Design a proportional controller to limit pitch rate to 5 deg/s for 1gimbal command

Exercise 16.51. A quadrotor must perform a roll maneuver from level flight. It has mass $m = 2$ kg, arm length $\ell = 0.3$ m, and $I_{xx} = 0.025$ kg·m^2.

(a) For a rolling maneuver command of 0.5 rad/s^2, calculate the required roll torque

(b) Find the force differential between motors: $\Delta F = F_1 - F_2$

(c) If available motor thrust is 10 N each, what is the maximum roll acceleration?

Exercise 16.52. Consider a satellite attitude control system using a three-axis reaction wheel assembly. The satellite inertia matrix is diagonal: $I_{xx} = 80$, $I_{yy} = 100$, $I_{zz} = 120$ kg·m^2.

(a) Write the decoupled single-axis control law for each axis: $\tau = I\ddot{\theta}$

(b) If maximum wheel torque is 0.2 N·m, find the maximum angular acceleration for each axis

(c) Design a PD controller for the yaw axis: $\tau = K_p e + K_d \dot{e}$, with desired natural frequency 0.1 rad/s and damping ratio 0.7

Exercise 16.53. A drug has one-compartment kinetics with $V = 35$ L and $t_{1/2} = 4$ hours.

(a) Calculate the elimination rate constant

(b) Find the loading dose to achieve 20 mg/L immediately

(c) Find the maintenance infusion rate

(d) Write the transfer function from infusion rate to concentration

Exercise 16.54. A patient's arterial compliance is $C = 2$ mL/mmHg and peripheral resistance is $R = 1$ mmHg·s/mL.

(a) Find the time constant of the arterial system

(b) If cardiac output steps from 80 to 100 mL/s, find the new steady-state pressure

(c) Sketch the pressure response to the step change

Exercise 16.55. A mechanical ventilator delivers pressure to a patient with $R_{aw} = 10$ cmH$_2$O·s/L and $C_L = 0.05$ L/cmH$_2$O.

(a) Find the time constant

(b) If inspiratory pressure is 20 cmH$_2$O above PEEP, find the tidal volume

(c) Find the inspiratory time needed to achieve 95% of the tidal volume

Exercise 16.56. An artificial pancreas controller must regulate glucose. The simplified model has:

$$\frac{G(s)}{U(s)} = \frac{-2}{(10s+1)(50s+1)} \tag{16.2}$$

where G is glucose (mg/dL) and U is insulin (U/hr), with time in minutes.

(a) Find the steady-state glucose change for 1 U/hr insulin increase

(b) Find the system poles and time constants

(c) Sketch the glucose response to a step insulin input

Exercise 16.57. A two-compartment drug delivery system has: $V_1 = 10$ L, $V_2 = 40$ L, $k_{10} = 0.2$ hr^{-1}, $k_{12} = 0.15$ hr^{-1}, and $k_{21} = 0.1$ hr^{-1}.

(a) Calculate the drug concentration in both compartments at 4 hours following a 500 mg IV bolus dose

(b) Find the half-lives for the distribution and elimination phases

(c) If the drug is continuously infused at 10 mg/hr, find the steady-state concentrations in both compartments

Exercise 16.58. A patient receiving mechanical ventilation has measured respiratory mechanics: $R_{aw} = 8$ cmH$_2$O·s/L, $C_L = 0.04$ L/cmH$_2$O, $I_r = 0.03$ L/cmH$_2$O (chest wall compliance).

(a) Find the total respiratory system compliance and time constant

(b) For a constant inspiratory pressure of 15 cmH$_2$O (above PEEP), calculate tidal volume after 5 time constants

(c) Design a pressure control strategy to deliver 0.5 L tidal volume in 1 second

Exercise 16.59. A pacemaker rate-response control system regulates heart rate based on activity. The transfer function from activity sensor voltage to pacing rate is:

$$\frac{R(s)}{A(s)} = \frac{K}{0.5s+1}, \quad K = 100 \text{ bpm/V} \tag{16.3}$$

(a) Calculate the steady-state pacing rate for activity input of 0.5 V

(b) Find the time to reach 90% of steady-state rate

(c) Design a controller to achieve 0 to 120 bpm range with sensor output of 0 to 1 V

Exercise 16.60. An infusion pump delivers IV medications with first-order actuator dynamics: $\frac{Q(s)}{V_{\mathrm{cmd}}(s)} = \frac{K_p}{\tau_p s+1}$, where $K_p = 5$ mL/min/V and $\tau_p = 0.1$ s.

(a) For a step command of 2 V, find the steady-state flow rate

(b) Calculate the actual flow rate at time $t = 0.2$ s

(c) If a patient needs 50 mL of drug infused in exactly 10 minutes, determine the required command voltage accounting for actuator dynamics

Exercise 16.61. A prosthetic limb controller uses proportional myoelectric control, with muscle EMG signal driving joint angle:

$$\theta(s) = \frac{G(s)}{1 + G(s)H(s)}E(s) \tag{16.4}$$

where $G(s) = \frac{K}{\tau s+1}$ (actuator), $H(s) = K_f s$ (feedback with damping), $K = 45$ deg/V, $\tau = 0.2$ s, and $K_f = 0.1$ V·s/deg.

(a) Find the closed-loop transfer function $\theta(s)/E(s)$

(b) For a step EMG input of 0.5 V, calculate steady-state joint angle

(c) Sketch the response and identify if overshoot or oscillation occurs

(d) Design a proportional controller to add to $G(s)$ that achieves 1 deg/V steady-state sensitivity with damping ratio 0.8

Exercise 16.62. A glucose-insulin model for artificial pancreas design is based on the Bergman minimal model. Patient parameters are: $p_1 = 0.03$ min^{-1}, $p_2 = 0.02$ min^{-1}, $p_3 = 1.2 \times 10^{-5}$ mL/(μU·min), $n = 0.1$ min^{-1}, $\gamma = 0.5$ mg/dL$/min/(\mu$U$/mL), with$G$_b = 95$ mg/dL and $I_b = 12$ μU/mL.

(a) Linearize the system about the basal operating point $(G_0, I_0) = (G_b, I_b)$

(b) Find the transfer functions $G(s)/U(s)$ and $G(s)/D(s)$

(c) Design a proportional insulin infusion controller: $u(t) = K_p(G_{\mathrm{ref}} - G(t))$ that regulates glucose to 100 mg/dL with settling time under 3 hours

Exercise 16.63. A conveyor belt has equivalent inertia $J = 2$ kg·m^2, friction $B = 0.5$ N·m·s/rad, drive roller radius $r = 0.1$ m, and motor constant $K_t = 0.5$ N·m/A.

(a) Find the transfer function from motor current to belt velocity

(b) Calculate the current needed for 0.5 m/s belt speed

(c) Find the time constant

Exercise 16.64. A single-link robot arm has $J = 0.5$ kg·m^2, $B = 0.1$ N·m·s/rad, $m = 2$ kg, and $\ell = 0.5$ m.

(a) Write the nonlinear equation of motion

(b) Linearize about $\theta_0 = 0$ (horizontal)

(c) Find the gravity compensation torque

(d) Derive the transfer function from torque to angle (with gravity compensated)

Exercise 16.65. An M/M/1 queue has arrival rate $\lambda = 80$ jobs/s and service rate $\mu = 100$ jobs/s.

(a) Calculate the utilization

(b) Find the mean queue length

(c) Find the mean response time (waiting + service)

(d) If the arrival rate increases to 90 jobs/s, how do the metrics change?

Exercise 16.66. A data center has thermal capacitance $C = 10^6$ J/K, heat load $Q_{\text{IT}} = 100$ kW, and air flow rate that removes heat according to $\dot{Q}_{\text{cool}} = 5000(T - 18)$ W where T is in °C.

(a) Find the steady-state temperature

(b) Calculate the thermal time constant

(c) If IT load suddenly increases to 120 kW, find the new steady-state and time to reach it

Exercise 16.67. A traffic intersection has arrival rate 600 vehicles/hour and saturation flow 1800 vehicles/hour. The cycle length is 90 seconds.

(a) Find the minimum green time to prevent queue buildup

(b) If green time is 40 seconds, calculate the average queue at the start of green

(c) Model the queue dynamics as a hybrid system

Exercise 16.68 (CNC Positioning Control)**.** A CNC machine has proportional position controller gain $K_p = 50$ V/mm and velocity controller gain $K_i = 10$ A/(mm/s). The motor has torque constant $K_t = 0.8$ N·m/A, axis inertia $J = 0.2$ kg·m^2, and viscous damping $B = 2$ N·m·s/rad. The lead screw couples the motor to the axis with a 5 mm pitch.

(a) Convert the velocity command from the position controller into screw rotational velocity (rad/s)

(b) Write the motor dynamics in state-space form with velocity feedback

(c) Find the steady-state position error for a constant velocity ramp command of 10 mm/s

(d) Design the proportional gain K_p to achieve a settling time of 0.5 s for a 1 mm step command

(e) Explain how cutting force disturbance $F_{\text{cut}} = 500$ N affects steady-state accuracy

Exercise 16.69 (Queueing System Design)**.** A server farm handles web requests with M/M/1 queue dynamics. Requests arrive at $\lambda = 50$ req/s and each server processes requests at $\mu = 60$ req/s.

(a) Calculate the server utilization and assess stability

(b) Compute the average number of requests in the system

(c) Find the average response time (waiting + processing)

(d) If we want to maintain $W < 50$ ms, how many additional servers are needed?

(e) Model the queue length as a first-order system: $\dot{q} = \lambda - \mu(t)$, and find the characteristic time constant

Exercise 16.70 (Data Center Thermal Control)**.** A data center has thermal dynamics:

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = Q_{\mathrm{IT}} - Q_{\mathrm{cool}}(T, \dot{m}) \tag{16.5}$$

where $C = 5 \times 10^6$ J/K is the room thermal capacitance, $Q_{\mathrm{IT}} = 500$ kW is the IT heat load, and cooling is $Q_{\mathrm{cool}} = \dot{m}c_p(T - T_{\mathrm{supply}})$ with $c_p = 1005$ J/(kg·K), $T_{\mathrm{supply}} = 15\mathrm{C}$, and controlled mass flow $\dot{m}(t)$ in kg/s.

(a) Find the steady-state room temperature when $\dot{m} = 100$ kg/s

(b) Linearize the system about this operating point

(c) Design a proportional controller for mass flow: $\dot{m}(t) = K_p(T_{\mathrm{ref}} - T(t))$ with $T_{\mathrm{ref}} = 20\mathrm{C}$

(d) Find the time constant of the closed-loop system with $K_p = 5$ kg/(s·°C)

(e) Analyze stability: what maximum K_p prevents oscillation (critical damping)?

Exercise 16.71 (Network Congestion Window Dynamics)**.** TCP congestion control uses a window-based mechanism. The congestion window W (in packets) evolves according to:

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \frac{A}{W} - B \cdot p(W) \tag{16.6}$$

where A and B are constants, and $p(W)$ is the packet loss probability, modeled as $p(W) = 0.3 \cdot (W/100)^2$ when $W > 50$ and $p(W) = 0$ otherwise (no loss during congestion avoidance).

(a) Find the equilibrium window size when $A = 2$, $B = 1$

(b) Analyze stability of the equilibrium: is it stable or unstable?

(c) If the RTT suddenly increases by 20%, how does the equilibrium window change?

(d) Model a step increase in competing traffic (increase in B) and sketch the expected transient response

(e) Propose a controller to maintain target window size $W_{\mathrm{ref}} = 80$ packets

Exercise 16.72 (Two-Layer Queueing Network)**.** Consider a tandem queueing network where jobs arrive at Queue 1 (servers rate $\mu_1 = 100$ jobs/s), then proceed to Queue 2 (servers rate $\mu_2 = 80$ jobs/s). The arrival rate is $\lambda = 70$ jobs/s.

(a) Are both queues stable? Justify your answer

(b) Calculate the average delay at Queue 1 and Queue 2 separately

(c) Find the total average time a job spends in the network

(d) If Queue 2 becomes a bottleneck (μ_2 reduced to 65 jobs/s), what is the new stability boundary?

(e) Propose a feedback control strategy (admission control) to maintain both queues stable despite temporary traffic spikes

(f) Sketch a block diagram showing fluid-flow models of both queues with control feedback